Probability and Statistics

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CONTENTS		
Chapter no.	**CHAPTER NAME**	**Page no.**
6.	**Probability and Statistics**	**4**
	6.1 Introduction	
	6.2 Fundamental Principle of counting	
	6.3 Permutations	
	6.3.1 Restricted permutations	
	6.3.2 Permutations of alike things	
	6.3.3. Permutations of repeated things	
	6.3.4 Circular permutations	
	6.3.5 Clockwise and counter – clockwise permutations	
	6.4 Difference between permutation and combination	
	6.5 Combinations	
	6.5.1 The Pigeonhole principle (PHP)	
	Practice SET	
	6.6 Probability	
	6.6.1 Approaches to probability	
	6.6.2 Conditional Probability	
	6.6.3 Bayes Theorem: Total Law of Probability	
	6.6.4 Random variables	
	6.6.5 Probability density function	
	6.6.6 Binomial Distribution	
	6.6.7 Poisson Distribution	
	6.6.8 Normal Distribution	
	6.7 Statistics	
	6.7.1 Classification of data	
	6.7.2 Arithmetic Mean of a frequency distribution	
	6.7.3 Median	
	6.7.4 Mode	
	6.7.5 Relationship between Mean, Mode and Median	
	6.7.6 Understanding the relevance of Mean and Standard Deviation	
	6.7.7 Standard deviation	
	6.7.8 Variance	
	6.7.9 Range	
	6.7.10 Moments	
	6.7.11 Skewness	
	6.7.12 Kurtosis	
	6.8 Correlation and Regression	
	6.8.1 Correlation	
	6.8.2 Regression	

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	6.8.3 Pearson's Correlation Coefficient (r)	
	SET-16	
	SET-17	
	SET-18	
	SET-19	
	Past GATE Questions	

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Chapter-6 Probability and Statistics

6.1 Introduction

Did-You-Know

Basic combinatorial concepts have appeared throughout the ancient world. In 6th century BC, physician Sushruta asserts in Sushruta Samhita that 63 combinations can be made out of 6 different tastes, taken one at a time, two at a time, etc., thus computing all possibilities.

In the Middle Ages, combinatorics continued to be studied, largely outside of the European civilization. Notably, an Indian mathematician Mahavira (c. 850) provided the general formulae for the number of permutations and combinations.

The rule to determine the number of permutations of n objects was known in Hindu culture at least as early as around 1150: the Lilavati by the Indian mathematician Bhaskara II contains a passage that translates to:

The product of multiplication of the arithmetical series beginning and increasing by unity and continued to the number of places, will be the variations of number with specific figures.

This is an important chapter and must be studied by all students. This is a building block topic, and is used to solve questions in many parts of any competitive test.

Many problems in probability theory require that we count the number of ways that a particular event can occur. For this, we study the topics of permutations and combinations.

Before discussing permutations, it is useful to introduce a general counting technique that will enable us to solve a variety of counting problems, including the problem of counting the number of possible permutations of n objects.

Enjoy.

And BTW, the answer to the team selection question is 1365!

Did you know?
There are 400 ways of playing the first move on each side in a game of chess (one move each of the two players).
There are 197,281 ways of playing the first two moves on each side, an estimated 318,979,564,000 ways of playing the first four moves on each side, and an estimated 169, 518, 829, 100, 544, 000, 000, 000, 000, 000 ways of playing the first ten moves on each side!

What the topic covers?
- Fundamental principle of counting
- Permutations
- Combinations
- Pigeonhole concept

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6.2 Fundamental Principle of counting

Multiplication principle:
If a job can be done in 'm' different ways **AND** another job can be done independently in 'n' different ways then the two jobs can be done, in succession, in 'mn' different ways.

Let's say that you want to flip a coin and roll a dice. There are 2 ways that you can flip a coin and 6 ways that you can roll a dice. There are then 2 × 6 = 12 ways that you can flip a coin *and* roll a dice. i.e. outcomes possible are H1, H2…H6 and T1, T2…T6.

A listing of all the possible outcomes is called the sample space and is denoted by the capital letter S.

The sample space for the experiments of flipping a coin and rolling a die are S = { H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}. Sure enough, there are twelve possible ways. The fundamental counting principle allows us to figure out that there are twelve ways without having to list them all out.

If you want to draw 2 cards from a standard deck of 52 cards without replacing them, then there are 52 ways to draw the first and 51 ways to draw the second, so there are a total of 52 × 51 = 2652 ways to draw the two cards. i.e. outcomes possible or sample space is 2652.

The above principle can be extended to the case in which different operations can be performed in m, n, p,…. ways. In this case the number of ways of performing all the operations together would be m × n × p ×…….

Question
A field has 4 gates, in how many ways is it possible to enter the field by one gate and come out at another?

Answer
There are 4 ways of entering the field, and with of each of these there is a choice of 3 ways of coming out (since the same gate cannot be used to come out); hence the number of ways is 4 × 3, or 12.

Addition Principle
If a total event can be accomplished in two or more alternative events/ways, then the number of ways in which the total event can be accomplished is given by the sum of the number of ways in which each alternative-event can be accomplished.

Number of ways in which the total event can be accomplished
= (Number of ways in which the first alternative-event can be accomplished) + (Number of ways in which the second alternative-event can be accomplished) + (Number of ways in which the third alternative-event can be accomplished) + ...

nE = nEa + nEb + nEc + ...

In how many ways can a committee of atleast 2 members be chosen from a group of 4 men?
To answer the question, we need to find the
a) number of teams with 2 members
b) number of teams with 3 members
c) number of teams with 4 members.

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As per addition principle, answer will be a + b + c
Let the 4 people be numbered P, Q, R and S.
Teams with two people = PQ, PR, PS, QR, WS, RS = 6
Teams with three people = PQR, PQS, PRS, QRS = 4
Teams with four people = PQRS = 1

No. of ways in which a committee of atleast 2 members be chosen from a group of 4 men = 6 + 4 + 1 = 11

6.3 Permutations

Suppose there are k seats and n people are to be arranged in these k seats. Note that $n \geq k$ and the seats have a specified order. We need to find the total number of possible arrangements or permutations.

In the first place, any one of the n people can sit. For any given person sitting in the first seat, any one of the remaining (n –1) persons can sit in the second seat. So, for each of the n ways of filling up the first seat, there are (n–1) ways for filling up the second. Thus, total number of ways for filling up the first two seats is n(n – 1). For each of these n(n – 1) ways to fill up the first two there are (n –2) ways to fill up the third, as any of the remaining people can sit on the third seat.

$\therefore$ **Total number of ways to fill up k seats**
$= n (n – 1)(n – 2) ..... (n – k + 1)$

A permutation is an arrangement of objects, without repetition, and order being important. Another definition of permutation is the number of such arrangements that are possible. The two key things to notice about permutations are that there is no repetition of objects allowed and that order is important.

The number of ways of arranging 'n' things taken 'r' at a time is called the number of permutations of 'n' things taken 'r' at a time and is denoted by ${}^{n}p_{r}$ or p (n, r).

[Learning a new notation:
n(n –1)(n –2) ..... 3.2. 1 is expressed as n! where n is a positive integer
1! = 1
2! = 2 × 1 = 2
3! = 3 × 2 × 1 = 6
And by convention, 0! = 1]

Now P (n, k) or number of permutations of 'n' things take 'k' at a time
$= n(n – 1)(n – 2) ..... (n – k + 1)$

$$= \frac{n(n-1)(n-2)...(n-k+1)(n-k)(n-k-1)...3\times2\times1}{(n-k)(n-k-1)...3\times2\times1} = \frac{n!}{(n-k)!}$$

[Also note that p (n, n) = ways of arranging n things among themselves = $\frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$]

Now, suppose there are k seats and n people are to be arranged, but k > n

Now, the first person can occupy any of the k seats. For each position of the first person, the second can occupy any of the remaining (k –1) positions resulting in the first two people can be seated in k(k –1) ways. Proceeding thus, we find that the number of permutations = $\frac{k!}{(k-n)!}$

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Examples of permutations:

Example 1: List all permutations of the letters ABCD

ABCD	BACD	CABD	DABC
ABDC	BADC	CADB	DACB
ACBD	BCAD	CBAD	DBAC
ACDB	BCDA	CBDA	DBCA
ADBC	BDAC	CDAB	DCAB
ADCB	BDCA	CDBA	DCBA

Now, if you didn't actually need a listing of all the permutations, you could use the formula for the number of permutations. There are 4 objects and you're taking 4 at a time.

$${}^4P_4 = \frac{4!}{(4-4)!} = \frac{4!}{0!} = \frac{4!}{1} = \frac{24}{1} = 24$$

This also gives us another definition of permutations. A permutation when you include all n objects is n!. That is, P(n,n) = n!

Example 2: List all three letter permutations of the letters in the word HAND

HAN	AHN	NHD	DHA
HNA	ANH	NDH	DAH
HAD	AHD	NAH	DAN
HDA	ADH	NHA	DNA
HND	AND	NAD	DHN
HDN	ADN	NDA	DNH

Now, if you didn't actually need a listing of all the permutations, you could use the formula for the number of permutations. There are 4 objects and you're taking 3 at a time. ${}^4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = \frac{24}{1} = 24$

Question

In how many ways can 5 posts be assigned to 5 candidates selected after an interview?

Answer

The post can be assigned in P(n,k) ways, n=5; k=5

P(5,5) = $\frac{5!}{(5-5)!} = \frac{5!}{0!}$ = 5! = 5 x 4 x 3 x 2 x 1 = 120 ways.

Question

Find the permutations of four alphabets p, q, r, s taken 2 at a time.

Answer

Number of permutations = P(n,k), n = 4, k = 2

P (n,k) = P (4,2) = $\frac{4!}{(4-2)!} = \frac{4\times3\times2\times1}{2\times1}$ = 12.

Note: Suppose one object is to be included in all permutations of n things, taken k at a time. The item that is to be always included is kept in any one of the k places.

The remaining (n –1) things can be arranged in the remaining (k –1) places in p(n–1, k–1) = $\frac{(n-1)!}{(n-k)!}$ ways.

Since this is true for each of the k positions of the constant article,

total number of permutations = k $\frac{(n-1)!}{(n-k)!}$ or k, p(n -1, k -1).

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Do-it-yourself (DIY exercise)

1. How many different signals can be made by 5 flags from 8-flags of different colours?
2. How many words can be made by using the letters of the word "SIMPLETON" taken all at a time?

6.3.1 RESTRICTED PERMUTATIONS:

Type I: ALWAYS OCCUR / NEVER OCCUR

Example: (i) In how many permutations of 15 things taken 6 at a time will one thing (a) always occur (b) never occur?

Keeping aside the particular thing which will always occur, the number of permutations of 14 things taken 5 at a time is ${}^{14}P_5$. Now this particular thing can take up any one of the six places and so can be arranged in 6 ways. Hence the total number of permutations = ${}^{14}P_5 \times 6 = 14\times13\times12\times11\times10\times6 = 1441440$ ways.

Leaving aside the particular thing which has never to occur, the number of 14 things taken 6 at a time is ${}^{14}P_6 = 14\times13\times12\times11\times10\times9 = 2162160$ ways.

Type II: WHEN CERTAIN THINGS DO NOT OCCUR TOGETHER:

(a) When the number of things not occurring together is two:
Find the total number of permutations when no restriction is imposed on the manner of arrangement. Then find the number of arrangements when the two things occur together. The difference of the two results gives the number of permutations in which the two things do not occur together.

Example: Find the number of ways in which 10 books can be placed on a shelf when two particular books are never together.

The two particular books can be considered as one book, and there are (10 –1), i.e. 9 books now which can be arranged in ${}^{9}P_9$ i.e. 9! ways. Now these two books can be arranged amongst themselves in 2! ways. Hence the total number of permutations in which these two books are placed together is (9! × 2!). The number of permutation 10 books without any restrictions is 10!.

Therefore, the number of arrangements in which these two books never occur together = $10! - 2! \times 9! = 10 \times 9! - 2 \times 9! = 9!\,(10 - 2) = (9! \times 8)$ ways.

(b) When the number of things not occurring together are more than two:
Example: In how many ways can 4 boys and 3 girls be arranged in a straight line so that no two girls are together?

First we arrange the position of the four boys indicated by B_1, B_2, B_3, B_4.

$$\times B_1 \times B_2 \times B_3 \times B_4 \times$$

This can be done in 4! ways.

Now, if the position of girls are fixed at places (including the two ends) shown by crosses, the three girls will never come together. In any of these arrangements there are 5 places for 3 girls and so the girls can sit in ${}^{5}P_3$ ways.

Hence the required number of ways in sitting of 4 boys and 3 girls is :
${}^{5}P_3 \times 4! = 5 \times 4 \times 3 \times 4 \times 3 \times 2 \times 1 = 1440$ ways.

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Type III: FORMATION OF NUMBERS WITH DIGITS:

(a) There is no restriction:
Example: Find the number of 4 digit numbers that can be made from 1,2,3,4,5
The number of 4 – digit numbers = ${}^5P_4 = 120$.
(b) Number in which a particular digit occupies a particular place:
Example: Find the total number of 6-digit numbers that can be formed with the digits, 1,2,3,4,5,6 such that 5 always occupies the one's place.
In this case the one's place is fixed and the remaining five places can be filled in by the remaining 5 digits in ${}^5P_5$ i.e. 5! ways, i.e. 120 ways.

$\therefore$ The number of numbers in which 5 occurs in the unit's place = 120.

(c) Numbers divisible by a particular number:

Example: Find the total number of 6-digit numbers that can be formed with the digits 2,3,4,5,6,7 which are divisible by 2.

These numbers will have 2 or 4 or 6 in the unit's place. Thus the unit's place can be filled in 3 ways. After filling up the unit's place in any one of the above ways, the remaining five places can be filled in ${}^5P_5$ ways i.e. 5! i.e. 120 ways.
$\therefore$ The total number of numbers divisible by $2 = 120 \times 3 = 360$.

(d) Numbers having particular digits in the beginning and the end:
Example: Find the total number of 6-digit numbers that can be formed with 0,1,2,3,4,5 such that the number begins with 2 and ends with 0.

Here the first and the last places are fixed and the remaining four places can be filled in 4! i.e 24 ways by the remaining four digits. Hence, the total number of numbers beginning with 2 and ending with 0 is 24.

(e) Numbers which are smaller than or greater than a particular number:
Example: Find the total number of numbers that can be formed with 3,5,7,8,9 which are greater than 50000.

In these numbers there will be 5 or a digit greater than 5 i.e. 7,8 or 9 in the ten thousand's place. Thus this place can be filled in 4 ways. The remaining four places can be filled in 4! = 24 ways.
Hence the total number of numbers = $4 \times 24 = 96$.

(f) 5 digit even numbers which are greater than 50000 formed from 4, 5, 6, 7 and 8.

Here the ten-thousandth can be 5, 6, 7 or 8. Also, the last digit can be 4 or 6 or 8.

Here, both conditions overlap i.e. say a number has 6 in the ten-thousandth place and hence cannot have 6 in the units place.

So a simple approach will not work here. We will perhaps have to split the cases as per the digit in the ten-thousandth place.

Case 1: Let digit in ten-thousandth place be 5 or 7. Then the units place, there are 3 choices. The remaining 3 digits can be arranged in 3! ways. This total number of ways = $2 \times 3 \times 3! = 36$

Case 2: Let digit in ten-thousandth place be 6. Then the units place, there are 2 choices, i.e. 4 or 8. The remaining 3 digits can be arranged in 3! ways. This total number of ways = $1 \times 2 \times 3! = 12$

Case 3: Let digit in ten-thousandth place be 8. Then the units place, there are 2 choices, i.e. 4 or 6. The remaining 3 digits can be arranged in 3! ways. This total number of ways = $1 \times 2 \times 3! = 12$

So Total number of ways = Case 1 + Case 2 + Case 3 = 60

Note : *It could have also been done by fixing units place first and then splitting accordingly.*

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Type IV: WORD BUILDING:

(a) There is no restriction on the arrangement of the letters:
Example: Form words with the letters of the word 'Problem'.
The seven different letters can be arranged in ${}^{7}P_{7} = 7! = 5040$ ways.
$\therefore$ The total number of words = 5040.

(b) All words begin with a particular letter:
Example: form words with the letters of the word 'Publish' such that all words begin with 'b'.
All words begin with 'b'. The remaining 6 places can be filled with remaining 6 letters in 6! = 720 ways.
Hence, the total number of words = 720.

(c) All words begin and end with particular letters:
Example: Form words with the letters of the word 'Century' such that the word begins or ends with 'u' or 't'.
The first and last position could have been filled in ${}^{2}P_{2} = 2$ ways.
The remaining positions are filled in ${}^{5}P_{5} = 5! = 120$ ways.
Hence the number of words = $2 \times 120 = 240$.

(d) Vowels occur together:
Example: Form words with the letters of the word 'wheat' such that the vowels occur together.
The vowels 'e' and 'a' can be regarded as one letter. The 4 letters can be arranged in 4! = 24 ways. These two vowels can be arranged amongst themselves in 2! = 2 ways.
Hence, the total number of words = $24 \times 2 = 48$.

(e) Consonants occur together:
E*xample:* Form words with the letters of the word 'Chlorine' such that the consonants occur together.
The consonants 'c', 'h', 'l', 'r', 'n' can be regarded as one letter and the 4 letters can be arranged in 4! = 24 ways.
These 5 consonants can be arranged amongst themselves in 5! = 120 ways.
Hence, the number of words in which consonants occur together = $24 \times 120 = 2880$.

Do-it-yourself (DIY exercise)

3. In how many ways can the letters of the word "Pre-University" be arranged?
4. A child has 3 pocket and 4 coins. In how many ways can he put the coins in his pocket.

6.3.2 PERMUTATIONS OF ALIKE THINGS:

The different permutations of the letters A, B and C are ABC, ACB, BAC, BCA, CAB, CBA.

To get the different permutations of the letters A, B and A, we replace all C with A. We get the permutations as ABA, AAB, BAA, BAA, AAB, ABA. However, there is now a repetition of ABA, AAB and also of BAA. Hence, actual number of permutations is 3.

As the letter A featured twice in the three digits, you divide the final answer by 2! as that is the number of permutation of the letters 2 As.

Similarly, to get the different permutations of the letters A, B, A and B, we find total permutations of 4 letters, i.e. 4! and divide it by 2! × 2!.

The number of permutations of n things taken all at a time where p of the things are alike and of one kind, q others are alike and of another kind, r others are alike and of another kind, and so on is $\frac{n!}{p!q!r!....}$

Example: Find the number of permutations of arranging the letters of the word 'INDEPENDENCE' amongst themselves? In how many arrangements will the two D's be together?

The word INDEPENDENCE has 12 letters of which 'N' is repeated 3 times, 'D' is repeated twice, 'E' is repeated 4 times, and the rest are different.

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Hence the number of arrangements is given by $\frac{12!}{3!\ 2!\ 4!}$.

If the two D's must be together, they can be considered as one letter.

Hence there are 11 letters of which 'N' is repeated 3 times and 'E' four times.

$\therefore$ Number of arrangements = $\frac{11!}{3!\ 4!}$

6.3.3. PERMUTATIONS OF REPEATED THINGS:

The number of permutations of n different things taken r at a time, when each thing may occur any number of times is n^r.

Example: How many numbers each containing five digits can be formed?

There are in all ten digits including zero. As the first digit of the number cannot be zero, so it can be chosen in 9 ways. As repetition is permitted, the second, third, forth and fifth digits of the numbers can be any of the ten digits and so each of the remaining four places can be filled in 10 ways.

Hence the total number of 5 digit numbers $= 9 \times 10^4 = 90000$.

A note about Permutations with repetition

Suppose there are k posts, and they are to be nominated for n people. If one person can be nominated to only one post, then the number of ways = $\frac{n!}{(n-k)!}$. But if repetition is allowed, i.e. one person can be nominated to more than one posts, then the first post can be filled in n ways, the second post can also be filled in 'n' ways and so on.

$\therefore$ Total number of ways = $n \times n \times$ .....n(k times) = n^k

Note that the logic is NOT that each person can be nominated in k ways and therefore number of ways = k^n. Do not commit this mistake.

Question

There are 3 letters which have to be put into 4 boxes. In how many ways can be this done?

Answer

Each letter can be put in any of the 4 boxes.

$\therefore$ Total number of ways = $4 \times 4 \times 4 = 64$.

6.3.4 CIRCULAR PERMUTATIONS:

Two ways of arranging five digits 1, 2, 3, 4, 5 would be 12345, 54321. Now, if these arrangements are written along the circumference of a circle, the two arrangements are one and the same. Thus, circular permutations are different only when the relative order of the objects is changed; otherwise they are the same. As the number of circular permutations depend on the relative position of objects, we fix the position of one object and arrange the remaining (n – 1) objects in all possible ways. This can be done in (n – 1)! ways.

Example: (i) In how many ways can 8 people sit around a table?
Fixing the position of one of the persons, the remaining 7 people can sit around the table in 7!, i.e. 5040 ways.

(ii) In how many ways can 3 boys and 3 girls be seated at a table so that no 2 boys are adjacent?
Let the girls first take up their seats. They can sit in 2! ways. When they have been seated, there remain 3 places for the boys each between two girls. Therefore the boys can sit in 3! ways. Therefore there are 2!× 3!, i.e. 12 ways of seating.

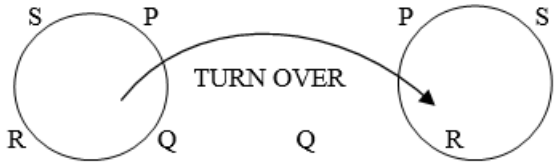
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6.3.5 **CLOCKWISE AND COUNTER – CLOCKWISE PERMUTATIONS**:

Case (i) In which counter-clockwise and clockwise are distinguishable.
While seating 4 persons P, Q, R, S around a table, the permutations are considered different.



Case (ii) Those in which counter clockwise and clockwise are not, distinguishable.
(a) While forming a garland, or stringing beads for a chain, the arrangements are not distinguishable, as the arrangements are not disturbed if we turn the garland over.



Question
Find the number of ways in which 20 beads can be arranged to form a necklace.

Answer
Fixing the position of one bead, the remaining beads can be arranged in (20 – 1)! i.e 19! ways.
As there is no distinction between the clockwise and anti-clockwise arrangements, the required number of ways = $\frac{1}{2}(19)!$

(b) The distinction between clockwise and anti-clockwise is ignored when a number of people have to be seated around a table, so as not to have the same neighbors.

Question
In how many ways can 5 girls sit around a table so that all shall not have the same neighbors in any two arrangements.

Answer
5 girls can sit around a table in 4! ways. As each person shall not have the same neighbors, the required arrangements = $\frac{1}{2}(4!) = \frac{24}{2}$ = 12 ways

Do-it-yourself (DIY exercise)

5. How many necklace of 12 beads each can be made from 18 beads of different colours?
6. How many 5-lettered words can be formed with the letters of the word 'OMEGA' when:
(i) 'O' and 'A' occupying end places.
(ii) 'E' being always in the middle
(iii) Vowels occupying odd-places
7. In how many ways can the letters of the word INDIA be arranged?
8. In how many ways can the letters of the word MISSISSIPI be arranged so that the two letters M & P are never together?
9. How many four digit numbers can be formed using 0,1,2,3,4 and 5?

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6.4 DIFFERENCE BETWEEN PERMUTATION AND COMBINATION

Before we get to Combinations, let us understand the difference between permutation and combination.

Each of the selections which can be made by taking some or all of a number of things (without regard to the order of the things in each group) is called a COMBINATION.

Example: The combination which can be made by taking the four letters p, q, r, s two at a time are six in number; namely pq, pr, ps, qr, qs, rs, each of these presenting a different selection of two letters. Thus when r things are selected out of n, each selection is called a nr – combination, and the number of ways in which such a selection can be made is called the number of combination of n things taken r at a time, and is denoted by the symbol ${}^{n}C_{r}$.

If the things in each selection are arranged in all possible orders, each of such arrangements is called a PERMUTATION.

Example: Each of the foregoing selections of the letters p, q, r, s taken two at a time, are twelve in number: namely, pq, pr, ps, qr, qs, rs, qp, rp, sp, rq, sq, sr.

Each arrangement which can be made by taking r things out of n is called a nr-permutation, and the number of ways in which such an arrangement can be made is called the number of permutations of n things taken r at a time, and denoted by the symbol ${}^{n}P_{r}$.

6.5 COMBINATIONS:

Each of the different groups or selections which can be made by taking some or all of a number of things at a time (irrespective of order) is called as combination.

Thus ${}^{5}C_{3}$ denotes the number of combinations of 5 things taken 3 at a time.

$${}^{n}C_{r} = \frac{n!}{r!(n-r)!}\ ,\ \therefore {}^{n}C_{r} = \frac{{}^{n}p_{r}}{r!} \qquad \left(\text{since } {}^{n}P_{r} = \frac{n!}{(n-r)!}\right)$$

Note: (i) ${}^{n}C_{r} = {}^{n}C_{n-r}$

(ii) ${}^{n}C_{n} = {}^{n}C_{o} = 1$

(iii) If ${}^{n}C_{x} = {}^{n}C_{y}$ then either $x = y$ or $x + y = n$

(iv) ${}^{n+1}C_{r} = {}^{n}C_{r} + {}^{n}C_{r-1}$

Logical interpretation of the Notes:

1. ${}^{n}C_{r} = {}^{n}C_{n-r}$ means that if we select r objects from n, automatically n – r objects are left out. Thus selecting r objects deselects n – r objects. Example, selecting 3 students out of 10 for a race is the same as selecting 7 out of 10 not to participate in the race.

2. (ii) ${}^{n}C_{n} = {}^{n}C_{o} = 1$ means if the number of ways of selecting all members to create a team is 1 (in this case all will be selected), and it can be also perceived as the number of ways of deselecting all members who would not be a part of the team (in this case 0 will be deselected).

3. (iii) If ${}^{n}C_{x} = {}^{n}C_{y}$ then either $x = y$ or $x + y = n$ can be inferred from Note 1.

4. (iv) ${}^{n+1}C_{r} = {}^{n}C_{r} + {}^{n}C_{r-1}$. Let us take the case that while we are making a selection of r objects from n, and after making the first selection, an object gets added to n.

If we select 1 object out of n, automatically n – 1 are left out, and 1 object has been selected. This can be selected in ${}^{n}C_{r}$ or ${}^{n}C_{1}$ way. Now if 1 more object is added to the list of items from which the selection is being made, we again have n objects. But now to make r selections, we need to further select r – 1, as 1 has already been selected. So the number of ways of selecting r – 1 from n is ${}^{n}C_{r-1}$. i.e. ${}^{n+1}C_{r} = {}^{n}C_{r} + {}^{n}C_{r-1}$

Career Avenues

Question
If ${}^{16}C_r = {}^{16}C_{r+2}$, find the value of ${}^{r}C_4$.

Answer
Since $r \neq r + 2$, so $r + r + 2 = 16$
$\therefore 2r = 14 \Rightarrow r = 7.$
$\therefore {}^{7}C_4 = \frac{7!}{4!3!} = \frac{7\times6\times5}{3\times2} = 35.$

Question
Find the number of diagonals that can be drawn by joining the angular points of a pentagon?

Answer
A pentagon has 5 angular points and 5 sides. The join of 2 angular points is either a side or a diagonal.
The number of lines joining the angular points = ${}^{5}C_2 = \frac{5!}{2!3!} = \frac{5\times4}{2} = 5\times2 = 10$
But the number of sides = 5. $\therefore$ The number of diagonals = 10 – 5 = 5.

Challenge question
A committee of 4 is to be selected from 7 boys and 5 girls. In how many ways can this be done so as to include (a) exactly 1 girl, (b) at least 1 girl.

Answer
(a) In this case we have to select 1 girl out of 5 girls and 3 boys out of 7 boys.
The number of ways of selecting 3 boys = ${}^{7}C_3 = \frac{7!}{4!3!} = \frac{7\times6\times5}{3\times2} = 35$
The number of ways of selecting 1 girl = ${}^{5}C_1 = 5$
$\therefore$ The required committee can be formed in 35 × 5 i.e 175 ways.

(b) The committee can be formed with
(i) four girls (ii) three girls and one boy (iii) two girls and two boys (iv) one girl and three boys.
$\therefore$ The committee formed in (i) ${}^{5}C_4$ ways, (ii) ${}^{5}C_3 \times {}^{7}C_1$ ways (ii) ${}^{5}C_2 \times {}^{7}C_2$ ways (iv) ${}^{5}C_1 \times {}^{7}C_3$ ways.
Hence the required number $= 5 + \left(\frac{5!}{3!\times2!}\times7\right) + \left[\left(\frac{5!}{3!\times2!}\right)\times\left(\frac{7!}{2!5!}\right)\right] + \left[5\times\frac{7!}{4!3!}\right]$
$= 5 + \frac{5\times4}{2}\times7 + \frac{5\times4}{2}\times\frac{7\times6}{2} + \frac{5\times7\times6\times5}{3\times2} = 5 + 70 + 210 + 175 = 460$ ways.

<u>Type 1: To find the total number of combinations of n dissimilar things taking any number at a time</u>:

(A) When all things are different:
Each thing maybe disposed of in two ways i.e. it maybe included or rejected.
$\therefore$ The total number of ways of disposing of all the things = 2 × 2 × 2 … n times = 2^n.

But this includes the case in which all things are rejected.
Hence the total number of ways in which one or more things are taken = $2^n - 1$

Example: There are 6 subjects in an examination. In how many ways can a boy pass in one or more subject?

The boy can either pass or fail in one subject.
Thus the number of ways of doing either = 2^6.

But this includes the case of failing in all subjects.
Hence, the total number of ways of passing in one or more subject = $2^6 - 1 = 64 - 1 = 63$.

(B) When all things are not different:
Suppose, out of (p + q + r ….+ x) = p are of one kind, q are of a second kind, r are of a third kind and the rest x are different:

Career Avenues

Out of p things we may take, 0, 1, 2 … or p.

Hence they maybe disposed in (p + 1) ways. Similarly q alike things maybe disposed of in (q + 1) ways and r alike things in (r + 1) ways. The x different things maybe disposed in 2^x ways.

This includes that case in which all are rejected.
$\therefore$ The total number of selections = $(p + 1)(q + 1)(r + 1)\,2^x - 1$

Example: Find the number of ways in which one or more letters from the phrase, 'Ten Tired Turtles' be selected.

In the phrase there are 4 t's, 3 e's, 1n, 1i, 2r, 1d, 1u, 1l, 1s
$\therefore$ The total number of selections = $(4+1)(3+1)(1+1)(1+1)(2+1)(1+1)(1+1)(1+1)(1+1) - 1$
$= 5 \times 4 \times 2 \times 2 \times 3 \times 2 \times 2 \times 2 \times 2 - 1 = 3840 - 1 = 3839$

Type 2: Division into groups:
To find the number of ways in which x + y different things can be divided into two groups containing x and y different things respectively.

Every time a set of x things are taken, a second set of y things are left behind. The number of combinations of (x+y) things taking x things at a time = $^{x+y}C_x = \frac{(x+y)!}{x!(x+y-x)} = \frac{(x+y)!}{x!\,y!}$

In general, $^{x+y+z}C_x \times {}^{y+z}C_y \times {}^{z}C_z = \frac{(x+y+z)!}{x!(y+z)!} \times \frac{(y+z)!}{y!\,z!} \times 1 = \frac{(x+y+z)!}{x!\,y!\,z!}$

Note: (i) The number of ways in which 4p things can be divided equally into 4 distinct groups in $\frac{(4p)!}{(p!)^4}$

(ii) The number of ways in which 4p things can be divided into four identical groups is $\frac{(4p)!}{4!(p!)^4}$

Examples:

(i) In how many ways can 20 things be divided into groups of 8, 7 and 5 things respectively?

The number of ways = $\frac{20!}{8!\,7!\,5!}$

(ii) In how many ways can 20 different sweets be equally divided among 4 students?

The number of ways = $\frac{(20)!}{(5!)^4}$

Type 3: Permutations and Combinations occurring simultaneously:

Example: How many different 5 letter words, each containing 2 vowels and 3 consonants can be formed with 5 vowels and 17 consonants?

The vowels can be selected in $^5C_2$ ways. Three consonants can be selected in $^{17}C_3$ ways.
$\therefore$ 2 vowels and 3 consonants can be selected in $^5C_2 \times {}^{17}C_3$ ways.

Now each group of 2 vowels and 3 consonants can be arranged in 5! ways.
$\therefore$ The total number of words = $^5C_2 \times {}^{17}C_3 \times 5! = 816000$.

Career Avenues

Do-it-yourself (DIY exercise)

10. If ${}^{n}p_{4} = 12 \times {}^{n}P_{2}$, find n **.**

11. A gentleman has 6 friends to invite. In how many ways can he send invitation cards to them if he has three servants to carry the cards, assuming two friends can be invited by the same servant.

12. How many 3-digit number divisible by 5 can be formed using any numerals from 0 to 9 without repetition?

Challenger question 1

Question

One red flag, three white flags and two blue flags are arranged in a line such that,

(a) no two adjacent flags are of the same colour

(b) the flags at the ends of the line are of different colours.

In how many different ways can the flags be arranged?

1. 6 2. 4 3. 10 4. 2

Answer

The possible arrangements of the six flags are:

R W B W B W

W B W B W R

W B W R W B

W R W B W B

B W B W R W

B W R W B W

Hence the answer is [1]

Challenger question 2

Question

Sam has forgotten his friend's seven-digit telephone number. He remembers the following: the first three digits are either 635 or 674, the number is odd, and the number nine appears once. If Sam were to use a trial and error process to reach his friend, what is the minimum number of trials he has to make before he can be certain to succeed?

1] 10000 2] 2430 3] 3402 4] 3006

Answer

Let us consider the last 4 digits as x x x x.

<u>case i</u> the last digit is 9.

$\therefore$ the remaining 3 digits can be filled by any of the 9 digits (i.e. 0 to 8) in 9^3 ways.

$\therefore$ total number of ways = 729.

<u>case ii</u> The last digit is not 9.

$\therefore$ the last digit can be filled in 4 ways.

Let any of the other 3 digits be 9. This can happen in 3 ways.

$\therefore$ the remaining two digits can be filled in 9^2 i.e. 81 ways.

$\therefore$ the total number of ways = $9^2 \times 3 \times 4 = 972$

$\therefore$ the minimum number of trials = 2 [729 + 972] = 3402

Hence the answer is [3]

Challenger question 3

Question

Let S be the set of five-digit numbers formed by the digits 1, 2, 3, 4 and 5, using each digit exactly once such that exactly two odd positions are occupied by odd digits. What is the sum of the digits in the rightmost position of the numbers in S?

1] 228 2] 216 3] 294 4] 192

Career Avenues

Answer
The 5 digit number is has 3 odd places. Odd Even Odd Even Odd
Of this, two places need to be filled with odd numbers from 1, 3 or 5. The third odd place can be 2, 4 or the remaining odd number.
Case 1
Let the rightmost digit be odd.
Case 1a : The rightmost digit is the number 1.
Then either the left most digit or the center digit can be odd.
Case 1a(i). The left most digit is odd.

Therefore, the number of ways with left most place odd, center place even and rightmost place as 1 is 2 × 2 × 2 × 1 × 1, i.e. 8. (Steps: Right most place is in 1 way i.e. 1, Leftmost place in 2 ways i.e. 3 or 5, Center digit in 2 ways i.e. 2 or 4, and the remaining two places can be filled with the remaining two numbers in 2 × 1 way).

So with 1 in the units place, 8 numbers can be created. Sum of numbers in the unit place = **8 × 1 = 8**.

Case 1a(ii). The center most digit is odd. It would be in the same format as above. Hence sum of numbers in the unit place = **8 × 1 = 8.**

Case 1b : The rightmost digit is 3. Again there would be two cases as center digit odd or left most digit is odd. We will again get 8 ways in which numbers can be created in the two different cases. Sum of numbers in the unit place = **8 × 3 = 24** and **8 × 3 = 24** for the two cases.

Case 1c: The rightmost digit is 5. Again there would be two cases as center digit odd or left most digit is odd. We will again get 8 ways in which numbers can be created in the two different cases. Sum of numbers in the unit place = **8 × 5 = 40** and **8 × 5 = 40** for the two cases.

Case 2: Rightmost digit is even. Then the left most and the center digits will both be odd, chosen from 1, 3 or 5.
Case 2a : Rightmost digit is 2.
Number of ways of rightmost digit being 2 is 3 × 2 × 2 × 1 × 1 ways = 12 ways. (Steps: Rightmost digit in 1 way, i.e. digit 2. Then the leftmost digit in 3 ways i.e. any of the 3 odd numbers, and the center digit in 2 ways i.e. any of the remaining 2 odd numbers. Now the remaining two places are filled by the remaining two numbers in 2 × 1 way.)

Sum of digits with 2 in the units place = **12 × 2 = 24.**

Case 2a : Rightmost digit is 4.
Number of ways of rightmost digit being 4 is 3 × 2 × 2 × 1 × 1 ways = 12 ways. (Steps: Rightmost digit in 1 way, i.e. digit 4. Then the leftmost digit in 3 ways i.e. any of the 3 odd numbers, and the center digit in 2 ways i.e. any of the remaining 2 odd numbers. Now the remaining two places are filled by the remaining two numbers in 2 × 1 way.)

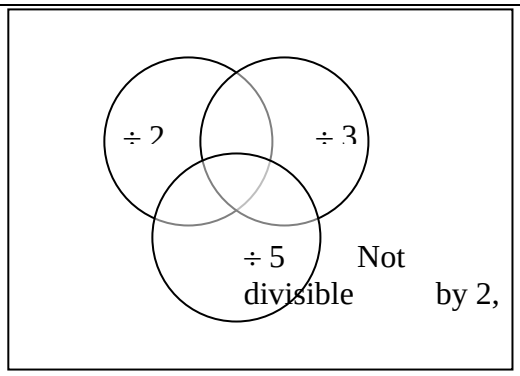
Sum of digits with 2 in the units place = **12 × 4 = 48.**

Total sum of digits of the right most digit = 8 + 8 + 24 + 24 + 40 + 40 + 24 + 48 = 216

Challenger question 4
Question
Find the number of numbers between 200 and 300, both inclusive, which are not divisible by 2, 3, or 5.
1] 25 2] 24 3] 75 4] 26

Answer
Number of numbers not divisible by 2, 3 or 5 = Total numbers – number of numbers divisible by 2, 3 or 5.

Career Avenues



Number of numbers divisible by 2, 3 or 5, i.e. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$.

Divisible by	From 1 to 300 (X)	From 1 to 199 (Y)	From 200 to 300 (X – Y)
2	150	99	51
3	100	66	34
5	60	39	21
2 and 3 i.e. 6	50	33	17
2 and 5 i.e. 10	30	19	11
3 and 5 i.e. 15	20	13	7
2, 3 and 5 i.e. 30	10	6	4

$n(A \cup B \cup C) = 51 + 34 + 21 - 17 - 11 - 7 + 4 = 75$.
Total number of numbers from 200 to 300 = 101
$\therefore$ Number of numbers not divisible by 2, 3 or 5 = 101 – 75 = 26. Hence [4]

Challenger question 5
Question
In how many ways can 12 one rupee coins be put in five different purses so that none of them are empty ?
1] 330 2] 462 3] 495 4] 792

Answer
Let us arrange the 1 Re coins in a row



To determine the distribution of the coins in the purses we must partition this now in 5 groups of coins using 4 walls. The 1st group for the 1st purse, 2nd for 2nd purse and so on. Thus, the number of ways to distribute twelve 1 Re coins in 5 purses equals the number of ways to put 4 walls into gaps between the coins in the row. Any wall can be in any of the 11 gaps (there are 11 = 12 – 1 gaps between 12 coins), and no two of them can be in the same gap(this would mean that one of the groups is empty.
$\therefore$ the number of possible partitions = ${}^{11}C_4 = 330$
Hence (1)

Challenger question 6
Question
Let A = {(x, y, z) : x, y, z are positive and x+ y + z = 12}. How many elements are in set A?
1] 55 2] 78 3] 45 4] None of these

Answer
First let us place 1 in all the three variables as they are positive.

Career Avenues

So we have nine 1's that have to be placed in three variables such that all the nine 1's could also go to any of the
variables.
This can be done in $^{11}C_2$ ways = 55 ways. Hence, [1].

Do-it-yourself (DIY exercise)

13.What is the sum of all possible three-digit numbers that can be formed using the digit 1, 2 and 3, such that the digits are not repeated?
14. What is the sum of all four-digit numbers that can be formed using the 1, 2, 3 and 4 exactly once?
15. What is the sum of all valid four-digit numbers that can be formed using the digits 0, 1, 2, and 3 exactly once?

6.5.1 The Pigeonhole principle (PHP)

Suppose that a flock of pigeons fly into a set of pigeonholes to roost. The PHP states that if there are more pigeons than pigeon holes ,then there must be at least one pigeonhole with at least two pigeons in it.

Theorem I
If k + 1 or more objects are placed into k boxes, then there is at least one box containing two or more objects.

Example
(i) Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.
(ii) In any group of 27 English words, there must be at least two that begin with the same letter, since there are 26 letters in the English alphabet.

Theorem II
If N objects are placed in k boxes, then there is at least one box containing at least [N/k] objects. Note that the symbol [ ] represents an integer greater than N/k.

Example
Among 100 people there are at least [100/12] i.e. 9 who are born in the same month.

Solved Examples

1. 5 points are marked randomly in a square plate of length 2 units. Show that a pair of them are apart by not more than $\sqrt{2}$ units.

By taking the midpoints of the edges and joining the points on opposite edges, we get a grid of 4 unit squares. By PHP, two of the points lie in one of these grids and the maximum distance between these points is $\sqrt{2}$ units (the length of the diagonal of the unit square).

2. What is the least number of area codes needed to guarantee that the 25 million phones in a state have distinct 10-digit telephone numbers ? Assume that telephone numbers are of the form NXX – NXX – XXXX, where the first three digits form the area code, N represents a digit from 2 to 1 inclusive, and x represents any digit.

There are 8 million (i.e. 8×10^6) different phone numbers of the form NXX – XXXX.

Hence, by PGP, among 25 million telephones, at least [25×10^6 / 8×10^6) of them must have identical numbers.

Hence, at least four area codes are required to ensure that all 10-digit numbers are different.

Example 1.5. A chess master who has 11 weeks to prepare for a tournament decides to play at least one game every day but, in order not to tire himself, he decides not to play more than 12 games during any calendar

Career Avenues

week. Show that there exists a succession of consecutive days during which the chess master will have played exactly 21 games.

Proof. Let $a1$ be the number of games played on the first day, $a2$ the total number of games played on the first and second days, $a3$ the total number games played on the first, second, and third days, and so on. Since at least one game is played each day, the sequence of numbers $a1, a2, \ldots\ldots, a77$ is strictly increasing, that is, $a1 < a2 < \ldots < a77$.

Moreover, $a1 \geq 1$, and since at most 12 games are played during any one week, $a77 \leq 12 \times 11 = 132$.
Thus
$1 \leq a1 < a2 < \ldots < a77 \leq 132$.

Note that the sequence $a1 + 21, a2 + 21, \ldots a77 + 21$ is also strictly increasing, and
$22 \leq a1 + 21 < a2 + 21 < \ldots < a77 + 21 \leq 132 + 21 = 153$.
Now consider the 154 numbers
$a1, a2, \ldots a77, a1 + 21, a2 + 21, \ldots\ldots, a77 + 21$,
each of them is between 1 and 153. It follows that two of them must be equal. Since $a1, a2, \ldots\ldots, a77$ are distinct and

$a1 + 21, a2 + 21, \ldots\ldots, a77 + 21$ are also distinct, then the two equal numbers must be of the forms ai and $aj + 21$.

Since the number games played up to the ith day is $ai = aj + 21$, we conclude that on the days $j + 1, j + 2, \ldots, i$ the chess master played a total of 21 games.

Solutions of DIY Questions

1. Number of ways taking 5 flags out of 8-flags = ${}^8P_5$
   $= \frac{8!}{(8-5)!} = 8 \times 7 \times 6 \times 5 \times 4 = 6720$
2. There are '9' different letters of the word "SIMPLETON"
   Number of Permutations taking all the letters at a time = ${}^9P_9 = 9! = 362880$.
3. $\frac{13!}{2! \times 2! \times 2!}$. The denominator takes care of the repetition of the letters e, r and i.
4. First coin can be put in 3 ways, similarly second, third and forth coins also can be put in 3 ways.
   So total number of ways = $3 \times 3 \times 3 \times 3 = 3^4 = 81$
5. Here clock-wise and anti-clockwise arrangements are same.
   Hence total number of circular–permutations: $\frac{{}^{18}P_{12}}{2 \times 12} = \frac{18!}{6! \times 24}$
6. (i) When 'O' and 'A' occupying end-places
   => M.E.G. (OA)
   Here (OA) are fixed, hence M, E, G can be arranged in 3! ways
   But (O,A) can be arranged themselves is 2! ways.
   => Total number of words = 3! × 2! = 12 ways.
   (ii) When 'E' is fixed in the middle
   => O.M.(E), G.A.
   Hence four-letter O.M.G.A. can be arranged in 4! i.e 24 ways.
   (iii) Three vowels (O,E,A) can be arranged in the odd-places (1st, 3rd and 5th) = 3! ways.
   And two consonants (M,G) can be arranged in the even-place (2nd, 4th) = 2! ways
   => Total number of ways= 3! × 2! = 12 ways.
7. There are 5 letters out of which there are 2 similar alphabets (I's). Hence the no. of ways = $5P_2 = \frac{5!}{2!} = 60$.
8. Number of letters in MISSISSIPI = 10, with 4 S's & 4 I's

Career Avenues

$\therefore$ Total number of ways in which the letters can be arranged = $\frac{10!}{4!4!}$

Since M and P are to be together, we treat them together as one letter.

Total number of words = $\frac{9!}{4!4!}$ (now then are 9 letters with M and P being arranged).

However, for each of these $\frac{9!}{4!4!}$ ways, M and P can be interchanged (MP and PM)

$\therefore$ Total number of ways in which M and P are side by side = $\frac{2\times 9!}{4!\times 4!}$.

$\therefore$ Total number of ways in which the letters M and P are never together $= \frac{10!}{4!4!} - \frac{2\times 9!}{4!4!} =$ 5040.

9. There are 6 digits. As zero cannot take the significant place, the thousand's place can be filled with one of the remaining 5 digits. Second, third and fourth places can be filled up in 6, 5 and 4 ways. Therefore total number of ways = 5 × 6 × 5 × 4 = 600 ways.

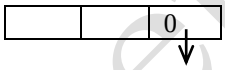
10. ${}^{n}p_{4} = 12\times {}^{n}p_{2}$

$$\frac{n!}{(n-4)!} = 12\frac{n!}{(n-2)!} \Rightarrow \frac{(n-2)!}{(n-4)!} = 12$$

$$(n-2)(n-3)\frac{(n-4)!}{(n-4)!} = 12 \Rightarrow (n-2)(n-3) = 12$$

$\Rightarrow n^2 - 5n + 6 = 12 \Rightarrow n^2 - 5n - 6 = 0 \Rightarrow n^2 - 6n + n - 6 = 0$,
i.e. $n(n - 6) + 1(n - 6) = 0 \Rightarrow n = 6$

11. Each friend can be called by any of the 3 servants in 3 ways.
Total number of ways = $3\times 3\ldots\times 3$ (6 times) $=3^6 = 729$.
Case 1: 3-digit number ending with 0
Restricted (1 way)



Non-restricted 1way
${}^{9}P_2$ ways

The number to be divisible by 5, must end with zero (at unit place).
So, permutation of restricted place (unit place) = 1 Now, tenth and hundredth place can be filled by using any two digits from the remaining of 9 digits (repetition not allowed).
So, permutation of non-restricted places (without repetition) = ${}^{n}p_{r} = {}^{9}p_{2} = 72$

Case 2:3 digit number ending with 5

Restricted	Non restricted	Restricted
1,2, 3, 4, 6, 7, 8 or 9		5
8 ways	${}^{8}P_1$	1 way

The number, divisible by 5, may also end with 5 (at unit place). So, permutation of unit place (restricted)=1
Now, hundredth place can be filled up by using any digit except 0 and 5. So, permutation of hundredth place (restricted) = 8.

Tenth place can be filled up by using any digit from the digits remaining after filling the above two places, because repetition of digits is not allowed.

Career Avenues

So, permutation of tenth place (non-restricted) = ${}^{8}p_1 = 8$

Total number of permutations = $1 \times 8 \times 8 = 64$

No. of 3-digit number ending with 5 = 64

Hence, required numbers of 3-digit number divisible by 5 = 72 + 64 [case I + Case II] = 136

13. Let the numbers be in the units, tens and hundreds column.
Now, the total number of numbers that can be created is $3 \times 2 \times 1 = 6$ numbers.
Each of the numbers 1, 2 and 3 appear in each of the columns exactly 2 times.
So sum of all nos. in the units place = $2 \times 1 + 2 \times 2 + 2 \times 3 = 2\ (1 + 2 + 3) = 12$
The sum of tens and hundreds is also = 12.
So the sum of the 6 nos. is 1332. (2 in the units place, and 1 carried over to tens place.
∴ 3 in the tens place and 1 carried over. Finally 13 is the total of the hundred's place.
Logic: For the sake of visualization, look at this table. It will help visualize other similar questions).

Hundreds	Tens	Unit
1	2	3
1	3	2
2	1	3
2	3	1
3	1	2
3	2	1

14. Using same logic as previous question.
No. of numbers = $4 \times 3 \times 2 \times 1 = 24$
So sum of the nos. in any column = $6 \times 1 + 6 \times 2 + 6 \times 3 + 6 \times 4 = 6\ (1 + 2 + 3 + 4) = 60$
∴ The sum of the 24 no's = 66660. (Use same method as prev question to get to the sum).

15. A four digit number cannot start with 0.
So we can first add all four digit numbers possible with 0, 1, 2 and 3 and then subtract those that have 0 in the thousand column.
Sum of nos. with 0,1,2,3 occupying all places = $6 \times 0 + 6 \times 1 + 6 \times 2 + 6 \times 3 = 6(1 + 2 + 3) = 36$
∴ Sum of the 24 nos. = 39996.
∴ Sum of nos. that have 0 in the thousandth column = 1332 (From question 16)
∴ Answer = 39996 – 1332 = 38664.

Career Avenues

Practice Set

(1) If 10Pr = 720, find the value of r

(2) A telegraph has 5 arms and each arm is capable of 4 distinct positions, including the position of rest. Find the total number of signals that can be made.

(3) How many numbers each lying between 100 and 1000, can be formed with digits 2, 3, 4, 0, 8, 9, (if repetition of digits not allowed)?

(4) Find the total number of signals that can be made by using 5 different coloured flags when any number of them may be hoisted at a time?

(5) Three dice are rolled. Find the number of possible outcomes in which at least 1 dice shows 4.

(6) How many natural numbers not exceeding 1000 can be made using the digits 1,2,3,4,5,6,7,8,9 if repetition is allowed?

(7) Find the number of ways in which 6 identical balls can be distributed among 10 identical boxes, if not more than one ball can go into a box.

(8) There are 5 copies of University Physics, 8 copies of University Chemistry, 3 different books on Law and 2 different books on History in a library. Find the number of ways in which one or more than 1 book can be selected.

(9) There are 5 different green dyes, 4 different blue dyes and 3 different red dyes. How many combinations of dyes can be chosen taking at least 1 green and 1 blue dyes?

(10) Radha has 7 friends. Each evening of 5 days in the week she invites 3 of her friends for dinner taking care that the same group of friends do not turn up once again. In how many ways can the groups be formed?

(11) A fruit seller sells 4 types of mangoes - Neelam, Malgova, Sendur and Daseri. In how many ways is it possible to purchase 7 mangoes?

(12) A cricket team of 11 players is to be formed from 16 players including 4 bowlers and 2 wicket keepers. In how many different ways can a team be formed so that the team consists of at least 3 bowlers and at least one wicket keeper?

(13) Find the number of ways in which 12 students may be equally divided into 3 groups.

(14) Rahul invites a party of 15 guests to a dinner and places 10 of them at a round table and the other 5 at another round table. Find the total number of ways in which he can arrange all the guests.

(15) Find the total number of ways in which 5 crows can perch on 3 trees when there is no restriction as to the choice of tree. In how many of these ways will a particular crow be alone on a tree?

(16) How many numbers greater than 40,000 can be made using one 1, one 3, two fours and one 5 (i.e. 1,3,4,4,5)?

(17) A letter lock consists of 4 rings each marked with 5 different letters; how many unsuccessful attempts can be made to open the lock?

(18) ${}^{28}C_{r+4} = {}^{28}C_{r-2}$, find r.

(19) Out of the 26 letters of the alphabet how many words can be made consisting of 4 letters, only one of which must be 'a'?

(20) Find the number of ways in which 6 different books can be arranged (a) if 3 specified books are always together, (b) if the 3 specified books are always separated.

Career Avenues

(21) There are 10 points in a plane, 4 of which are collinear. Find the number of (a) straight lines, and (b) triangles which result from joining them.

(22) Find the number of words which can be formed by taking two alike and two different letters from the word "COMBINATION".

(23) A railway carriage will accommodate 5 passengers on each side: In how many ways can 10 persons take their seats when two of them decline to face the engine, and a third cannot travel backwards?

(24) In how many ways can 5 chairs be occupied by 3 men and 2 boys taken from 6 men and 5 boys?

(25) In how many ways can n men be arranged in a row if two specified men are neither of them to be at either extremity of the row?

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ANSWER KEY TO MORE PRACTICE QUESTIONS

(1) 3
(2) 1023
(3) 100
(4) 325
(5) 91
(6) 819
(7) 210
(8) 1727
(9) 3720
(10) 35
(11) 120
(12) 2472
(13) $\frac{12!}{(4!)^3}$
(14) $\frac{15!}{50}$
(15) 243, 48
(16) 48
(17) 624
(18) 13
(19) 55200
(20) (a) 144, (b) 144
(21) (a) 40 (b) 116
(22) 756
(23) 504000
(24) 24000
(25) (n – 2) (n – 3) (n – 2)!

Career Avenues

6.6 Probability

Probability is the chance that something will happen - how likely it is that some event will happen.

Sometimes you can measure a probability with a number: "10% chance of rain", or you can use words such as impossible, unlikely, possible, even chance, likely and certain.

Example: "It is unlikely to rain tomorrow".

In the real world events can not be predicted with 100% certainty. The best we can do is say how likely they are to happen, using the concept of probability.

Tossing a Coin
When a coin is tossed, there are two possible outcomes:
• heads (H) and
• tails (T).

We say that the probability of the coin landing H is 1/2.
Similarly, the probability of the coin landing T is 1/2.

Throwing Dice
When a single die is thrown, there are six possible outcomes: 1, 2, 3, 4, 5, 6.
The probability of throwing any one of these numbers is 1/6.

In general:

Probability of an event happening = Number of ways it can happen / Total number of outcomes

Question
There are 5 marbles in a bag: 3 are red, and 2 are blue. What is the probability that a blue marble will be picked?
Answer
Number of ways it can happen: 2 (there are 2 blues) Total number of outcomes: 5 (there are 5 marbles in total) So the probability is 2/5 = 0.4

Experiment
Experiment: an action where the result is uncertain.
Tossing a coin, throwing dice, seeing what pizza people choose are all examples of experiments

Random Experiment
It is an experiment whose all possible results are known and which can be repeated under identical conditions, but it is not possible to predict the result of any particular experiment in advance.
Example: tossing a coin, throwing a dice.

Sample Space
Each performance in a random experiment is called a trial. The result of a trial in a random experiment is called an outcome, an elementary event, or a sample point. The totality of all possible outcomes (i.e. sample points) of a random experiment constitute the sample space (S).

Career Avenues

Note:
(i) Each element of S denotes a possible outcome of the experiment.
Example: When a coin is tossed there are two possible outcomes i.e. head up (H) or tail up (T) Then the set {H, T} is the set of all possible distinct outcomes of this experiment. This set is called the sample space of the experiment.
(ii) Any trial results in an outcome that corresponds to one and only one element of the set S.

Event
Event: a single result of an experiment
Example Events:
Getting a Tail when tossing a coin is an event
Rolling a "5" is an event.
Rolling an "even number" (2, 4 or 6) is also an event.

The null set ϕ is known as the 'Impossible Event' and S is called 'Sure Event'
Example: when a dice is thrown, there are six possible out comes the sample space S = {1,2,3,4,5,6}.
The event E: the outcome is even is {2,4,6}.
The event G: the outcome is 7 is ϕ.

Since the events are sets, the statements concerning events can be translated into set theory. Thus if A and B are events:
(i) $A \cup B$ is the event 'either A or B or both'.
(ii) $A \cap B$ is the event 'both A and B'.
(iii) A' is the event 'not A'.
(iv) A – B is the event 'A but not B', i.e. only A.
(v) Events A and B are mutually exclusive if $A \cap B = \phi$.

Mutually Exclusive Events
Two or more events are mutually exclusive if the occurrence of one of the events excludes the occurrence of the other events, i.e. if one event occurs, other cannot occur at the same time.
Example: In tossing a coin, the events "Head" and "Tail" are mutually exclusive.
note: If A is any event and A' is the complement of A then $A \cap A' = \phi$ i.e. any event and its complement are mutually exclusive events.

Equally Likely Events
Two or more than two events are equally likely if no one of the events can occur in preference to the other events.
Example: In tossing a coin, "Head" and "Tail" are equally likely events. On throwing a dice, all the six faces are equally likely events.

Exhaustive Events
Events are said to be exhaustive if at least one of them must necessarily occur.
Example: In tossing a coin, the two exhaustive events are "Head" and "Tail". In throwing a dice, the exhaustive events are 1, 2, 3, 4, 5 or 6.

Compound Events
It signifies the simultaneous occurrence of two or more than two events.

Conditional or Dependent Events
Two or more events are dependent events if one event occurs only when the other event(s) has already occurred.
Example: If 2 cards are drawn without replacement, then the above events A and B are dependent because the number of available cards in the second draw will be different from the number of cards in the first draw.

Definition of probability of an Event
If a trial results in 'n' exhaustive, mutually exclusive and equally likely cases and 'm' of them are favourable to the happening of an event E, then the probability 'p' of happening of E is given by,

Career Avenues

Probability of any event E = $p = \dfrac{m}{n} = \dfrac{\text{number of arrangements/combination favourable to event}}{\text{number of arrangements/combination available (possible)}}$

The number of cases favourable to the 'non-happening' of the event E are (n – m), the probability 'q' that E will not happen is given by,

$$q = \frac{n - m}{n} = 1 - \frac{m}{n} = 1 - p$$

$\Rightarrow$ **p + q = 1**

Note:
(i) p and q are non – negative and cannot exceed unity, i.e. $0 \le p \le 1$, $0 \le q \le 1$.
(ii) The probability 'p' of the happening of an event is also known as the Probability of Success and the probability (q) of the non-happening of the event as the Probability of Failure.
(iii) If P(A) = 1, A is called a Certain Event and if P(A) = 0, A is called an Impossible Event.
(iv) In the solution of some problem on probability, it is necessary to use results from theory of combinations and permutations. Thus,

(a) the number of ways to choose 'r' objects from 'n; distinct objects, is ${}^{n}C_r = \dfrac{n!}{r!(n-r)!}$.

(b) the number of arrangements of 'r' objects in a line out of 'n' distinct objects is ${}^{n}P_r = \dfrac{n!}{(n-r)!}$.

(v) If the odds in favour of an event A is a:b then Probability of success of A is given by,

$$P(A) = \frac{a}{a+b} \text{ and } P(\overline{A}) = 1 - \frac{a}{a+b} = \frac{b}{a+b}$$

(vi) P(at least one event occurs) = 1 – P (no event occurs)

PROBABILITY THEOREM: For any two events A and B, the probability of at least one event (i.e. either A or B, or both occurs) is given by $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
Note: When events A and B are Mutually Exclusive, $P(A \cup B) = P(A) + P(B)$.

The above theorem may be extended for more than two events also. For any three events A, B, and C, Probability of at least one event is given by :
$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$.

Question
What is the probability that one card drawn at random from the pack of playing cards may either be a King or an Ace?

Answer
One card can be drawn from a pack of cards in ${}^{52}C_1$ ways = 52 ways.
Let K = The card be a King
A = The card be an Ace
$\therefore$ n(S) = 52, n(K) = 4, n(A) = 4
Since K and A are mutually exclusive events,

$$P(A \cup B) = P(A) + P(B) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$$

Career Avenues

Question
In the previous example, what is the probability that one card drawn at random is a king or a spade?

Answer
n(S) = 52,
Let K = event of getting a king
S = event of getting a spade
$\therefore$ $K \cap S$ = event of getting a king and a spade.

$\therefore$ $P(K) = \frac{4}{52} = \frac{1}{13}$, $P(S) = \frac{13}{52} = \frac{1}{4}$

$P(K \cap S) = \frac{1}{52}$

$\therefore$ $P(K \cup S) = P(K) + P(S) - P(K \cap S)$

$= \frac{1}{13} + \frac{1}{4} - \frac{1}{52} = \frac{4+13-1}{52} = \frac{16}{52} = \frac{4}{13}$

Probability of occurrence of at least one of the 'r' independent events

Let $A_1, A_2 \ldots A_r$ be independent events and let $P_1, P_2 \ldots P_r$ be the probability of the happening of the events $A_1, A_2 \ldots A_r$

$P(A_1) = P_1$, $P(A_2) = P_2 \ldots$, $P(A_r) = P_r$

$\therefore$ $P(\overline{A}_1) = 1 - P_1$, $P(\overline{A}_2) = 1 - P_2 \ldots$, $P(\overline{A}_r) = 1 - P_r$

$\therefore$ Probability of none of the events happen = $(1 - P_1)(1 - P_2) \ldots (1 - P_r)$

Hence probability that at least one of the events happens $= 1 - (1 - P_1)(1 - P_2) \ldots (1 - P_r)$

$= 1 - P(\overline{A}_1) P(\overline{A}_2) \ldots P(\overline{A}_r)$

Question
The probability that at least one of the events A and B occurs is 0.7 and they occur simultaneous is 0.2. Find $P(\overline{A}) + P(\overline{B})$.

Answer
$P(A \cup B) = 0.7$ and $P(A \cap B) = 0.2$

$\therefore$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.7 = P(A) + P(B) - 0.2$

$\therefore$ $P(A) + P(B) = 0.9$

$\therefore$ $1 - P(\overline{A}) + 1 - P(\overline{B}) = 0.9$

$\therefore$ $P(\overline{A}) + P(\overline{B}) = 2 - 0.9 = 1.1$

6.6.1 Approaches to probability

There are mainly four approaches to probability. We will discuss the first two approaches.

1) The classical or priori approach
2) The relative frequency or empirical approach
3) The axiomatic approach
4) The personalistic approach

The Classical or Priori Approach

Probability is the ratio of the number of favourable cases as compared to the total likely cases. Suppose an event can occur in N ways out of a total of M possible ways. Then the probability of occurrence of the event is denoted by

$p = Pr(N) = \frac{N}{M}$. Probability refers to the ratio of possible outcomes to all possible outcomes.

The probability of non-occurrence of the same event is given by {1 – p(occurrence)}. The probability of occurrence plus non-occurrence is equal to one.

If probability occurrence; p(O) and probability of non-occurrence (O'), then p(O)+p(O')=1.

Career Avenues

Empirical Probability (Relative Frequency Probability)

Empirical probability arises when frequency distributions are used. For example:

Observation (X)	0	1	2	3	4
Frequency (f)	3	7	10	16	11

The probability of observation (X) occurring 2 times is given by the formulae

$$P(2)=\frac{\text{freuency of } 2}{\text{sum of frequencies}}=\frac{f(2)}{\sum f}=\frac{10}{3+7+10+16+11}=\frac{10}{47}$$

Rules of Probability : ***Addition rules :***

Rule 1: When two events A and B are mutually exclusive, then P(A or B)=P(A) + P(B)

Question
When a is tossed, find the probability of getting a 3 or 5.
Answer
P(3) =1/6 and P(5) =1/6. Therefore P(3 or 5) = P(3) + P(5) = 1/6+1/6 =2/6=1/3.

Rule 2: If A and B are two events that are NOT mutually exclusive, then P(A or B) = P(A) + P(B) – P(A and B), where A and B means the number of outcomes that event A and B have in common.

Question
When a card is drawn from a pack of 52 cards, find the probability that the card is a 10 or a heart.
Answer P(10) = 4/52 and P(heart)=13/52 P (10 that is Heart) = 1/52 P(A or B) = P(A) +P(B) – P(A and B) = 4/52 + 13/52 – 1/52 = 16/52.

Rules of Probability: ***Multiplication rules :***

Rule 1: For two independent events A and B, then P(A and B) = P(A) × P(B).

Question
Determine the probability of obtaining a 5 on a die and a tail on a coin in one throw.
Answer P(5) =1/6 and P(T) =1/2. P(5 and T)= P(5) × P(T) = 1/6 × 1/2 = 1/12.

Rule 2: When to events are dependent, the probability of both events occurring is P(A and B)=P(A) × P(B|A), where P(B|A) is the probability that event B occurs given that event A has already occurred.

Question
Find the probability of obtaining two Aces from a pack of 52 cards without replacement.
Answer P(Ace) =4/52 and P(second Ace if NO replacement) = 3/51 Therefore P(Ace and Ace) = P(Ace) × P(Second Ace) = 4/52 × 3/51 = 1/221

Do-it-yourself (DIY exercise)

1. A bag contains 3 orange, 3 yellow and 2 white marbles. Three marbles are selected without replacement. Find the probability of selecting first two yellow and a third white marble.
2. In a class, there are 8 girls and 6 boys. If three students are selected at random for debating, find the probability that all are girls.

Career Avenues

3. In how many ways can 3 drama officials be selected from 8 members?
4. A box has 12 bulbs, of which 3 are defective. If 4 bulbs are sold, find the probability that exactly one will be defective.

6.6.2 Conditional Probability

Let A and B be two events defined on a sample space S. Let P(B) > 0.
Then the conditional probability of A provided B has occurred, denoted by P(A/B) is defined as

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \; P(B) > 0$$

Similarly $P(B/A) = \frac{P(A \cap B)}{P(A)}$, P(A) > 0.

Example: A dice is rolled. If the outcome is an odd number, what is the probability that it is a prime?
S = {1, 2, 3, 4, 5, 6}
Let A = The event of getting an odd number
B = The event of getting a prime number
Then A = {1, 3, 5} and B = {2, 3, 5}
$A \cap B$ = {3, 5}
we have to find the conditional probability of B when A has occurred

i.e. $P(B/A) = \frac{n(A \cap B)}{n(A)} = \frac{2}{3}$

Hence, The conditional probability of two events A and B is $P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$, where P(A and B) means the probability of the outcomes that events A and B have in common.

Question
When a die is rolled once, find the probability of getting a 4 given that an even number occurred in an earlier throw.
Answer
P(4 and an even number) = 1/6 i.e. P(A and B) =1/6. P(even number) =3/6 =1/2.
$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{1/6}{1/2} = \frac{1}{3}$

Independent Events

Two events A and B are said to be independent if the occurrence of one of them, say A, does not affect the probability of occurrence of the other event B.

Thus, A and B are independent if and only if P(B/A) = P(B) and P(A/B) = P(A).
Otherwise, they are said to be dependent events. For dependent events, the occurrence or non-occurrence of one of them affects the probability of occurrence of the other events.

Example: when a coin is tossed twice, the result of the first throw does not affect the result of the second throw.

note: (i) If A and B are independent events if and only if $P(A \cap B) = P(A) \times P(B)$.

(ii) If A and B are two dependent events, then $P(A \cap B) = P(A)\, P(B/A) = P(B)\, P(A/B)$.

i.e. the probability of the simultaneous occurrence of two events, A and B, is the product of the probability of A by the conditional probability of B given that A has occurred.

note: (i) when A and B are independent events, we have, P(A/B) = P(A) and P(B/A) = P(B)
$\therefore P(A \cap B) = P(A)\, P(B)$

(ii) If A and B are independent events, then $\overline{A}$ and $\overline{B}$ are also independent.
$\therefore P(\overline{A} \cap \overline{B}) = P\left[(A \cup B)'\right] = 1 - P(A \cup B)$

Career Avenues

$= 1 - \{P(A) + P(B) - P(A \cap B)\}$
$= 1 - P(A) - P(B) + P(A \cap B)$
$= [1 - P(A)] - P(B) + P(A) \cdot P(B)$
$= [1 - P(A)] - P(B)\,[1 - P(A)]$
$= [1 - P(A)]\,[1 - P(B)]$
$= P(\overline{A}) \cdot P(\overline{B})$

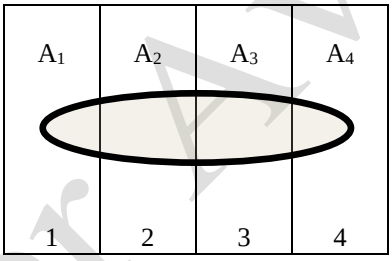
Question
For two events A and B let P(A) = P, P(B) = 0.4, and $P(A \cup B) = 0.6$. (a) Find P so that A and B are independent events (b) for what value of P are A and B mutually exclusive?

Answer
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$0.6 = P + 0.4 - P(A \cap B)$
$\therefore P(A \cap B) = P - 0.2$

(a) Since A and B are independent events, $P(A \cap B) = P(A) \cdot P(B)$
$\therefore P - 0.2 = P(0.4)$,
i.e. $0.6P = 0.2$, i.e. $P = \frac{0.2}{0.6} = \frac{1}{3}$

(b) Since A and B are mutually exclusive events, $P(A \cap B) = 0$
$\therefore P - 0.2 = 0 \Rightarrow P = 0.2$

6.6.3 Bayes' Theorem : Total Law of Probability



$A_1, A_2, A_3, A_4, \ldots A_i$, are mutually exclusive and exhaustive events.

You know $P(A|A_i)$ for all i and $P(A_i)$

To find P(A)

$P(A) = \sum_{i=1}^{A} P(A \mid Ai).P(Ai)$

In the above example, **$P(A) = P(A|A_1)P(A_1) + P(A|A_2)P(A_2) + P(A|A_3)P(A_3) + P(A|A_4)P(A_4)$**

One may think as follows. Event A is composed of parts in $A_1, A_2, A_3,$ and A_4. For A_1 part, it is probability that A will occur given A_1 has occurred and for that A_1 must have occurred.

Thus area marked 1 in above figure is $P(A|A_1)P(A_1)$, area 2 is $P(A|A_2)P(A_2)$…

Bayes Theorem
Say $A_1, A_2, A_3, A_4, \ldots A_i$, are mutually exclusive and exhaustive events. Then any event A can be written as

$P(A) = \sum_{i=1}^{A} P(A \mid Ai).P(Ai)$ as we saw in previous section of total law of probability.

Now, if I ask the question that suppose if event A has occurred, then what is the probability that it occurred from A_1 region?

Career Avenues

$$\mathbf{P(A_1|A)} = \frac{P(A_1 \cap A)}{P(A)} = \frac{P(A \mid A_1).P(A_1)}{\sum_{i=1}^{n} P(A \mid A_i)P(A_i)}.$$

Thus Bayes theorem will answer that given that an event has occurred which can be done in n different regions (events) which are mutually exclusive and exhaustive, it says what is the probability that a event A has occurred due to a particular region (event A).

Question
Given: G = Having good seats, B = Having bad seats, A_1 = Seeing *Comedy Movie*, A_2 = Seeing *Scary Movie!*, the probability of having good seats given that you are seeing *Comedy Movie* $[P(G/A_1)] = 0.98$, the probability of having good seats given that you are seeing *Scary Movie!* $[P(G/A_2)] = 0.95$, $P(A_1) = 0.65$, and $P(A_2) = 0.35$… Please find the probability of a) Seeing *Comedy Movie* given you have bad seats, and b) Seeing *Scary Movie!* given you have bad seats.

Answer
$P(B/A_1) = 1 – P(G/A_1) = 1 – 0.98 = 0.02$; $P(B/A_2) = 1 – P(G/A_2) = 1 – 0.95 = 0.05$

a) $P(A_1/B) = [(0.65)(0.02)]/[(0.65)(0.02) + (0.35)(0.05)] = 0.0130/(0.0130 + 0.0175) = 0.0130/0.0305 = 0.4262$

b) $P(A_2/B) = [(0.35)(0.05)]/[(0.65)(0.02) + (0.35)(0.05)] = 0.0175/(0.0130 + 0.0175) = 0.0175/0.0305 = 0.5738$

Question
A lot of fifteen items contains five defective items. Two items are drawn at random. What is the probability that the second item drawn will be defective?

Answer
Let A = event that first item is defective
Let A' = event that first item is good
Let B = event that second item is defective

The question stated in probability terms: what is P(B) = ?

P(A) = 5/15, P(A') = 10/15
P(B) = P(A & B) or P(A' & B) → P(first item defective & second item defective) or P(first item good & second item defective)
P(B) = P(B/A) P(A) or P(B/A') P(A')
P(B) = P(B/A) P(A) + P(B/A') P(A')
P(B) = (4/14)(5/15) + (5/14)(10/15)
P(B) = (20/210) + (50/210) = 70/210 = 0.333

Question
It has been found that 10% of certain relays have bent covers and will not work. If 40% have bent covers, what is the probability that a relay with a bent cover will not work?

Answer
A and B – bent AND will not work – 0.1
A = 0.4
Let A = event that relays have bent covers
Let B = event that relays will not work
Given: P(A & B) = .10, P(A) = .40
The first formula of the conditional probability formulas, Bayes formulas, gives the following solution:
P(B/A) = P(A and B) / P(A) = 10/40 = 0.25

Career Avenues

6.6.4 Random variables

Definition: A **probability distribution function**, or pdf, is a description that lists the possible outcomes of an experiment and assigns probabilities to each outcome.

Example: The pdf for a coin flip is:

Heads	½
Tails	½

Notice that we should be able to add up all the probabilities for a pdf and get 1, since $P(\Omega) = 1$.
(In this case, ½ + ½ = 1, so we're happy.)

Definition: A **random variable** is a variable which assigns numbers to outcomes of some random experiment.

In general, once we get out of this probability mess and into the real world, a variable will be some particular characteristic of the experimental unit we're studying. For example, a psychologist will want to study people, and some characteristics, or variables, he might be interested in are gender, age, political status (Democrat, Republican, Independent), favorite color, and education. A doctor might be interested in height, weight, age, gender, smoking status, and eating habits. A biologist studying mating habits of newts might like to know the length, weight, age, gender, species, eating habits, and sleeping habits of the newts under consideration. All of the things I listed are variables, and we can easily assign numbers to the different characteristics within each variable.

Definition: A random variable is discrete if it only takes on a countable number of values. A random variable is continuous if it is defined on some interval. (We're not going to deal with random variables that are half discrete and half continuous.)

Examples:
a.) Let X be a random variable describing the number of heads I get in 5 flips of a coin. Since X can only be equal to 0, 1, 2, 3, 4 or 5, it is discrete.

b.) Let X be a random variable describing the number of times I have to flip a coin until I get a head. I could get a head on the first try, so X could be 1. I could get a tail and then a head, so X could be 2. I could get tails, tails, heads, so X could be 3. I could get tails, tails, tails, tails, tails, tails, heads, so X could be 7.

Theoretically, I could keep flipping for a really long time (though the probability that I did is very, very small), so X can take on any positive integer value all the way up to infinity. This is what we call countable infinity, though, since I can count the values that X takes on. In this case, then X is discrete. You might want to think about it this way: if you can find a number between two values of X where X is not defined, then X is not continuous. In our example, X can be equal to 1 and 2, but X cannot be equal to 1.5 – that doesn't make sense. So X is not continuous.

c.) Let X be a random variable which takes on the height of a randomly selected person from the population of a country's citizens. Then X is defined on an interval between about new born babies (say of 12 inches) to tall adults, say 8 feet. Any value between the two extremes is a possible value of X, so we say that X is continuous.

Example: We might, say, assign the value 1 to the outcome of head of a coin flip, and assign the value 0 to the outcome tails. Then if X is my random variable, X takes on the values 0 and 1 with probability ½ each. Then I might write out the pdf as follows:

X	**P(X)**
0	½
1	½

Career Avenues

Definition: The **expected value** of X, E[X], or the mean of X, μ, for discrete random variables is defined to be $\sum_i X_i P(X_i)$.

Fun Gyan
The Greek letter sigma, Σ, is like our Roman letter "S". (The lower case version is σ.) If it helps you remember, "S" is the first letter of the word "sum," so we use Σ to denote summation. For continuous random variables, E[X] = $\int xf(x)dx$ *where f(x) is the pdf. Incidentally, the integral symbol is just a stretched-out letter S, because it also stands for sum; it's just an uncountably infinitely large sum.*

Example: Roll a single fair, six-sided die. What value would you expect to get on average, after a large number of rolls?
First, decide what X is and get its probability distribution function. If X is the value that comes up when you roll the die, we know that since every value is equally likely (the die is fair), the probability of finding any particular value is 1/6. So the pdf is as follows:

X	P(X)
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Then E[X] = $\sum_i X_i P(X_i) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = 3.5.$

- Note that 3.5 in this example is NOT an observable value of X! In other words, you can't roll a 3.5! Averages are like that – sometimes you end up with a value that wasn't observable in the first place. Another place you can find this crazy phenomenon is when you calculate your Average Marks. If you make 4 subjects and in 2, you scored 4 out of 5 and in other 2 you scored 3 out of 5 , then your average will be (4 + 4 + 3 + 3) / 4 = 3.5. You didn't make a 3.5 – halfway between 4 and 3 – on any of your paper, but that's your average.

Definition: The **variance** of a random variable X is $\sigma^2 = E[(X-E[X])^2] = E[X^2] - E[X]^2$, where $E[X^2] = \Sigma X^2P(X)$ if X is discrete. (If X is continuous, $E[X^2] = \int x^2 f(x)dx$.)

Definition: The standard deviation of a random variable is the square root of its variance.

Conceptually, the first way of writing variance is easier to handle; the second way, $E[X^2] - E[X]^2$, is easier to compute. If we just look at the inside of the first formula, $[(X-E[X])^2]$, we see that we're finding a difference between the random variable X and its mean. We square the difference rather than taking an absolute value because it makes our lives easier mathematically, but the concept of finding out how far a variable goes from its mean is the basic notion of variance. More variance means more of a hassle for statisticians.

Example: Let's use the die roll we had in the previous example. $E[X^2] = 1(1/6) + 4(1/6) + 9(1/6) + 16(1/6) + 25(1/6) + 36(1/6) = 15.1666666666667$. Another way of writing $E[X]^2$ is $(E[X])^2$; it's just the mean squared, so the variance of X is $\sigma^2 = 15.1666666666667 - 3.5^2 = 2.9166666666666667$.

6.6.5 Probability density function

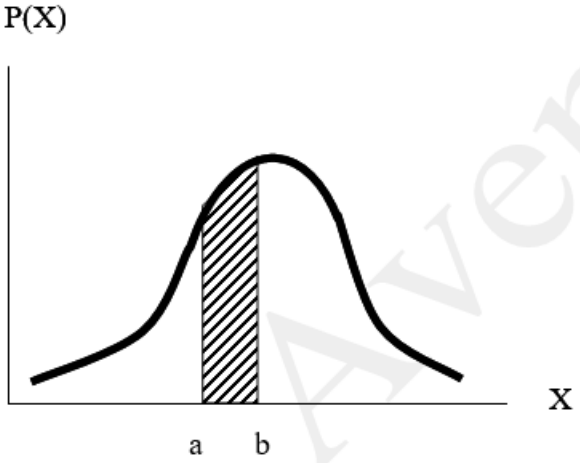
Discrete Distribution

If X is a variable that can assume a discrete set of values $X_1, X_2, X_3, \ldots\ldots, X_k$ with respect to probabilities $p_1, p_2, p_3, \ldots\ldots, p_k$, where $p_1 + p_2 + p_3, \ldots\ldots, + p_k = 1$, we say that a discrete probability distribution for X has been defined. The function p(X), which has the respective values $p_1, p_2, p_3, \ldots\ldots, p_k$ for $X= X_1, X_2, X_3, \ldots\ldots, X_k$ is called the probability function, or frequency function, of X. Because X can assume certain values with given probabilities, it is often called a discrete random variable. A random variable is also known as a chance variable or stochastic variable.

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Continuous Distribution

Suppose X is a continuous random variable. A continuous random variable X is specified by its probability density function which is written f(x) where $f(x) \geq 0$ throughout the range of values for which x is valid. This probability density function can be represented by a curve, and the probabilities are given by the area under the curve.



The total area under the curve is equal to 1. The area under the curve between the lines x=a and x=b (shaded) gives the probability that X lies between a and b, which can be denoted by P(a<X<b). p(X) is called a probability density function and the variable X is often called a ***continuous random variable***

Since the total area under the curve is equal to 1, it follows that the probability between a range space a and b is given by

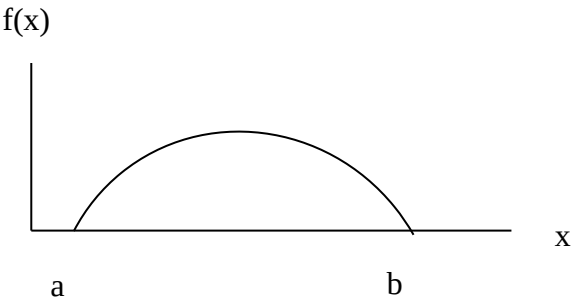
$P(a \leq X \leq b) = \int_a^b f(x)dx$, which is the shaded area.

Note: when computing area from a to b, we need not distinguish ($\leq$ and $\geq$) and (< and >) inequalities. We assume the lines at a and b have no thickness and its area is zero.

Solved Examples: The continuous random variable X is distributed with probability density function f defined by $f(x) = kx(16 - x^2)$, for $0 < x < 4$.

Evaluate:
The value of constant k
The probability of range space $P(1 < X < 2)$
The probability $P(x \geq 3)$

Solution:



For any function f(x) such that $f(x) \geq 0$, for $a \leq x \leq b$, and $\int_a^b f(x)dx = 1$

may be taken as the probability density function (p.d.f) of a continuous random variable in the range space $a \leq x \leq b$.

Career Avenues

Procedure:

Step 1: In general, if X is a continuous random variable (r.v) with p.d.f. f(x) valid over the range $a \le x \le b$, then

$\int_{\text{all } x} f(x)dx = 1$ i.e.

$\int_a^b f(x)dx = 1$

Step 2: a). To determine k, we use the fact that in $f(x) = kx(16 - x^2)$, for $0 < x < 4$, then

$\int_0^4 kx(16 - x^2)dx = 1$

$\Rightarrow \quad k\int_0^4 16x - x^3)dx = 1$

$\Rightarrow \quad k = \frac{1}{64}$

Step 3: b) Find $P(1 < X < 2)$

Solution:

$P(1<X<2) = \int_1^2 f(x)dx$

$= \quad \frac{1}{64}\int_1^2 (16x - x^3)dx = \frac{81}{256}$

Step 4: c). To find $P(x \ge 3)$

$P(x \ge 3) = \frac{1}{64}\int_3^4 (16x - x^3)dx = \frac{49}{256}$

Example 2:

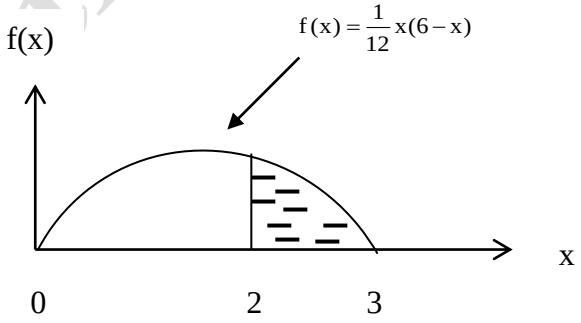
2) X is the continuous random variable 'the mass of a substance, in kg, per minute in an industrial production process', where

$$f(x) = \begin{cases} \frac{1}{12}x(6-x) & (0 \le x \le 3) \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that the mass is more than 2 kg.

Solution:

X can take values from 0 to 3 only. We sketch f(x), and shade the area required.



$P(x > 2) = \int_2^3 \frac{1}{12}x(6-x)dx$

Career Avenues

$$= \frac{1}{12}\int_2^3 (6x - x^2)dx$$
$$= \frac{1}{12}\left[3x^2 - \frac{x^3}{3}\right]_2^3$$
$$= 0.722 \quad (3\text{ d.p})$$

The probability that the mass is more than 2 kg is 0.722.

Worked Example:

3) The continuous random variable (r.v) has a probability density function(p.d.f) where

$$f(x) = \begin{cases} k & 0 \le x < 2 \\ k(2x-3) & 2 \le x \le 5 \\ 0 & otherwise \end{cases}$$

a) Find the value of the constant k

b) Sketch y=f(x)

c) Find P($X \le 1$)

d) Find P(X>2.5)

Solution:

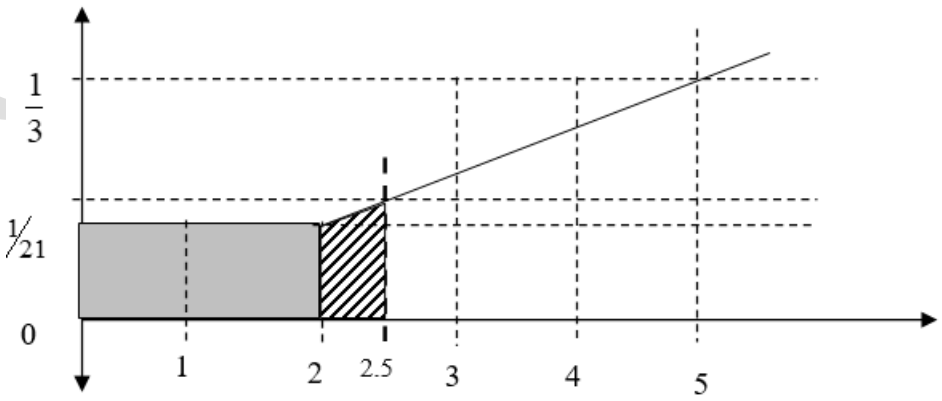
a) Since X is a r.v, then $\int_{\text{all x}} f(x)dx = 1$

Therefore

$$\int_0^2 k dx + \int_2^5 k(2x-3)dx = 1$$
$$kx \;\Big|_0^2 + k\left[x^2 - 3x\right]_2^5$$
$$2k + 19k = 1$$
$$\Rightarrow k = \frac{1}{21}$$

b) So the p.d.f of X is

$$f(x) = \begin{cases} \frac{1}{21} & 0 \le x < 2 \\ \frac{1}{21}(2x-3) & 2 \le x \le 5 \\ 0 & otherwise \end{cases}$$



Career Avenues

c) $P(x \leq 1)$ = area between zero and 1 = L × W= 1 × $\frac{1}{21} = \frac{1}{21}$ = 0.048

d) Find P(X>2.5) = $\int_{2.5}^{5} \frac{1}{21}(2x-3)dx$

$= \frac{1}{21}\left(x^2 - 3x\right)_{2.5}^{5}$

$= \frac{1}{21}\left[(25-15)-(6.26-7.5)\right]$

= 11.25/21 = 0.535

Expectation

Definition: If X is a continuous variable (r.v) with probability density function (p.d.f) f(x), then the expectation of X is E(X) where $E(X) = \int_{all\ x} x \quad f(x)dx$

NB: **E(X) is often denoted by μ and referred to as the mean of X**

Example:

1) If X is a continuous variable (r.v) with a p.d.f $f(x) = \frac{1}{16}x^2, \quad 0 \leq x \leq 3,$ find E(X).

Solution:

$$E(X) = \int_{all\ x} x \quad f(x)dx$$

$$\Rightarrow \int_0^3 \frac{1}{16}\{x\}\ x^2 dx$$

$$= \frac{1}{16}\left[\frac{x^4}{4}\right]_0^3 = \frac{81}{64} = 1.265$$

2) If the continuous random variable X has p.d.f $f(x) = \frac{2}{5}(3+x)(x-1), \qquad 1 \leq x \leq 3$, find E(X).

$$E(X) = \int_{all\ x} x \quad f(x)dx$$

$$E(x) = \int_1^3 \frac{2}{5}\{x\}\ (3+x)(x-1)dx$$

$$= \frac{2}{5}\left[\frac{x^4}{4} + \frac{2x^3}{3} - \frac{3x^2}{2}\right]_1^3$$

$$= \frac{608}{60}$$

$$= 10.13$$

Career Avenues

Generalization

If g(x) is any function of the continuous random variable r.v X having p.d.f f(x), then

$$E\left[g(X)\right] = \int_{\text{all x}} g(x)f(x)dx$$

and in particular

$$E(X^2) = \int_{\text{all x}} x^2 f(x)dx$$

The following conclusions hold

1. $E(a) = a$
2. $E(aX) = aE(X)$
3. $E(aX + b) = aE(X) + b$
4. $E\left[(f_1(X) + f_2(X)\right] = E\left[f_2(X)\right]$

Example:

1) The continuous random variable X has p.d.f f(x) where f(x)= $\frac{1}{2}x, \quad 0 \le x \le 3.$

Find
a) E(X)
b) $E(X^2)$
c) E(2X +3)

Solution:

a) $$E(X) = \int_{all\ x} x \ \ f(x)dx$$
$$= \int_0^3 \frac{1}{2}x^2 dx$$
$$= \frac{1}{2}\left[\frac{x^3}{3}\right]_0^3$$
$$= 4.5$$

b) $$E(X^2) = \int_{\text{all x}} x^2 f(x)dx$$
$$= \frac{1}{2}\int_0^3 x^3 dx$$
$$= \frac{1}{2}\left[\frac{x^4}{4}\right]_0^3$$
$$= \frac{81}{8} = 10.125$$

c) E(2X +3) = E (2X) + 3
= 2E(X) +3
= 2(10.125)+5
= 25.25 (from (b) above)

Career Avenues

6.6.6 Binomial Distribution

In probability theory and statistics, the binomial distribution is the discrete probability distribution of the number of successes in a sequence of n independent yes/no experiments, each of which yields success with probability p. Such a success/failure experiment is also called a Bernoulli experiment or Bernoulli trial. In fact, when n = 1, the binomial distribution is a Bernoulli distribution. The binomial distribution is the basis for the popular binomial test of statistical significance.

Examples

An elementary example is this: roll a die ten times and count the number of 1s as outcome. Then this random number follows a binomial distribution with n = 10 and p = 1/6.

For example, assume 5% of the population is green-eyed. You pick 500 people randomly. The number of green-eyed people you pick is a random variable X which follows a binomial distribution with n = 500 and p = 0.05 (when picking the people with replacement).

Examples which follow binomial distribution are :

* I flip a coin once. The outcome is either Heads or Tails.
* I randomly select a student and find the student's gender. The outcome is either male or female.
* I randomly select a resident of Athens and find the student's favorite color. In some cases, I might be interested in all possible favorite colors, but in other cases, I could be interested in whether or not the person's favorite color is red. In that case, there are two outcomes: red and not red.

A **binomial random variable** has 4 characteristics:
i.) It consists of Bernoulli trials.
ii.) The Bernoulli trials are independent of one another. (In other words, the outcome of a trial does not affect the outcome of any of the other trials.)
iii.) The number of Bernoulli trials is fixed before the experiment begins.
iv.) The probability of a success must be the same for all of the Bernoulli trials.

If four coins are flipped, each with P(H) = 0.7, what is the probability of having two H(Heads) total? One outcome that realises this event is $H_1H_2T_3T_4$ (i.e. Heads on 1st and 2nd coins, Tails on 3rd and 4th), with probability $P(H_1H_2T_3T_4) = P(H_1)\,P(H_2)\,P(T_3)\,P(T_4)$ ($\because$ s.i. between coins) $= 0.7\times0.7\times0.3\times0.3 = 0.7^2 0.3^2$. But many other (mutually exclusive) outcomes also count, such as $H_1T_2H_3T_4$, which all have the same probability $0.7^2 0.3^2$. How many are there? It is the number of ways to pick 2 out of 4 places to be labelled "H" (and the rest "T"), which is $\frac{4!}{2!(4-2)!} = 6$, so, since the outcomes are all mutually exclusive,

$P(\text{two heads}) = P(H_1H_2T_3T_4 \text{ or } H_1T_2H_3T_4 \text{ or } \ldots\ldots) = P(H_1H_2T_3T_4) + P(H_1T_2H_3T_4) + \ldots\ldots = C_2^4\, 0.7^2 0.3^2$

Thus, in the general case, to have x "success" among n independent trials, with P(success) = p in each trial, we have the binomial distribution

$$P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \quad (x = 0,1,2,\ldots,n)$$

A series of independent trials which can result in one of the two mutually exclusive possibilities success or failure such that the probability of success (or failure) in each trial is constant, then such repeated independent trials are called Bernoullian trials (named after James Bernoulli who discovered this distribution)

If we perform a series of n **Bernoullian trials** such that for each trial p is the probability of success and q is the probability of failure (p + q = 1), then the probability of 'r' success in a series of n independent trials is given by,

$P(r) = {}^nC_r\ p^r\ q^{n-r}$ where p+q = 1 and r = 0,1,2 ….n

Career Avenues

Properties of Binomial Distribution

(i) n and p are the two parameters of the binomial distribution. As soon as the values of n and p are known, the binomial distribution is completely determined.

(ii) The mean of the binomial distribution is np

(iii) The standard deviation and the variance of the binomial distribution are $\sqrt{npq}$ and (npq) respectively. Since q < 1, therefore n p > n p q. Thus the mean of the binomial distribution is always greater than the variance.

(iv) If $p = q = \frac{1}{2}$, then the binomial distribution is a symmetrical distribution.

Question

A die is thrown 8 times. If getting an "odd number" is success, Find the probability of getting at least 6 success.

$p = \frac{1}{2}$, $q = \frac{1}{2}$, n =8.

Answer

P(at least 6 success) = P (6 success) + P(7 success) + P(8 success)

$= {}^8C_6\, p^6\, q^2 + {}^8C_7\, p^7 q + {}^8C_8\, p^8\, q^0$

$= \frac{8\times 7}{2}\times\left(\frac{1}{2}\right)^6\left(\frac{1}{2}\right)^2 + 8\left(\frac{1}{2}\right)^7\left(\frac{1}{2}\right) + 1\left(\frac{1}{2}\right)^8\left(\frac{1}{2}\right)^0$

$= \left(\frac{1}{2}\right)^8 [28+8+1] = 37\left(\frac{1}{2}\right)^8$

Do-it-yourself (DIY exercise)

5. A coin is tossed five times. (This is the same as a sample size of five). What is the probability of obtaining exactly two heads in the five tosses?

6. In manufacturing screwdrivers, it was empirically determined that the process yields, on average, 5% defective product. What is the probability that in a sample of ten screwdrivers there are exactly three defective units?

6.6.7 Poisson Distribution

It is a distribution related to the probabilities of events which are extremely rare but which have a large number of independent opportunities for occurrence. Examples are the number of people corn blind per year in a large city, or the number of soldiers in an army being injured due to the kick of a horse.

This distribution can be derived as the limiting case of the binomial distribution by making 'n' very large and 'p' very small, keeping n×p fixed (say = m).

The probability of r successes in a binomial distribution is

$P(r) = {}^nc_r p^r q^{n-r} = \frac{n(n-1)(n-2)...(n-r+1)}{r!} p^r q^{n-r}$

$= \frac{np(np-p)(np-2p)...(np-\overline{r-1}p)}{r!}(1-p)^{n-r}$

As $n \to \infty$, $p \to 0$ and np = m, we have

$$P(r) = \frac{m^r}{r!}\lim_{n\to\infty}\frac{\left(\frac{1-m}{n}\right)^n}{\left(\frac{1-m}{n}\right)^r} = \frac{m^r}{r!}e^{-m}$$

so that the possibilities 0, 1, 2..., r, ...successes in a poisson distribution are given by

$e - m,\ me - m,\ \frac{m^2e^{-m}}{2!},.....\frac{m^re^{-m}}{r!},...$

The sum of these possibilities is unity, as it should be.

Career Avenues

Consonants of a poisson distribution : These consonants can be easily derived from the corresponding consonants of the binomial distribution simply by making $n \to \infty$, $p \to 0$, $(q \to 1)$ and noting that np = m

Mean Lt (np) = m

Variance = Lt npq) = m Lt(q) = m

$\therefore$ Standard Deviation = $\sqrt{m}$

Question

In a certain factory that makes blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10. Use poisson distribution to calculate the approximate number of packets containing no defective, one defective or two defective blades respectively in a consignment of 10000 packets.

Answer

We know that $m = np = 10 \times 0.002 = 0.02$

$e^{-0.02} = 1 - 0.02 + \frac{(0.02)^2}{2!} - ... = 0.9802$ approximately

Probability of no defective blade is $e^{-m} = e^{-0.02} = 0.9802$

$\therefore$ No. of packets containing no defective blade is = $10{,}000 \times 0.9802 = 9802$

Similarly the number of packets containing one defective blade = $10{,}000 \times me^{-m}$

$= 10{,}000 \times (0.02) \times 0.9802 = 196.$

Finally the number of packets containing two defective blades

$= 10{,}000 \times \frac{m^2 e^{-m}}{2!} = 10{,}000 \times \frac{(0.02)^2}{2!} \times 0.9802 = 2$ approximately.

Do-it-yourself (DIY exercise)

7. In making switches, it has been determined by empirical studies that there is, on average, one defect per switch. What is the probability of selecting a sample of five switches that contains zero defects?

6.6.8 Normal Distribution

Now we consider a continuous of fundamental importance, namely the normal distribution. Any quantity whose variation depends on random causes is distributed according to the normal distribution.

Let us define a variate $z = \frac{x - np}{\sqrt{(npq)}}$ …(1)

Where x is a binomial variate with mean np and S.D. $\sqrt{(npq)}$ so that z is a variate with mean zero and variance unity. In the limit as n tends to infinity, the distribution

(1) for large values of n when neither p nor q is very small, is the normal distribution. The normal curve is of the form

$$y = \frac{1}{\sigma\sqrt{(2\pi)}} e^{-(x-\mu)^2/2\sigma^2} \quad \ldots(2)$$

where μ and σ are the mean and standard deviation respectively.

(2) Properties of the normal distribution

1. The normal curve (2) is bell-shaped and is symmetrical about its mean. It is unimodal with ordinates decreasing rapidly on both sides of the mean. The maximum ordinate is $\frac{1}{\sigma\sqrt{(2\pi)}}$, found by putting $x = \mu$ in (2).

As it is symmetrical, its mean, median and mode are the same.

Career Avenues

The probability of x lying between x_1 and x_2 is given by the area under the normal curve from x_1 and x_2, i.e.

$P(x_1 \le x \le x_2)$

$= \frac{1}{\sigma\sqrt{(2\pi)}} \int_{x_1}^{x_2} e^{-(x-\mu)^2/2\sigma^2} dx$

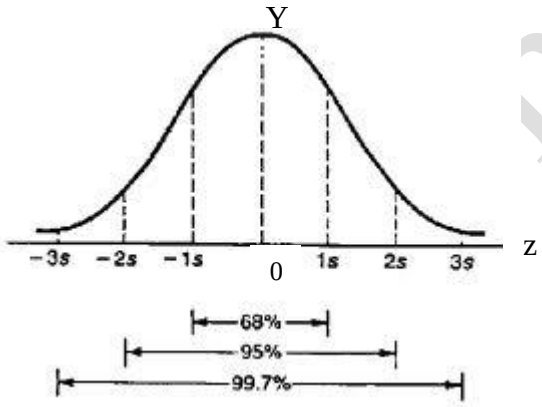
$= \frac{1}{\sqrt{(2\pi)}} \int_{z_1}^{z_2} e^{-z^2/2} dz$ where $z = (x-\mu)/\sigma, dz = dx/\sigma$ and $z_1 = (x_1-\mu)/\sigma, z_2 = (x_2-\mu)/\sigma.$

$= \frac{1}{\sqrt{(2\pi)}} \left\{ \int_0^{z_2} e^{-z^{2/2}} dz - \int_0^{z_1} e^{-z^{2/2}} dz \right\} = p_2(z) - p_1(z)$

The values of each of the above integrals can be found from the standard table which gives the values of

$$p(z) = \frac{1}{\sqrt{(2\pi)}} \int_0^z e^{-z^2/2} dz$$

For various values of z. This integral is called the probability integral or the error function due to its use in the theory of sampling and the theory of errors.



From figure, we see that the area under the normal curve from z = 0 to z = 1, i. e. from x = μ to $\mu + \sigma$ is 0.3413.

$\therefore$ (i) The area under the normal curve between the ordinates x = $\mu - \sigma$ and x = $\mu + \sigma$ is 0.6826, - 68% nearly. Thus approximately 2/3 of the values lie within these limits.

(ii) The area under the normal curve between x = $\mu - 2\sigma$ and x = $\mu + 2\sigma$ is 0.9544 – 95.5%, which implies that about $4\frac{1}{2}\%$ of the values lie outside these limits.

(iii) 99.73% of the values lie between x = $\mu - 3\sigma$ and x = $\mu + 3\sigma$ i. e. only a quarter % of the whole lies outside these limits.

Career Avenues

Question
If there are 100 typographical errors randomly distributed. In 500 pages manuscripts find the probability that any given page has exactly 4 errors.

Answer
Find the mean number of errors $m = \frac{100}{500} = \frac{1}{5} = 0.2$

In other words there is an average of 0.2 errors per page. In this case $\lambda = 4$ so the probability of selecting a page with exactly 4 errors

$$\frac{e^{-m}.m^{x}}{x!} = \frac{(2.7183)^{-0.2}(0.2)^{4}}{4!}$$

= 0.00168 = Approx. 0.2%

DIY Answers

1. P(1st Y) =3/8, P(2nd Y) = 2/7 and P(W)= 2/6
P(Y and Y and W)=P(Y) x P(Y) x P(W) = 3/8 x 2/7 x 2/6 = 1 / 28

2. P(G) =8/14 and P(B) =6/14. P(1st G)=8/14, P(2nd G) 7/13 and P(3rdG)= 6/12.
P(three girls) 8/14 x 7/13 x 6/12= 2/13

3. ${}^{8}C_{3}$ = 56 ways.

4. Ways of selecting one defective bulb = ${}^{3}C_{1}$ and
Ways of selecting three non defective bulb = ${}^{9}C_{3}$
Thus, total number of ways of selecting one defective bulb and three non defective bulb=

${}^{3}C_{1} \times {}^{9}C_{3} = \frac{3!}{(3-1)!1!} \times \frac{9!}{(9-3)!3!} = 25$

Ways of selecting four bulbs from 12 = ${}^{12}C_{4}$ = 495.
P(1 defective bulb and 3 okay bulbs) = 295/495=0.509.

5. It is known, by prior knowledge, that the probability of a single success (probability of a head in one toss of a coin) is fifty percent. The question is looking for two successes or two heads in five tosses of a coin. A success is the outcome that is desired to occur.
For this example:
The number of trials = n = 5
The probability of a single event = p = 1/2
The number of successes that the question is seeking (x = 2).
To arrive at the answer to the question the values are entered in the binomial formula.

P(2 Heads) = $\binom{5}{2}\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{3} = 10 \times \frac{1}{4} \times \frac{1}{8} = 0.3125$ or 31.25%

6. n = 10, p = .05, x = 3

P(3 defective units) = $\binom{10}{3}(0.05)^{3}(0.95)^{7} = 120 \times 0.000125 \times 0.6983 = 0.0105$ or 1.05%

7. To solve this problem, use the formula where x = 0, n = 5, and p = 1, therefore
np = 5 × 1 = 5.

P(0) = $\frac{e^{-5}(5)^{0}}{0!} = \frac{0.0674}{1} = 0.0674 \text{ or } 0.674\%$

Career Avenues

6.7 Statistics

Some interesting stats trivia

1. In the year 8,000 B.C., there were only 5 million people on Earth. Four thousand years later, the population had only risen by 2 million people, to 7 million people. The current population of Earth is over 6,910,000,000. Around 1900 there were only 1,600,000,000 people, meaning that Earth's population has more than quadrupled in slightly over 100 years' time.

2. Only 1% of the population has a "genius" IQ, one of 140 or higher.

3. Over 88% of the world's population lives north of the Equator.

4. Half of the people killed by bombs are those trying to make or set the bombs.

5. Around one in three murder cases are never solved.

The above trivia reads like fun. However, to create each of these trivia, enormous amount of statistical research and analysis would have happened. Statistics (*stats* for short) is everywhere. From making a prediction on which party is likely to win the election to making weather announcements, it is all about statistics. Infact, there is also research currently on in various parts of the world on which two star signs are most likely to fall in love at first sight, or on which day are you most likely to win a lottery!

This book will walk you through many of the concepts that you need to know for competitive tests. Make sure that all concepts in this book are well understood, especially those on probability.

There is a good chance that most of you would have already studied these chapters somewhere during your education. In our experience, this topic is better learnt if you aim at understanding the concepts and their applications. Without applications, it is highly unlikely that you would be able to solve higher level questions. So enjoy your prep with this book, solve all the practice questions, and make sure that you do not proceed further until you have understood the 'why' and 'how' of the concepts.

Enjoy. And as a parting note, here is a statistical joke.

A statistics professor was describing sampling theory to his class, explaining how a sample can be studied and used to generalize to a population. One of the students in the back of the room kept shaking his head. "What's the matter?" asked the professor. "I don't believe it," said the student, "why not study the whole population in the first place?" The professor continued explaining the ideas of random and representative samples. The student still shook his head. The professor launched into the mechanics of proportional stratified samples, randomized cluster sampling, the standard error of the mean, and the central limit theorem. The student remained unconvinced saying, "Too much theory, too risky, I couldn't trust just a few numbers in place of ALL of them." Attempting a more practical example, the professor then explained the scientific rigor and meticulous sample selection of the Nielsen television ratings which are used to determine how multiple millions of advertising dollars are spent. The student remained unimpressed saying, "You mean that just a sample of a few thousand can tell us exactly what over 250 MILLION people are doing?" Finally, the professor, somewhat disgruntled with the skepticism, replied, "Well, the next time you go to the clinic and they want to do a blood test...tell them that a syringe full of blood is not good enough ...tell them to TAKE IT ALL!!"

What the book covers?
- Sampling
- Mean, median, mode and standard deviation
- Correlation and regression analysis.
- Probability
- Conditional probability, Bayes Theorem
- Random variables
- Discrete and continuous distributions
- Poisson, Normal and Binomial distribution

Career Avenues

Statistics is the art of learning from data. It is concerned with the collection of data, its subsequent categorization and its analysis. This analysis often leads to the drawing of conclusions.

Sometimes, statistics begins with data already available. At other times, the data needs to be captured. For example, to answer the question "How many students in the class are boys?" may be easy to answer as such information may be available with the teacher. However, if the question was "How many students in the class are boys who are above 5 feet and weigh more than 50 kgs?", the relevant data may not be directly available. The teacher then needs to find the height and weight of all boys in the class, and then she would be able to answer the question.

What if the question is "What is the percentage of students in Class X across schools of India who weigh more than 50 kgs and are above 5 feet?". While this can be accurately answered by collecting data from all schools in India, it is not practical to do that due to the time, effort and costs involved. Also, as the data may take several months to collect, by the time you get the data, many more students whose data has already been captured but who earlier did not meet the criteria, may now meet the criteria.

To answer the question, you may not need the data of all the students (in statistical parlance, called the 'population'), but of a **representative** sample of the population. The word representative is critical as if the sample is not a true (unbiased) indicator of the population, then your confidence on the final outcome is not likely to be high. For example, if you only take sample data from students living in major metro cities, then it may be biased as due to better income in general, nutritional intake of students in metro cities may be higher than rural areas, leading to better overall growth in height and weight. So a correct sample would comprise a correct mix of students from different parts of the country. That mix may be based on geographical region (city, town, rural), nature of school (govt., private, etc), and so on. This could be further divided into other sub-categories such as English or Local Language schools, or schools in remote areas, hilly or desert regions, etc.

Sampling, therefore, is a subject by itself, and any statistical analysis should take into account the nature of samples, and be aware of likely sampling errors.

To recap,
A *population* is a finite or infinite collection of individuals or objects. Often it is impossible or impractical to get data on all the members of the population and a small part of it or *sample* is examined instead.

If a sample is representative of the population, a lot can be learned about the population by studying the sample.
Descriptive statistics deals with the description or simple analysis of population or sample data.

Perhaps the simplest analysis of data is to classify individual observations into two or more *categories* according to some *attribute* that they possess, e.g. people can be classified into employed/unemployed, or into several groups according to religion, etc.

A *variable* is a measurable characteristic which changes from one member of a sample or a population to another, e.g. age of a person, number of defective items in a standard box of screws, GDP of a country, etc.

6.7.1 Classification of data

Let us take the performance in the CAT exam of 20 students in Math. The Math section had a maximum score of 50 marks. The scores were

Student No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Marks	45	35	30	25	28	32	41	17	19	8	22	27	31	24	16	19	28	32	37	30

By themselves, these scores do not aid us in anyway. But when we classify the data, we may get more information on the scores.

In most real-life cases, one does not look at raw data as above, but classified data which is a representation of the raw data in different forms. Only using classified data, you can derive a lot of information about the raw data as well.

Career Avenues

Range of scores	No. of students
1-10	1
11-20	4
21-30	8
31-40	5
41-50	2

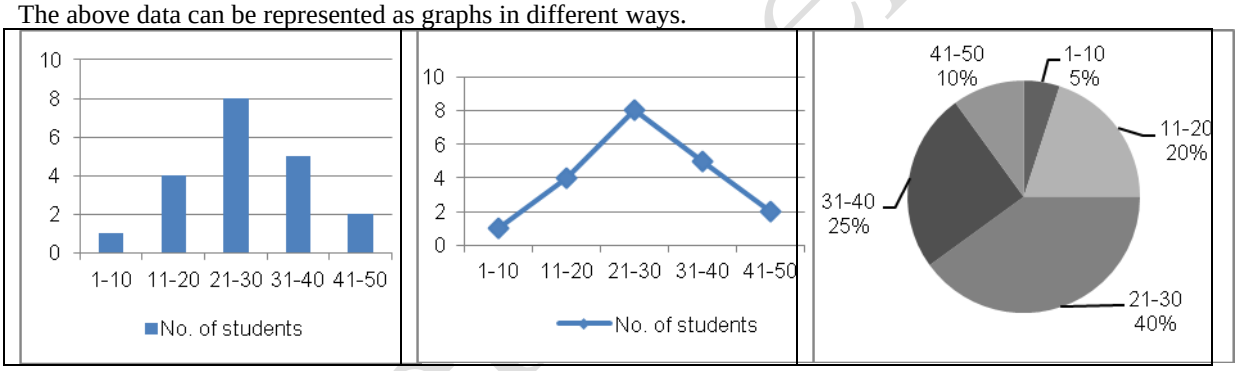
The above table gives us an idea that the scores are well spread in the 5 ranges, and that most scores are in 21-30 bracket. It also tells us that scores are more inclined towards higher values than lower values as there are 7 scores of greater than 30 compared to 5 of less than 20.

The **'range of scores'** above is called as **'class'** and the 'number of students' is called as 'frequency of the class'.

The width of each class is 10 marks, and the table is called as frequency distribution table.

Mid value of the class is calculated as $\frac{\text{starting value of range} + \text{ending value of range}}{2}$. In the above case, it would be 5.5, 15.5, 25.5, etc.

The above data can be represented as graphs in different ways.



More information can be retrieved by calculation of a few further measures from the data. We first try to evaluate what is the data pointing to in broad terms. This can also be called as the central tendency of the data. We then look at how well is the data spread across the range (dispersion).

Measures of central tendency : Mean, Median, Mode
Measures of dispersion : Range, Quartiles, Standard Deviation, Variance.

ARITHMETIC MEAN OR ARITHMETIC AVERAGE

Arithmetic Mean **(AM)** is the generic usage of the word average. AM is used when all values are given equal weightage and it is the mathematically central value.

For example, Average (A.M) of 1, 2, 3 and 4 is $\frac{1+2+3+4}{4} = \frac{10}{4} = 2.5$

Average of a group of entire data is defined as, $\bar{x} = \frac{\text{Sum of all data in the group}}{\text{Number of data in the group}}$

Sum of all data in the group means sum of all items in the group.

Hence, the average of 'n' quantities $X_1, X_2 \ldots X_n$, Average, $\bar{x} = \frac{X_1 + X_2 + X_3 + X_4 + \ldots + X_n}{n}$

Career Avenues

AVERAGE BY 'DEVIATION METHOD'	
METHOD	EXAMPLE
This is an easy method to calculate the average of a group of data. When large numbers are involved, the average is generally assumed to facilitate calculation. The average of n quantities $X_1, X_2 \ldots X_n$, Average, $\bar{x} = A + \frac{\sum_{i=1}^{i=n}(X_i - A)}{n}$ Where • $\bar{x}$ is the Average of the group • 'A' stands for Assumed Average • $(X_i - A)$ is the deviation of the X_i from the Assumed Average A • $\sum_{i=1}^{i=n}(X_i - A)$ is the sum of the deviations of all the items in the group • n is the number of items in the group	Find the average of 52, 56, 58, 62, 64 and 60. We know that Average lies between the smallest and the largest value in the group. Smallest number in the group = 52 Largest number in the group = 64 Let us take the Assumed Average as 60 A = 60. By deviation method, $\bar{x} = A + \frac{\sum_{i=1}^{i=n}(X_i - A)}{n}$ The deviation between the Assumed Average and each number in the group are (52 – 60), (56 – 60), (58 – 60), (62 – 60), (64 – 60) and (60 – 60) $\bar{x} = 60 + \frac{-8-4-2+2+4+0}{6}$ $\bar{x} = 60 - \frac{8}{6} = 58.67$

6.7.2 Arithmetic Mean of a frequency distribution

In a frequency distribution, where $x_1, x_2, \ldots, x_n$ are the mid values of the class, and $f_1, f_2, \ldots, f_n$ are frequencies for the values, then Average or $\bar{X} = \frac{f_1x_1 + f_2x_2 + \ldots + f_nx_n}{f_1 + f_2 + \ldots f_n} = \frac{\sum f_i x_i}{\sum f_i}$

In our previous example on students scores, it would be
$\frac{5.5\times1+15.5\times4+25.5\times8+35.5\times5+45.5\times2}{1+4+8+5+2} = 27$.

Note that this is not too far from 27.3 which is the actual arithmetic mean if raw data was used.

6.7.3 Median

Median is used to define the central value in a set. The central value is identified in terms of its magnitude. i.e. Median is the value which divides the frequency into two parts.
Examples
1. The set of numbers 1, 3, 5, 6, 8, 9, 10 has median as 6
2. The set of numbers 1, 3, 4, 5, 7, 9 has median $\frac{4+5}{2} = 4.5$

Note: a) If there are even number of items in the group, median is found by taking the average of the two middle values. b) The numbers should be arranged in ascending or descending order before the median is calculated.

Median can also be taken as second quartile. A quartile divides the data into 4 parts (quarters). The median divides the data into two parts.

Career Avenues

Question
Find the median of the following set of numbers: 57, 58, 61, 42, 38, 65, 72 and 66.
1] 58 2] 61 3] 59.5 4] 60.5

Answer
Arranging the data in ascending order we get, 38, 42, 57, 58, 61, 65, 66, 72 (N = 8)

$\therefore$ Median = Average of $\left(\frac{N}{2}\right)^{th}$ and $\left(\frac{N}{2}+1\right)^{th}$ item

= Average of $\left(\frac{8}{2}\right)^{th}$ i.e 4th and $\left(\frac{8}{2}+1\right)$ i.e 5th items

$= \frac{58+61}{2} = 59.5$

6.7.4 Mode

Mode is used to define the most repeated value in the group of items.
Example
1. Mode of the numbers 2, 3, 4, 4, 5, 5, 5, 6, 7, 9 is 5 since 5 is repeated most number of times
2. Mode of the numbers 2, 3, 4, 4, 5, 5, 7, 9 are 4 and 5, since 4 and 5 both repeated the same number of times.

In a grouped distribution, it is given by the formula Mode = $L + \frac{\Delta_1}{\Delta_1 + \Delta_2} h$

where L is the lower limit of the class containing the mode (highest frequency class), h is the size of the modal class, Δ_1 is the excess of modal frequency of mode class compared to preceding class, and Δ_2 is the excess of modal frequency of mode class compared to following class.

6.7.5 Relationship between Mean, Mode and Median

In a symmetrical distribution, the mean, mode and median coincide.
E.g. 1, 2, 2, 3, 3, 3, 4, 4, 5, the mean, mode and median are all 3.

In non-symmetrical distributions, they are different. We will discuss this in the next section.

6.7.6 Understanding the relevance of Mean and Standard Deviation

The **mean** is the sum of a list of numbers, divided by the total number of numbers in the list.
The following table represents the different marks obtained by 7 students:

Mark	33%	42%	51%	59%	67%	75%	83%

So the mean or average mark is $\frac{33+42+51+59+67+75+83}{7} = \frac{410}{7} \approx 58.57$

A list of number can be represented in a different way. For example, say that in a group of ten students, seven students got 46% and three students got 61%. In this case you'd have to calculate the mean as follows:

$$\frac{7\times 46 + 3\times 61}{10} = 50.5$$

Now, say that in a group of ten students, five students got 17% and five students got 83%. The mean is, in this case,

$$\frac{5\times 17 + 5\times 83}{10} = 50 .$$

Even though the average mark is 50%, all the marks are far from 50% so the mean does not tell us much about the list of numbers, here.

Career Avenues

We need another tool to tell us whether the numbers are close to their mean or far from it. This tool is the **standard deviation**.

6.7.7 Standard deviation

The **standard deviation** measures how far, on average, the numbers are from their mean (the standard deviation is usually represented by the greek letter σ).

We need a mathematical formula to better explain the concept.

Say you are given a list of numbers $x_1,\ldots,x_N$. Then the mean is $\bar{x} = \frac{x_1 + \ldots + x_N}{N}$

and the standard deviation is $\sigma = \sqrt{\frac{1}{N}\sum_{k=1}^{k=N}\left(x_k - \bar{x}\right)^2}$. Please note that this is the formula when SD for the entire population is calculated. If we have data on only of a sample, and not the entire population, then the formula to calculate SD would be $\sigma = \sqrt{\frac{1}{N-1}\sum_{k=1}^{k=N}\left(x_k - \bar{x}\right)^2}$

Let's go back now to one of the previous example: five students got 17% and five students got 83%.
In terms of a list of numbers, you can write it as follows: $x_1 = x_2 = \ldots = x_5 = 17$ and $x_6 = x_7 \ldots = x_{10} = 83$.

The mean, as before, is 50% but since we are trying to stick with mathematical notations we say $\bar{x} = 50$

The standard deviation is now $\sigma = \sqrt{\frac{1}{10}\left(5\times(17-50)^2 + 5\times(83-50)^2\right)} = 33.$

It works really well, since each mark is precisely 33% away from the average mark $\bar{x} = 50\%$.

The meaning of Standard Deviation

The standard deviation is kind of the "mean of the mean," and often can help you find the story behind the data. To understand this concept, it can help to learn about what statisticians call **normal distribution** of data.

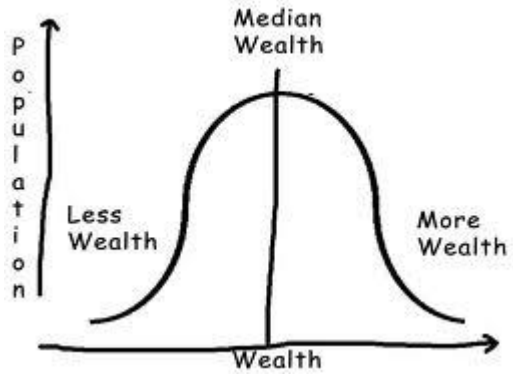
A normal distribution of data means that most of the examples in a set of data are close to the **"average,"** while relatively few examples tend to one extreme or the other.

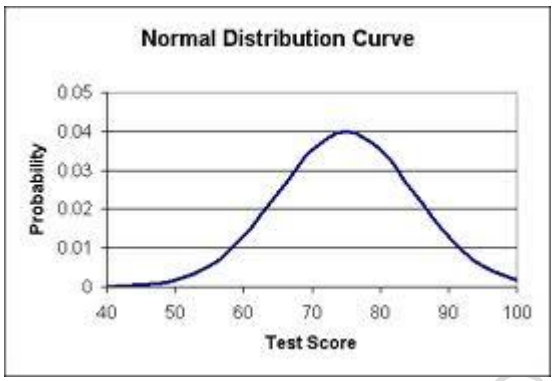
Let's say you are writing a story about nutrition. You need to look at people's typical daily calorie consumption. Like most data, the numbers for people's typical consumption probably will turn out to be normally distributed. That is, for most people, their consumption will be close to the Mean, while fewer people eat a lot more or a lot less than the Mean.

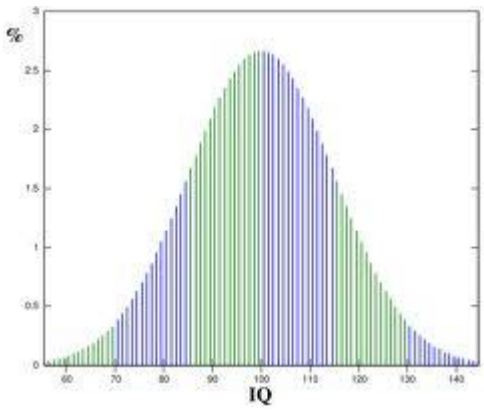
When you think about it, that's just common sense. Not that many people are getting by on a single serving of dal and roti. Or on eight meals of burgers and milkshakes. Most people lie somewhere in between.

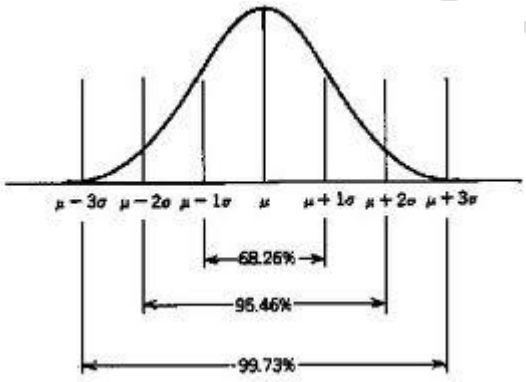
If you looked at normally distributed data on a graph, it would look something like this:

Career Avenues



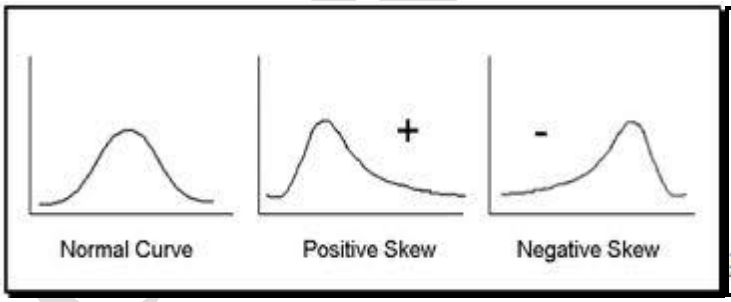


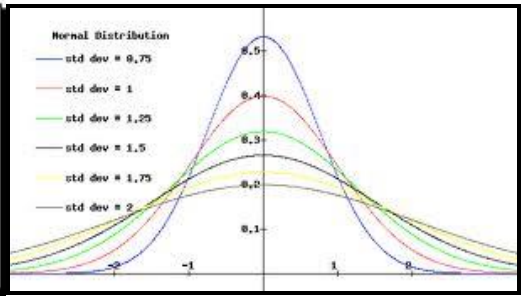




The **x**-axis (the horizontal one) is the value in question... calories consumed, wealth earned or marks scored, for example. And the **y**-axis (the vertical one) is the number of datapoints for each value on the **x**-axis... in other words, the number of people who eat **x** calories, the number of households that earn **x** salary, or the percentage of students with **x** as IQ score.

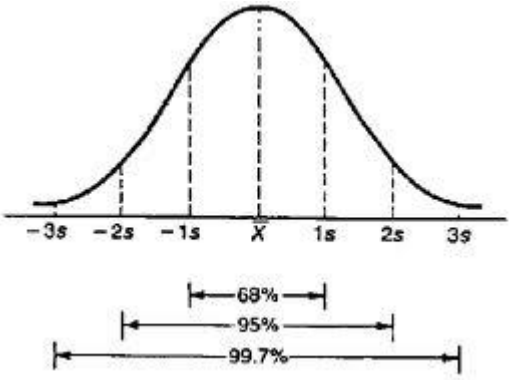
Now, not all sets of data will have graphs that look this perfect. Some will have relatively flat curves, others will be pretty steep. Sometimes the mean will lean a little bit to one side or the other. But all normally distributed data will have something like this same "bell curve" shape.





The **standard deviation** is a statistic that tells you how tightly all the various examples are clustered around the mean in a set of data. When the examples are pretty tightly bunched together and the bell-shaped curve is steep, the standard deviation is small. When the examples are spread apart and the bell curve is relatively flat, that tells you that you have a relatively large standard deviation.

Graphically what a standard deviation represents...



One standard deviation away from the mean in either direction on the horizontal axis (the central area on the above graph) accounts for somewhere around 68 percent of the people in this group. Two standard deviations away from the mean account for roughly 95 percent of the people. And three standard deviations account for about 99 percent of the people.

If this curve were flatter and more spread out, the standard deviation would have to be larger in order to account for those 68 percent or so of the people. So that's why the standard deviation can tell you how spread out the examples in a set are from the mean.

Simple Example of Calculating Standard Deviation

Let's say we wanted to calculate the standard deviation for the amounts of gold coins pirates on a pirate ship have.

There are 100 pirates on the ship. In statistical terms this means we have a population of 100. If we know the amount of gold coins each of the 100 pirates have, we use the **standard deviation equation for an entire population:**

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N}}$$

Where

σ = standard deviation

x = each value in population

$\bar{x}$ = the mean of the values

N = number of values (population)

What if we don't know the amount of gold coins each of the 100 pirates have? For example, we only had enough time to ask 5 pirates how many gold coins they have. In statistical terms this means we have a sample size of 5 and in this case we use the **standard deviation equation for a sample of a population:**

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N - 1}}$$

where

σ = standard deviation

x = each value in population

$\bar{x}$ = the mean of the values

N = number of values (the sample size)

The rest of this example will be done in the case where we have a sample size of 5 pirates, therefore we will be using the standard deviation equation for a sample of a population.

Here are the amounts of gold coins the 5 pirates have:

4, 2, 5, 8, 6.

Career Avenues

Now, let's calculate the standard deviation:

1. Calculate the mean: 4, 2, 5, 8, 6 i.e. $\bar{x} = 5$
2. Calculate $x - \bar{x}$ for each value of the sample. i.e. – 1, – 3, 0, 3, 1
3. Calculate $\sum (x - \bar{x})^2 = (-1)^2 + (-3)^2 + 0^2 + 3^2 + 1^2 = 20$
4. Calculate the standard deviation using the formula

4. Calculate the standard deviation: $\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N - 1}} = \sqrt{\frac{20}{5 - 1}} = 2.24$

The standard deviation for the amounts of gold coins the pirates have is 2.24 gold coins.

6.7.8 Variance

Standard Deviation is the square root of variance. Hence Variance of a sample = $\sigma^2 = \frac{\sum (x - \bar{x})^2}{N - 1}$

Difference between SD and Variance

Standard Deviation shows how much variation or "dispersion" there is from the average (mean, or expected value). A low standard deviation indicates that the data points tend to be very close to the mean, whereas high standard deviation indicates that the data are spread out over a large range of values. A useful property of standard deviation is that, unlike variance, it is expressed in the same units as the data.

6.7.9 Range

Range is the difference between the **greatest and the least** value of the variable.

Question

Two players play in the IPL. In 10 matches, their performance in terms of runs scored is given.

A	45	40	35	38	27	46	55	12	35	50
B	112	72	36	3	24	7	62	14	12	90

Who is likely to score more? Who is more consistent?

Answer

Mean of A = 38. 3 and SD of A = 12.3
Mean of B = 43. 2 and SD of B = 38.4

Hence A is more consistent and B is likely to be a better scorer.

6.7.10 Moments

The first few central moments have intuitive interpretations:

$$\mu_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^r$$

- The "zeroth" central moment μ_0 is one.
- The first central moment μ_1 is zero.
- The second central moment μ_2 is called the variance, and is usually denoted σ^2, where σ represents the standard deviation.
- The third and fourth central moments are used to define the standardized moments which are used to define skewness and kurtosis, respectively.

Career Avenues

6.7.11 Skewness

In everyday language, the terms "skewed" is used to refer to something that is out of line or distorted on one side. When referring to the shape of frequency or probability distributions, "skewness" refers to asymmetry of the distribution. A distribution with an asymmetric tail extending out to the right is referred to as "positively skewed" or "skewed to the right," while a distribution with an asymmetric tail extending out to the left is referred to as "negatively skewed" or "skewed to the left."

Karl Pearson (1895) first suggested measuring skewness by standardizing the difference between the mean and the mode, that is, $\text{sk} = \frac{\text{mean} - \text{mode}}{\sigma}$. This statistic ranges from –1 to +1. Absolute values above 0.2 indicate great skewness.

Great skewness should result in paying a closer look at outliers (data at the extremes). When making decisions about which measure of location to report (means being drawn in the direction of the skew) and which inferential statistic to employ (one which assumes normality or one which does not), one should take into consideration the estimated skewness of the population. Normal distributions have zero skewness.

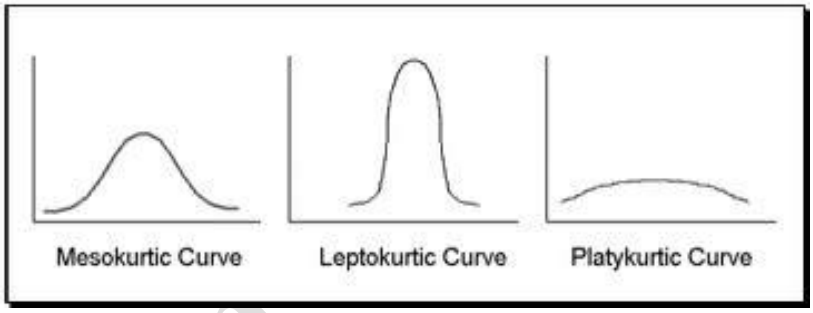
6.7.12 Kurtosis

Kurtosis is usually of interest only when dealing with approximately symmetric distributions. It is represented by the symbol γ_2.

Kurtosis is commonly defined as the fourth cumulant divided by the square of the second cumulant, which is equal to the fourth moment around the mean divided by the square of the variance of the probability distribution minus 3, which is also known as *excess kurtosis*.

Kurtosis = $\gamma_2 = \frac{\mu^4}{\sigma^4} - 3$

The "minus 3" at the end of this formula is often explained as a correction to make the kurtosis of the normal distribution equal to zero.



Pearson introduced kurtosis as a measure of how flat the top of a symmetric distribution is when compared to a normal distribution of the same variance. He referred to more flat-topped distributions ($\gamma_2 < 0$) as "platykurtic," less flat-topped distributions ($\gamma_2 > 0$) as "leptokurtic," and equally flat-topped distributions as "mesokurtic" ($\gamma_2 \approx 0$). Kurtosis is actually more influenced by scores in the tails of the distribution than scores in the center of a distribution.

Career Avenues

6.8 Correlation and Regression

6.8.1 Correlation

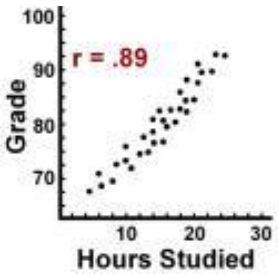
Scatter Plots

A scatter plot is designed to show a relationship between two variables by graphing a series of observations on a two-dimensional graph – one variable on the X-axis, the other on the Y-axis.

Definitions

Correlation: A measure of the degree of relatedness of two or more variables (r). The results can range from -1 to 1. A correlation of -1 is a perfectly negative correlation. A perfect negative correlation dictates that an "inverse relationship" exists between two variables: as one variable becomes larger, the other becomes smaller. A correlation of +1 is a perfectly positive correlation. A correlation of 0 has no correlation, or what we would consider a weak relationship.

Example of Positive Correlation:



Example of Negative Correlation:



r, the coefficient of linear correlation (also called as "The Pearson product-moment correlation coefficient"), is the numerical measure of the strength of the linear relationship between two variables.

Sample Covariance

To quantify a linear relationship between two variables, we start by finding the **covariance** of a sample of paired observations. A sample covariance between two random variables X and Y is the average value of the cross-product of all observed deviations from each respective sample mean. A cross-product, for the ith observation in a sample, is found by this calculation: (ith observation of X – sample mean of X) * (ith observation of Y – sample mean of Y). The covariance is the sum of all cross-products, divided by (n – 1).

To illustrate, take a sample of five paired observations of two variables, which we will label X and Y:

Year	X	Y	Cross-Product: $(X_i - X_{mean}) \times (Y_i - Y_{mean})$
1st	+15.5	+9.6	(15.5 – 6.6) × (9.6 – 7.3) = 20.47
2nd	+10.2	+4.5	(10.2 – 6.6) × (4.5 – 7.3) = -10.08
3rd	-5.2	+0.2	(-5.2 – 6.6) × (0.2 – 7.3) = 83.78
4th	-6.3	-1.1	(-6.3 – 6.6) × (-1.1 – 7.3) = 108.36
5th	+12.7	+23.5	(12.7 – 6.6) × (23.5 – 7.3) = 196.02
Sum	32.9	36.7	398.55
Average	6.6	7.3	398.55/(n – 1) = 99.64 = Cov (X,Y)

Average X and Y were found by dividing the sum by n or 5, while the average of the cross-products is computed by dividing the sum by n – 1, or 4. The use of n – 1 for covariance is done to ensure an unbiased estimate.

The fact that **Cov (X,Y)** of 99.64 was greater than 0 does indicate a positive or linear relationship between X and Y. Had the covariance been a negative number, it would imply an inverse relationship, while 0 means no relationship. Thus 99.64 indicates that the returns have positive co-movement (when one moves higher so does the other), but doesn't offer any information on the extent of the co-movement.

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Sample Correlation Coefficient

By calculating a **correlation coefficient**, we essentially convert a raw covariance number into a standard format that can be more easily interpreted to determine the extent of the relationship between two variables. The formula for calculating a sample correlation coefficient (r) between two random variables X and Y is the following:

$$r = \frac{\text{covariance between X, Y}}{\text{sample standard deviation of X} \times \text{sample standard deviation of X}}$$

Example: Correlation Coefficient

Return to our example from the previous section, where covariance was found to be 99.64. To find the correlation coefficient, we must compute the sample variances, a process illustrated in the table below.

Year	X	Y	Squared X deviations $(X-\bar{x})^2$	Squared Y deviations $(Y-\bar{y})^2$
1st	+15.5	+9.6	$(15.5 - 6.6)^2 = 79.21$	$(9.6 - 7.3)^2 = 5.29$
2nd	+10.2	+4.5	$(10.2 - 6.6)^2 = 12.96$	$(4.5 - 7.3)^2 = 7.84$
3rd	-5.2	+0.2	$(-5.2 - 6.6)^2 = 139.24$	$(0.2 - 7.3)^2 = 50.41$
4th	-6.3	-1.1	$(-6.3 - 6.6)^2 = 166.41$	$(-1.1 - 7.3)^2 = 70.56$
5th	+12.7	+23.5	$(12.7 - 6.6)^2 = 146.41$	$(23.5 - 7.3)^2 = 262.44$
Sum	32.9	36.7	544.23	369.54
Average	$\bar{x} = 6.6$	$\bar{y} = 7.3$	136.06 = X variance	99.14 = Y variance

Answer:

As with sample covariance, we use (n – 1) as the denominator in calculating sample variance (sum of squared deviations as the numerator) – thus in the above example, each sum was divided by 4 to find the variance. Standard deviation is the positive square root of variance: in this example, sample standard deviation of X is $(136.06)^{1/2}$, or 11.66; sample standard deviation of Y is $(99.14)^{1/2}$, or 9.96.

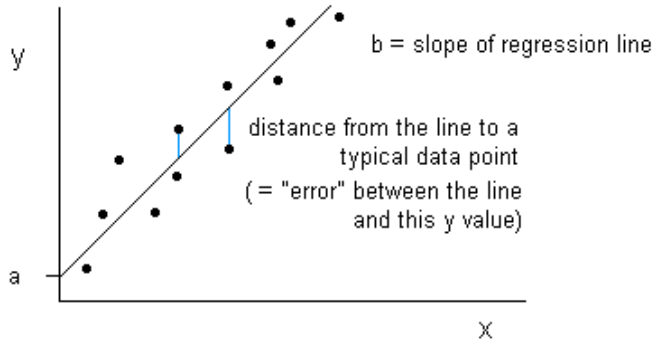
Therefore, the correlation coefficient is (99.64)/(11.66 × 9.96) = 0.858. A correlation coefficient is a value between –1 (perfect inverse relationship) and +1 (perfect linear relationship) and the closer it is to 1, the stronger the relationship. This example computed a number of 0.858, which would suggest a strong linear relationship.

6.8.2 Regression:

It is likely that the dots of the scatter diagram generally tend to cluster around a well defined direction which suggests a linear relationship between variables x and y. Such a line of best fit for the given distribution of dots is called as the line of regression. There are two such lines:

1) Regression of y on x: gives the best possible mean value of y for each specified value of x.
2) Regression of x on y: gives the best possible mean value of x for each specified value of y.

It's customary to use "a" or "alpha" for the intercept of the line, and "b" or "beta" for the slope; so linear regression gives you a formula of the form: $y = bx + a$



Regression analysis is used to understand the statistical dependence of one variable on other variables. The technique can show what proportion of variance between variables is due to the dependent variable, and what

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proportion is due to the independent variables. The relation between the variables can be illustrated graphically, or more usually using an equation.

Simple Regression is linear and involves only two variables where the independent variable (x) is used to predict the dependent variable (y). Multiple Regression involves one dependent variable with two or more independent variables.

By measuring exactly how large and significant each independent variable has historically been in its relation to the dependent variable, the future value of the dependent variable can be predicted. Essentially, regression analysis attempts to measure the degree of correlation between the dependent and independent variables, thereby establishing the latter's predictive value. For example, a manufacturer of baby food might want to determine the relationship between its sales and percentage of working women in the city workforce as part of a sales forecast. Using a scatter graph, it might plot on the X and Y axes the historical sales for ten years and the working woman percentage values for the same period. A line connecting the average dots, called the regression line, would reveal the degree of correlation between the two factors by showing the amount of unexplained variation-represented by the dots falling outside the line.

Thus, if the regression line connected all the dots, it would demonstrate a direct relationship between baby food sales and percentage of working women, meaning that one could be predicted on the basis of the other. The proportion of dots scattered outside the regression line would indicate, on the other hand, the degree to which the relationship was less direct, a high enough degree of unexplained variation meaning there was no meaningful relationship and that women working have no predictive value in terms of baby food sales. This proportion of unexplained variations is termed the *coefficient of determination*, and its square root the Correlation Coefficient. The correlation coefficient is the ultimate yardstick of regression analysis: a correlation coefficient of 1 means the relationship is direct-baby food and women working move together; -1 means there is a negative relationship-the more working women there are, the less baby food is sold; a coefficient of zero means there is no relationship between the two factors.

Line of Best Fit: This results from an analysis of two (or more) variables that are related quantitative variables. When the line of best fit is plotted on a graph, various information is represented. First, and most obvious, is the pictorial illustration. The graph line shows whether or not there is a linear relationship (either positive or negative) between the variables. It also shows the relationship of the variables as an equation that can be derived from the line of best fit. This equation can then be used to predict the outcomes of particular variables that may exist outside of the given sample population of data.

Regression analysis finds the equation of the line that best describes the relationship between the two variables. The use of this equation is to make predictions. The problem with forecasting, is that you can only predict a value that falls in the data you already know. The algebraic expression is the expression of y **(y-hat) = a + bx.** This is the **line of best fit**.

Steps in Calculating Regression
Correlation and regression are concerned with the investigation of two variables.

We look at two associated variables. We might want to know:

- If a relationship exists between those variables;
- if so, how strong that relationship is;
- what form that relationship takes.
- Can we make use of that relationship for predictive purposes i.e. forecasting?

Correlation describes the strength of the relationship. It is not concerned with 'cause' and 'effect'.

Regression describes the relationship itself in the form of a straight line equation which best fits the data.

Some initial insight into the relationship between two continuous variables can be obtained by plotting a scatter diagram and simply looking at the resulting graph.

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Does the relationship seem to be linear or curved?

If there appears to be a linear relationship, it can be quantified. A correlation coefficient is calculated as the measure of the strength of this relationship. Its symbol is 'r' and its value lies between -1 and +1.
Is the association between the two variables strong enough to be useful?

If the relationship is found to be significantly strong, then its nature can be found using linear regression, which defines the equation of the straight line of 'best fit' through the bi-variate data, $y = a + bx$. For example, £x spent on Advertising expected to increase Sales by £y.

The 'goodness of fit' can be calculated to see how well the line fits the data.

Once defined by an equation, the relationship can be used for predictive purposes.

Example
'Ice cream Sales' for a particular firm of manufacturers and 'Average Monthly Temperature'.

Month	Av.Temp. (°C)	Sales (Rs. '000)
January	4	73
February	4	57
March	7	81
April	8	94
May	12	110
June	15	124
July	16	134
August	17	139
September	14	124
October	11	103
November	7	81
December	5	80

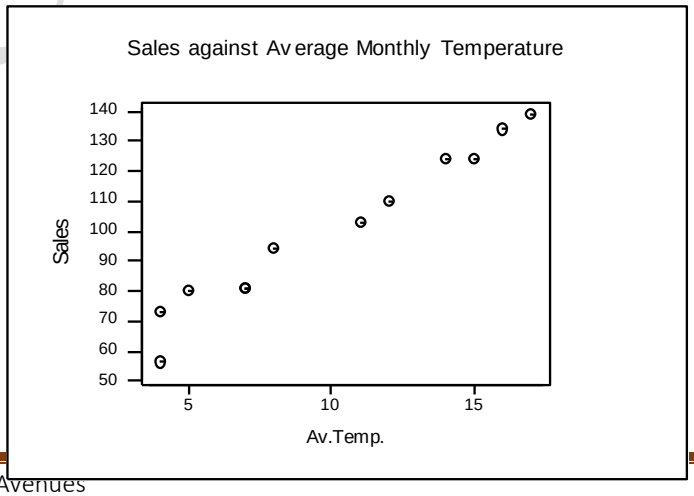
From this data we need:

- Scatter diagram of Sales against Temp.
- Correlation coefficient between them
- Regression line associating Sales with Temp.
- Goodness of fit of data to line
- Prediction of estimations for Sales
- Scatter diagrams

We look for a linear relationship with the bivariate points plotted being reasonably close to the, yet unknown, 'line of best fit'.

Plot the independent variable, x (Av.Temp), on the horizontal axis.

Plot the potentially dependent variable (Sales) on the vertical, y, axis.



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Looks promising: a straight line relationship, with all points fairly close to a 'line of best fit'.

6.8.3 Pearson's Correlation Coefficient (r)

(for quantitative data only)

The strength of the relationship will be quantified by calculating the correlation coefficient and tested for significance.

The formula used to find the Pearson's Product Moment Correlation coefficient is:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \qquad (-1 \le r \le +1)$$

where $S_{xx} = \sum x^2 - \frac{\sum x \sum x}{n}$; $S_{yy} = \sum y^2 - \frac{\sum y \sum y}{n}$; $S_{xy} = \sum xy - \frac{\sum x \sum y}{n}$

The following example gives ice-cream sales per month against mean monthly temperature (Fahrenheit)

Needed: $\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$, n.

Month	Average temp (x)	Sales (y)	x^2	y^2	xy
Jan.	4	73	16	5329	292
Feb.	4	57	16	3249	228
March	7	81	49	6561	567
April	8	94	64	8836	752
May	12	110	144	12100	1320
June	15	124	225	15376	1860
July	16	134	256	17956	2144
Aug.	17	139	289	19321	2363
Sept.	14	124	196	15376	1736
Oct.	11	103	121	10609	1133
Nov.	7	81	49	6561	567
Dec.	5	80	80	6400	400
Sum Σ	1200	1200	1450	127674	13362

Calculations

$$n = 12, \quad \sum x = 120, \quad \sum y = 1200$$
$$\sum x^2 = 1450, \quad \sum y^2 = 127674, \quad \sum xy = 13362$$

$$S_{xx} = 1450 - \frac{120 \times 120}{12} = 250$$
$$S_{yy} = 127674 - \frac{1200 \times 1200}{12} = 7674$$
$$S_{xy} = 13362 - \frac{120 \times 1200}{12} = 1362$$

$$r = \frac{2433}{\sqrt{792 \times 7674}} = 0.9869$$

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Regression equation

The gradient, b, is calculated from :

$$b = \frac{S_{xy}}{S_{xx}}$$

where $S_{xy} = \sum xy - \frac{\sum x \sum y}{n}$

and $S_{xx} = \sum x^2 - \frac{\sum x \sum x}{n}$

Since the regression line passes through the centroid, both means, its equation can be used to find the value of a, the intercept on the y-axis: $a = \bar{y} - b\bar{x}$

Sales now assumed to be dependent on temperature

$S_{xy} = 1362 \quad S_{xy} = 250$

$b = \frac{1362}{250} = \underline{5.448} \qquad a = \frac{1200}{12} - 5.448 \times \frac{120}{12} = \underline{45.52}$

The values of a and b are therefore 45.5 and 5.45 respectively giving the regression line: y = 45.5 + 5.45x

Career Avenues

SET-16

Permutation and Combination

1. In how many ways can 5 posts be assigned to 5 candidates selected after an interview?
2. Find the permutations of four alphabets p, q, r, s taken 2 at a time.
3. In how many ways can the letters of the word "Pre-University" be arranged?
4. A child has 3 pocket and 4 coins. In how many ways can he put the coins in his pocket.
5. There are 3 letters which have to be put into 4 boxes. In how many ways can be this done?
6. Find the number of ways in which 20 beads can be arranged to form a necklace.
7. How many necklace of 12 beads each can be made from 18 beads of different colors?
8. How many 5-lettered words can be formed with the letters of the word 'OMEGA' when?
   (i) 'O' and 'A' occupying end places.
   (ii) 'E' being always in the middle
   (iii) Vowels occupying odd-places
9. In how many ways can the letters of the word INDIA be arranged?
10. In how many ways can the letters of the word MISSISSIPI be arranged so that the two letters M & P are never together?
11. How many four digit numbers can be formed using 0,1,2,3,4 and 5?
12. Find the number of diagonals that can be drawn by joining the angular points of a pentagon?
13. A committee of 4 is to be selected from 7 boys and 5 girls. In how many ways can this be done so as to include (a) exactly 1 girl, (b) at least 1 girl.
14. In how many ways can 5 girls sit around a table so that all shall not have the same neighbors in any two arrangements?

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Solutions:

Permutation and Combinations (SET-1)

1. Ans:

The post can be assigned in P(n,k) ways, n=5; k=5

P(5,5) = $\frac{5!}{(5-5)!} = \frac{5!}{0!}$ = 5! = 5 x 4 x 3 x 2 x 1 = 120 ways.

2. Ans:

Number of permutations = P(n,k), n = 4, k = 2

P (n,k) = P (4,2) = $\frac{4!}{(4-2)!} = \frac{4\times3\times2\times1}{2\times1}$ = 12.

3. Ans:

$\frac{13!}{2!\times2!\times2!}$. The denominator takes care of the repetition of the letters e, r and i.

4. Ans:

First coin can be put in 3 ways, similarly second, third and forth coins also can be put in 3 ways.

So total number of ways = 3 × 3 × 3 × 3 = 3^4 = 81

5. Ans:

Each letter can be put in any of the 4 boxes.

∴ Total number of ways = 4 × 4 × 4 = 64.

6. Ans:

Fixing the position of one bead, the remaining beads can be arranged in (20 – 1)! i.e 19! ways.

As there is no distinction between the clockwise and anti-clockwise arrangements, the required number of ways = $\frac{1}{2}(19)!$

7. Ans:

Here clock-wise and anti-clockwise arrangements are same.

Hence total number of circular–permutations: $\frac{{}^{18}P_{12}}{2\times12} = \frac{18!}{6!\times24}$

8. Ans:

(i) When 'O' and 'A' occupying end-places

=> M.E.G. (OA)

Here (OA) are fixed, hence M, E, G can be arranged in 3! ways

But (O,A) can be arranged themselves is 2! ways.

=> Total number of words = 3! × 2! = 12 ways.

(ii) When 'E' is fixed in the middle

=> O.M.(E), G.A.

Hence four-letter O.M.G.A. can be arranged in 4! i.e 24 ways.

(iii) Three vowels (O,E,A) can be arranged in the odd-places (1st, 3rd and 5th) = 3! ways.

And two consonants (M,G) can be arranged in the even-place (2nd, 4th) = 2! ways

=> Total number of ways= 3! × 2! = 12 ways.

9. Ans:

There are 5 letters out of which there are 2 similar alphabets (I's). Hence the no. of ways = $5P_2$ = $\frac{5!}{2!} = 60$.

10. Ans:

Number of letters in MISSISSIPI = 10, with 4 S's & 4 I's

∴ Total number of ways in which the letters can be arranged = $\frac{10!}{4!4!}$

Since M and P are to be together, we treat them together as one letter.

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Total number of words = $\frac{9!}{4!4!}$ (now then are 9 letters with M and P being arranged).

However, for each of these $\frac{9!}{4!4!}$ ways, M and P can be interchanged (MP and PM)

$\therefore$ Total number of ways in which M and P are side by side = $\frac{2\times 9!}{4!\times 4!}$.

$\therefore$ Total number of ways in which the letters M and P are never together = $\frac{10!}{4!4!} - \frac{2\times 9!}{4!4!}$ = 5040.

11. Ans:

There are 6 digits. As zero cannot take the significant place, the thousand's place can be filled with one of the remaining 5 digits. Second, third and fourth places can be filled up in 6, 5 and 4 ways. Therefore total number of ways = 5 × 6 × 5 × 4 = 600 ways.

12. Ans:

A pentagon has 5 angular points and 5 sides. The join of 2 angular points is either a side or a diagonal.

The number of lines joining the angular points = ${}^5C_2 = \frac{5!}{2!3!} = \frac{5\times 4}{2} = 5\times 2 = 10$

But the number of sides = 5. $\therefore$ The number of diagonals = 10 – 5 = 5.

13. Ans:

(a) In this case we have to select 1 girl out of 5 girls and 3 boys out of 7 boys.

The number of ways of selecting 3 boys = ${}^7C_3 = \frac{7!}{4!3!} = \frac{7\times 6\times 5}{3\times 2} = 35$

The number of ways of selecting 1 girl = ${}^5C_1 = 5$

$\therefore$ The required committee can be formed in 35 × 5 i.e 175 ways.

(b) The committee can be formed with

(i) four girls (ii) three girls and one boy (iii) two girls and two boys (iv) one girl and three boys.

$\therefore$ The committee formed in (i) ${}^5C_4$ ways, (ii) ${}^5C_3 \times {}^7C_1$ ways (ii) ${}^5C_2 \times {}^7C_2$ ways (iv) ${}^5C_1 \times {}^7C_3$ ways.

Hence the required number = $5 + \left(\frac{5!}{3!\times 2!}\times 7\right) + \left[\left(\frac{5!}{3!\times 2!}\right)\times\left(\frac{7!}{2!5!}\right)\right] + \left[5\times\frac{7!}{4!3!}\right]$

$= 5 + \frac{5\times 4}{2}\times 7 + \frac{5\times 4}{2}\times\frac{7\times 6}{2} + \frac{5\times 7\times 6\times 5}{3\times 2}$ = 5 + 70 + 210 + 175 = 460 ways.

14. Ans:

5 girls can sit around a table in 4! ways. As each person shall not have the same neighbors, the required arrangements = $\frac{1}{2}(4!) = \frac{24}{2}$ = 12 ways

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SET- 17
Permutation and Combination

1. There are two books each of three volumes and two books each of two volumes. In how many ways can these books be arranged in a shelf so that the volumes of the same book remain together?
2. In how many ways can 7 different – colour balls be arranged in a hollow glass cylinder so that red, blue and green balls are never together?
3. A company has 11 computer engineers and 7 mechanical engineers. In how many ways can they be seated in a row so that no two of the mechanical engineers may sit together?
4. A company has 11 computer engineers and 7 mechanical engineers. In how many ways can they be seated in a row so that all the mechanical engineers do not sit together?
5. In how many ways can 6 professors and 5 engineers be seated in a row so that they are positioned alternately?
6. A company has 11 computer engineers and 7 mechanical engineers. In how many ways can they be seated in a round table so that no two of the mechanical engineers may sit together?
7. A company has 11 computer engineers and 7 mechanical engineers. In how many ways can they be seated in a round table so that all the mechanical engineers do not sit together?
8. In how many ways 5 professors and 5 engineers can be seated in a round table so that they are positioned alternately?
9. Find the number of six-digit telephone numbers in a city if at least one of their digits is repeated and
   (i) Zero is allowed at the beginning of the telephone number, and
   (ii) Zero cannot initiate the number.
10. A tea party is arranged for 16 people along the two sides of a long table with 8 chairs on each side. 4 men wish to sit on one particular side and two on the other side. In how many ways can they be seated?

11. How many words can be formed with the letter of the word 'MATHEMATICS' if
   (i) there is no restriction,
   (ii) all the vowels are together,
   (iii) the vowels are together and the consonants are together,
   (iv) two M's are together,
   (v) these start with 'H',
   (vi) these begin with M,
   (vii) these begin with C and end with E,
   (viii)the vowels are never together,
   (ix) the relative order of vowels and consonants remains unchanged,
   (x) Vowels occupy even places.

12. From 7 engineers and 4 doctors a committee of 5 members is to be formed. In how many ways can this be done
   (i) to include exactly one doctor,
   (ii) to include at least one doctor?

13. There are 7 men and 5 women who are eligible for being included in a committee of 4 men and 3 women. In how many ways can the committee be formed if
   (i) one women W_1 refuses to be in the committee if another particular woman W_2 is included?
   (ii) If one Man M_1 refuses to serve if another woman W_2 is included?
14. There are 10 white, 8 red and 6 green balls on a billiard board. Find the number of ways in which one or more balls can be put in the pocket.

Career Avenues

15. There are ‘n’ different video films and ‘p’ cassettes of each film in a video library. Find the number of ways in which one or more than one video cassette can be selected.
16. In how many ways can 20 different computers be divided equally among 5 persons?
17. A person has 12 friends and wants to invite 8 of them to a birthday party. Find how many times 3 particular friends will always attend the party.

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Solutions

Permutation and Combinations (SET-2)

1. (i) Each volume of the same book are different.
(ii) Since volumes of the same book will remain together so, this will be treated as the same group.
Now, number of different books in group-I = $n_1=3$
Number of different books in group-II = $n_2=3$
Number of different books in group III = $n_3=2$
Number of different books in group IV = $n_4=2$
Hence, total Number of different groups = N = 4
Using the formula, required no. of ways = $N!\times n_1!\times n_2!\times n_3!\times n_4! = 4!\times 3!\times 3!\times 2!\times 2! = 3456.$

2. Here, total number of different balls, n = 7.
No. of balls to be never together = t = 3.
Never together = General permutation – always together
$= n! - (n-t+1)!\ t!$
$= 7! - (7-3+1)!\times 3! = 7! - (5!\times 3!)$
$= (7\times 6)\,5! - 3!\times 5! = (7\times 6-6)5!$
$= 36\times 5! = 36\times 120 = 4320$

3. Here no two of the mechanical engineers should sit together, so the group of mechanical engineers are under restriction. The no, of mechanical engineers (on restriction) = t = 7 and the number of computer engineers (no restriction) = n = 11
Required number of ways = $n! \times {}^{n+1}P_t = 11!\ {}^{11+1}p_7 = 11! \times \frac{12!}{(12-7)!} = 11! \times \frac{12!}{5!}$ ways

4. Here all the mechanical engineers do not sit together implies that 1 mechanical engineer or 2 or 3 or 4 or 5 or 6 mechanical engineers can sit together but all the seven mechanical engineers cannot sit together. So the group of mechanical engineer are on restriction
Let t = number of mechanical engineer = 7
Number of computer engineers (no restriction) = n = 11
Since $N \neq t$ and
All never together = Total permutation – all together
(1) Total number of permutations of (n + t = 18) objects = 18!
(2) Total number of permutations of all mechanical engineers together
Treat all mechanical engineers as one object so totally there are 11 + 1 = 12 objects
Number of ways of arranging 12 objects = 12!
Number of ways of arranging 7 mechanical engineers among themselves = 7!
Total number of permutations of all mechanical engineers together = 12! × 7!
∴ Number of permutations of all mechanical engineers not sitting together = 18! - (12!×7!)

5. Here
(i) both groups have persons who are all different
(ii) Number of professors, n = 6
(iii) Number of engineers, t = 5
(iv) $n \neq t$

6. professors can be seated in a row with in 6! ways.
There are 5 engineers seated in 5 spaces in between with 5! ways.
Required number of permutations = 6! × 5! = 86400

6. Here

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(i) No two of the mechanical engineers should sit together, so the group of mechanical engineers are under restriction. The number of mechanical engineers = t = 7
(ii) Number of computer engineers = n = 11
11 computer engineers can be arranged in a circle in (11 – 1)! = 10! ways
There are 11 spaces and 7 mechanical engineers can be seated in $^{11}p_7$ ways,

required number of permutations $= 10! \times \frac{11!}{4!}$

7. Never together = Total permutation- all together
Total permutation = Total number of ways of arranging (11 + 7 = 18) people in a circle = (18 – 1)! = 17! ways
In number of ways of all mechanical engineers together, consider all mechanical engineers as one block object. Therefore there are totally 12 objects (11 computer engineers + 1).
These can be arranged in a circle in (12 – 1)! = 11! ways.
The 7 mechanical engineers can be arranged among themselves in 7! ways.
So total number of ways of arranging all mechanical engineers together = 11! ×7!
Required permutation = (17!) – (11! × 7!)

8. 5 professors can be seated in a circle in (5 – 1)! = 4! ways.
There will then be 5 spaces into which 5 engineers can be seated in 5! ways.
Required No: of ways = 4! × 5! = 2880.

9. (i) Here all the 10 digits 0,12..9 have equal importance and each of the six digits can be filled in 10 ways.
Hence, no. of six digit telephone numbers = 10^6
No. of six digit phone numbers with no repetition of digits = $^{10}p_6$
Number of 6 digit phone numbers with at least one digit repeating
= Total no. of ways — No digits repeating = $10^6 - {}^{10}p_6 = 8,48,800$
(ii) Here, Total number of ways = 9×10^5 as first place can be filled in 9 ways only
Number of six digit phone numbers with no repetition = $9 \times {}^9p_5$
Required number of telephone numbers = $(9 \times 10^5) - (9 \times {}^9p_5) = 7,63,920$

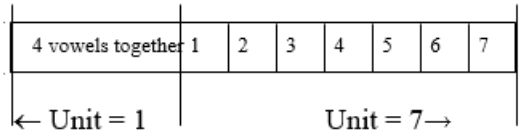
10. 4 Persons can be arranged in 8 chairs on one particular side in $^8p_4$ ways.
2 persons can be arranged in 8 chairs on the other side of the table in $^8p_2$ ways
Now there are 4 chairs remaining on one side. There are 10 people remaining. These can be seated in $^{10}p_4$ ways, and finally the remaining 6 people can be seated in the remaining 6 chairs on the other side in $^6p_6$ ways. $^{10}p_4 \times {}^{10}p_{10} = 10!$
This could also be viewed as follows: The rest 10 people can be arranged in 10 chairs in $^{10}p_{10} = (10!)$ ways
Total number of ways = $^8p_4 \times {}^8p_2 \times {}^{10}p_{10} = {}^8p_4 \times {}^8p_2 \times 10!$

11. In the word 'MATHEMATICS'
total number of letters = N= 11
number of consonants = 7
number of vowels = 4
Here all the letters are not different
'M' occurs twice ($S_1 = 2$)
'T' occurs twice ($S_2 = 2$)
'A' occurs twice ($S_3 = 2$)
(i) Using the formula

Required number = $\frac{N!}{S_1!S_2!S_3!} = \frac{11!}{2!2!2!}$ ways

Career Avenues

(ii)



'M' twice

'A' twice 'T' twice

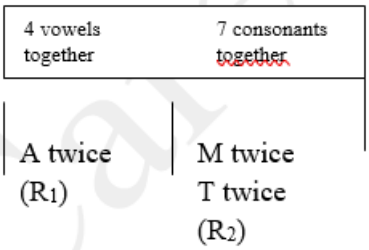
The restriction is all the 4 vowels are together (R_1)

4 vowels can be arranged in $\frac{4!}{2!}$ ways

Total units (1 + 7 = 8) can be arranged in $\frac{8!}{2!2!}$ ways

Required no. of words $\frac{4!}{2!} \times \frac{8!}{2!2!}$

(iii) 4 vowels are together (R_1)



4 vowels can be arranged in $R_1 = \frac{4!}{2!}$

7 consonants can be arranged in $R_2 = \frac{7!}{2!2!}$

The 2 units (vowels and consonants) can be arranged in 2! ways

Required no. of words = $\frac{4!}{2!} \times \frac{7!}{2!2!} \times 2! = \frac{4!\,7!}{2!\,2!}$

Career Avenues

(iv) 2 M's are together

2 M's can be arranged in $\frac{2!}{2!} = 1$ way

Total units (1+9 =10) can be arranged in $\frac{10!}{2!\ 2!}$ ways

Required no. of words = $1 \times \frac{10!}{2!\ 2!}$

(v) the starting letter is H

H	1	2	3	4	5	6	7	8	9	10

Fixed →

↔ ←10 units (M twice, A twice, T twice)

Location of H is fixed:

The balance 10 places can be arranged in $\frac{10!}{2!\ \ 2!\ 2!}$ ways

Required no. of words = $\frac{10!}{2!2!2!}$

(vi) Starting letter is M.
In the balance 10 units, A occurs twice and T twice.

The balance 10 places can be arranged in $\frac{10!}{2!\ 2!}$ ways.

Required no. of words = $\frac{10!}{2!2!}$

(vii) If the starting letter is C and ending letter is E,

C	1	2	3	4	5	6	7	8	9	E

→

'M' twice
'A' twice
'T' twice

Required number of ways = Number of ways of arranging the 9 letters = $\frac{9!}{2!\ \ 2!\ \ 2!}$

(viii) Here vowels are never together

Never together = Total number of ways — Always together = $\frac{11!}{2!2!2!} - \frac{8! \times 4!}{2! \times 2! \times 2!}$

(ix) The relative order of the vowels and consonants remains unchanged
This means, in the word "MATHEMATICS" in whichever positions vowels (A, A, E, I) come get interchanged only within those positions where vowels exist and consonants get interchanged only in those positions where consonants exist.

4 vowels can be arranged in their respective positions in $\frac{4!}{2!}$ ways

Career Avenues

7 consonants can be arranged in their respective positions in $\frac{7!}{2!2!}$

Required Total number of words = $\frac{4!}{2!} \times \frac{7!}{2! \times 2!} = \frac{4! \times 7!}{2! \times 2! \times 2!}$

(x) Vowels occupy even places.
In MATHEMATICS, there are 11 places.

Number of even places $= \frac{11-1}{2} = 5$.

4 Vowels can be arranged in these 5 places is $\frac{^5p_4}{2!}$ ways as A occurs twice.

Once the vowels are allocated, consonants are placed in balance places in $\frac{7!}{2!2!}$ ways

Required number of words $\frac{^5p_4}{2!} \times \frac{7!}{2!2!} = \frac{5! \times 7!}{2! \times 2! \times 2!}$

12. (i) Required number of ways to form a committee with 1 doctor = $^7C_4 \times {}^4C_1 = 35 \times 4 = 140$.
This problem involves selection of people (rather then arrangement and hence is a problem on combination).
(ii) To form a committee with at least one doctor =
Total number of ways – Number of ways of forming a committee with no doctors = $^{7+4}C_5 - {}^7C_5 = 462 - 21 = 441$

13. (i) Here W_1 is not a member if W_2 is member and vice-versa.
4 men can be chosen out of 7 men in $^7C_4$ ways. Of the 3 women, let number of ways in which W_1 is chosen and W_2 is not chosen be n_1.
Let number of ways in which W_2 is chosen and W_1 is not chosen be n_2
$n_1 + n_2$ = Total number of ways in which W_1 and W_2 are never together
= Total number of ways of selecting 3 women – Number of ways in which W_1 and W_2 are always together
$= {}^5C_3 - {}^3C_1 = 10 - 3 = 7$
Since the number of ways of selecting men and women are independent, number of ways of selecting the committee = $^7C_4 \times 7 = 245$.
(ii) Number of ways in which M_1 and W_2 are never together
= Total number of ways of forming a committee – Number of ways in which M_1 and W_2 are always together
$= (^7C_4 \times {}^5C_3) - (^6C_3 \times {}^4C_2) = (35 \times 10) - (20 \times 6) = 350 - 120 = 230$

14. Here the number of balls to be put is not mentioned, It could be a minimum of one ball or a maximum of 24 balls.
The pocketing of a white ball, red ball or green ball is independent of the colour of ball.
$\therefore$ Total number of ways of pocketing the balls = Number of ways of pocketing white balls × No. of ways of pocketing red balls × Number of ways of pocketing green balls
White balls could be pocketed in 0 or 1 or ...10 balls at a time in 10 + 1 = 11 ways
Red balls could be pocketed in 0 or 1 or 2 or …8 balls at a time in 8+1 = 9 ways
Green balls could be pocketed in 0 or 1 or 2 or ….6 balls at a time in 6+1= 7 ways.
Total number of ways of pocketing the balls = $11 \times 9 \times 7 = 693$
This includes one way in which no ball is pocketed (neither green nor white nor red).

Career Avenues

Hence required number of ways = 693 – 1 = 692.

15. Zero or more video cassettes of one film can be selected out of identical 'p' cassettes in (p + 1) ways
For n films, the number of ways possible for selecting a cassette is = (p + 1) (p + 1)……n times = $(p + 1)^n$
This includes one combination where no cassette of any film is chosen.
The required number of ways = $(p + 1)^n - 1$

16. Here each person gets 4 computers
Number of ways of choosing 4 computers out of 20 = $^{20}C_4$
Number of ways of choosing 4 computers out of balance = $^{16}C_4$
Number of ways of choosing 4 computers out of balance = $^{12}C_4$
Number of ways of choosing 4 computers out of balance = $^{8}C_4$
Number of ways of choosing the balance for computers = $^{4}C_4$
Total number of a ways of dividing the computers = $^{20}C_4 \times {}^{16}C_4 \times {}^{12}C_4 \times {}^{8}C_4 \times {}^{4}C_4$

$$= \frac{20!}{4!16!} \times \frac{16!}{4!12!} \times \frac{12!}{4!8!} \times \frac{8!}{4!4!} \times \frac{4!}{4!0!} = \frac{20!}{(4!)^5}$$

Alternate method,

No. of ways in which 20 computers can be divided into 5 distinct groups = $\frac{20!}{(4!)^5}$ ways.

17. If 3 particular friends always attend the party, balance 5 people will be chosen out of balance 9 people in
$^{9}C_5$ [= $^{12-3}C_{8-3}$] = 126 ways.

Career Avenues

SET- 18
Probability

1. A fair coin is tossed three times in succession. If the first toss produces a head then the probability of getting exactly two heads in three tosses is
   a) $1/8$
   b) $1/2$
   c) $3/8$
   d) $3/4$
2. Supose there are two coins. The first coin gives head with probability 5/8 when tossed, while the second coin gives heads with probabilty ¼. One of the two coins is picked up at random with equal probability and tossed. What is the probabilty of obtaing heads.
   a) 7/8
   b) 1/2
   c) 7/16
   d) 5/325
3. A single dice is thrown twice. What is the probability that the sum is neither 8 nor 9?
   a) 1/9
   b) 5/36
   c) 1/4
   d) 3/4
4. A fair dice is rolled twice. The probability that an odd number will follows an even number is
   a) 1/2
   b) 1/6
   c) 1/3
   d) 1/4
5. Two dice are thrown. What is the probability that the sum of the numbers on the two dice is eight?
   a) 5/36
   b) 5/18
   c) 1/4
   d) 1/2
6. Two dices are rolled simultaneously. The probability that the sum of digits on the top surface of two dice is even is,
   a) 0.5
   b) 0.25
   c) 0.167
   d) 0.125
7. Ram have two bags, one bag contains 5 white and 4 red balls and another bag contains 4 white and 2 red ball If one ball is drawn from each bag, the probability that one is white and one is red, is
   a) 14/27
   b) 13/27
   c) 5/27
   d) 8/27

Career Avenues

8. Shyam has two bag, One bag contains 4 white and 2 black balls and another bag contains 3 white and 5 black balls. If one ball is drawn from each bag, the probability that both are white is
   a) 1/4
   b) 3/4
   c) 5/24
   d) 19/24
9. A box contains 5 black and 5 red balls. The balls are randomly picked one after another from the box, without replacement, The probability for both balls being red is
   a) 1/90
   b) 1/5
   c) 19.90
   d) 2/9
10. Ram speaks truth in 75% and Shyam in 80% of the cases. In what percentage of cases are they likely to contradict each other narrating the same incident?
   a) 10%
   b) 25%
   c) 35%
   d) 45%
11. The odds against Ram who is 45 years old, living till he is 70 are 7:5 and the odds against his wife Rani who is 43, living till she is 61 are 5.36. The probability that at least one of them will be alive 25years hence, is
   a) 35/96
   b) 61/96
   c) 13/96
   d) 53/36
12. If the probabilities that Ram and Shyam will die within a year are p and q respectively, then the probability that only one of them will be alive at the end of the year is
   a) p + q-2pq
   b) p (1-q)
   c) q(1-p)
   d) pq
13. The probability that an event A occurs in one trails of an experiment is 0.4 . Three independent trails of experiment are performend. The probability that A occurs at least once is
   a) 0 0.432
   b) 0.784
   c) 0.859
   d) 0.964
14. An anti-aircraft gun of Indian army can take a maximum of 4 shots at an enemy plane moving away from it. The probabilities of hitting the plane at the first, second, third, and fourth shot are 0.4, 0.3, 0.2 and 0.1 respectively. The probabilty that the gun hits the plane is
   a) 0.3021
   b) 0.4096
   c) 0.5904
   d) 0.6976

15. A lot has 10% defective items. Ten items are chosen randomly from this lot. The probability that exactly 2 of the chosen items are defective is
   a) 0.0036
   b) 0.1937
   c) 0.2234
   d) 0.3874

Career Avenues

16. A box contains 20 defective items and 80 non-defective items. If two items are selected at random without replacement, what will be the probability that both items are defective?
    a) 1/5
    b) 1/25
    c) 20/99
    d) 19/495

17. An examination consists of two papers,paper 1 and paper 2. The probability of failing in Paper 1 is 0.3 and that in Paper2 is 0.2. Given that a student has a failed in Paper 2, the peobability of failing in Paper 1 is 0.6 The probibilty of a student failing in both the papers is
    a) 0.5
    b) 0.05
    c) 0.12
    d) 0.06

18. From a pack of regular playing catrd, two cards are drawn at random. What is the probability that both cards will be kings, if the first card is NOT replaced?
    a) 1/26
    b) 1/52
    c) 1/169
    d) 1/221

19. There are 25 calculators in box, two of them are defective. Suppose 5 calculators are randomly picked for inspection (i.e each has the same chance of being selected), what is the probability that only one of the defective calculators will be included in the selection?
    a) 1/2
    b) 1/3
    c) 1/4
    d) 1/5

20. The probability of a defective piece being produced in a manufacturing process is 0.01.
    The probability that out of 5 successive pieces, only one is defective is
    a) $(0.99)^4$ (0.01)
    b) (0.99) (0.01)
    c) 5 x $(0.01)^4$ (0.99)
    d) 5 x $(0.99)^4$ (0.01)

Career Avenues

Solutions:

Probability

1. Ans: b)

If first toss produces head,then for exactly two head in three tosses there one head in next two tosses. The probability of one head in two tosses. Hence B is correct option.

2. Ans: c)

Probability of selecting of one of two coins = $\frac{1}{2}$

Probability of being head of the first coin = $\frac{5}{8}$

Probability of being head of the second coin = $\frac{1}{4}$

So the probability of obtaining heads =(5/8 x 1/2) + (1/4 x 1/2) = 7/16

3. Solu :- d)

By throwing dice twice 6x6 =36 possibilities will occur

We have to take sum of outcome 8 or 9 as (2, 6) (3,5) (3,6) (4,4) (4,5) (5,3) (5,4) (6,2) (6,3) = 9 then n (E) = 9 and n(S) =36

then P (8 or 9) = $\frac{n(E)}{n(S)} = \frac{9}{36} = \frac{1}{4} -$

then P (not getting 8 or 9) = 1- $1/4 = \frac{3}{4}$

4. Ans: d)

P(Odd) = 3/6 = ½

P(Even) = 3/6 = ½

P (odd follows even) = ½ x ½ = ¼

5. Ans: a)

n(S) =6x6=36

n(E) = Sum of outcome is 8 = (2,6) (3,5) (4,4) (5,3) (6,2) =5

then P(Sum of 8) = $\frac{n(E)}{n(S)} = \frac{5}{36}$

6. Ans: a)

n(S) = 6x6=36

n(E) = Sum of outcome is even = (1,1) (1,3) (1,5) =18

(2,2) (2,4) (2,6)

(3,1) (3,3) (3,5)

(4,2) (4,4) (4,6)

(5,1) (5,3) (5,5)

(6,2) (6,4) (6,6)

Then required probability P(Sum is even) = n(E) / n(S) = 18/36 = ½ = 0.5

Career Avenues

7. Ans: b)

P(WR) = P (WR) + P (RW) = (5/2 x 2/6) + (5/27 x 8/27) = 13/27

8. Ans: a)

P (both balls are white) = P (1st ball white and 2nd ball white)

= P (1st ball white) x P (2nd ball white) = $\left(\frac{4}{6}\times\frac{3}{8}\right)=1/4$

9. Ans: d)

The probability of drawing a red ball = 5/10
If the ball is not replaced, the box will have a 9 ball,
Then probability of drawing the red ball in next chance = 4/9
Hence Probability of drawing 2 balls = 5/10 x 4/9 = 2/9

10. Ans: c)

Let A be the event that Ram speaks the truth and B be the event Shyam speaks truth
Then, P (A) = 75/100=3/4 and P (B) = 80/100=4/5

Then $P(\overline{A})=1-\frac{3}{4}=\frac{1}{4}$ *and* $P\overline{B}=1-\frac{4}{5}=1/5$

P (Ram and Shyam contradict) = P (Ram speaks truth and Shyam tells a lie) + P (Ram speaks lie and Shyam speaks truth) = P(A) $P(\overline{B})+P(\overline{A})P(B)=\frac{3}{4}\times\frac{1}{5}+\frac{1}{4}\times\frac{4}{5}=\frac{7}{20}=35\%$

11. Ans: b)

Let A = Event that Ram will be alive 25 years
B= Event that the his wife will alive 25 year hence
P(A) = 5/12 and P(B) = 3/8

$P\overline{A}=1-\frac{5}{12}=\frac{7}{12}$ *and* $P\overline{B}=1-\frac{3}{8}=5/8$

P (atleast one of them alive 25 years hence) = 1- P(none will alive 25 years hence)

= $1-P(\overline{A})P(\overline{B})=1-\frac{7}{12}\times\frac{5}{8}=61/96$

12. Ans: a)

Probability = P(Ram dies and Shayam Is alive) + P(Ram is alive and Shyam dies)

$=p(1-q)+q(1-p)\,p+q-2pq$

13. Ans: b)

Here p=0.4 and q= 0.6 and n =3
Required probabilty = P(A occuring at least once)

$={}^3C_1(0.4)0.6^2+{}^3C_2(0.6)0.4^2+{}^3C_3\,0.4^2$

$=784/1000=0.784$

14. Ans: d)

$p_1=0.4\quad p_2=0.3\quad p_3=0.2$ *and* $p_4=0.1$

P(The gun hits the palne) = P(the plane is hit atleast once)

= 1- P(the plane is hit in none of shots)

$=1-(1-p_1)(1-p_2)(1-p_3)(1-p_4)$

$=1-0.6\times0.7\times0.8\times0.9=0.6976$

Career Avenues

15. Ans: b)
Percentage of defective items = 10% or p = 0.1 then q = 1 – 0.1 = 0.9
Probability of exactly 2 item are defective = $^{10}C_2\ p^2\ q^8 = {}^{10}C_2 - 0.1^2\ 0.9^8 = 0.1937$

16. Ans: d)
No. of defective items = 20
No. of Non- defective items = 80, total items = 80-20=100
Then probability of both 2 are defective = 20/100 x 19 / 99 = 19/495

17. Ans: c)
Probaility of failing in Paper 1 = P(X) =0.3
Probaility of failing in Paper 2 =P(Y) = 0.2
Probability of failing in paper 1, when already failed in paper 2 is P(X/Y) = 0.6
Then by bayes rule
$P(X \cap Y) = P(Y)P(X/Y) = 0.6 \times 0.2 = 0.12$

18. Ans: d)
Probability that both cards are Kings = 4/52 x 2/51 = 1/221

19. Ans: b)
Ways of choosing 5 calculators = $^{25}C_5$
Ways of choosing 1 defective calculators = $^{2}C_1$
Ways of choosing 4 good calculators = $^{23}C_4$
Then Probability = $\frac{^{2}C_1 \times {}^{23}C_4}{^{25}C_5} = \frac{1}{3}$

20. Ans: d)
P=0.01 , q = 0.99 and n= 5
probability that only one is defective = $^{25}C_5\ p^1\ q^4 = {}^{25}C_5 = 0.01^1 0.99^4$

Career Avenues

SET- 19

Bayes' Theorem, Probability Density Function and Statistics

1. A dice is rolled. If the outcome is an odd number, what is the probability that it is a prime?

   a) 2/3
   b) 3/2
   c) 4/3
   d) 3/4

2. When a dice is rolled once, find the probability of getting a 4 given that an even number occurred in an earlier throw.
   a) 1/3
   b) 2
   c) 2/3
   d) 3/2

3. Given: G = Having good seats, B = Having bad seats, A_1 = Seeing *Comedy Movie*, A_2 = Seeing *Scary Movie!*, the probability of having good seats given that you are seeing *Comedy Movie* $[P(G/A_1)] = 0.98$, the probability of having good seats given that you are seeing *Scary Movie!* $[P(G/A_2)] = 0.95$, $P(A_1) = 0.65$, and $P(A_2) = 0.35$…
   Please find the probability of i) Seeing *Comedy Movie* given you have bad seats, and
   ii) Seeing *Scary Movie!* given you have bad seats.
   a) i) 0.42 ii) 0.57
   b) i) 0.82 ii) 0.28
   c) i)1.2 ii) 2.1
   d) i)0.72 ii) 0.27

4. It has been found that 10% of certain relays have bent covers and will not work. If 40% Have bent covers, what is the probability that a relay with a bent cover will not work?
   a) 0.25
   b) 0.5
   c) 0.75
   d) 0.08

5. The continuous random variable X is distributed with probability density function f defined by $f(x) = kx(16 - x^2)$, for $0 < x < 4$.
   Evaluate:
   i) The value of constant k
   ii) The probability of range space $P(1 < X < 2)$
   iii) The probability $P(x \geq 3)$

6. X is the continuous random variable 'the mass of a substance, in kg, per minute in an industrial production process', where

$$f(x) = \begin{cases} \frac{1}{12}x(6-x) & (0 \leq x \leq 3) \\ 0 & \text{otherwise} \end{cases}$$

   Find the probability that the mass is more than 2 kg.

Career Avenues

7. The continuous random variable (r.v) has a probability density function(p.d.f) where

$$f(x)=\begin{cases} k & 0\le x<2 \\ k(2x-3) & 2\le x\le 5 \\ 0 & otherwise \end{cases}$$

i) Find the value of the constant k
ii) Sketch y=f(x)
iii) Find P($X\le 1$)
iv) Find P(X>2.5)

8. If X is a continuous variable (r.v) with a p.d.f $f(x)=\frac{1}{16}x^2,\quad 0\le x\le 3$, find E(X).

9. If the continuous random variable X has p.d.f $f(x)=\frac{2}{5}(3+x)(x-1), 1\le x\le 3$, find E(X).

10. The continuous random variable X has p.d.f f(x) where f(x)= $\frac{1}{2}x,\quad 0\le x\le 3.$

a) Find
b) E(X)
c) $E(X^2)$
d) E(2X +3)

11. Five persons A, B, C, D and E occupy seats in a row at random. What is the probability that A and B will sit next to each other**.**

12. A box contains 4 green, 7 white and 5 red balls. If four balls are drawn one by one with replacement, what is the probability that none is red?

13. There are 10% defective items in a large bulk of items. Find the probability that a sample of 10 times will not include more than one defective item.

14. The mean number of success of a binomial distribution is 240 where p is the probability of success. The standard deviation is 12. Calculate the value of n, p and q.

15. A and B are two events and P(A) = P_1, P(B) = P_2 and P(A∩B) = P_3. Find P($\overline{A}\,|\,\overline{B}$)

16. The probability is 0.02 that an item produced by a factory is defective. A shipment of 10000 items is sent to its ware house. Calculate the expected number (E) of the defected items and standard deviation.
a) 200, 14 respectively
b) 100,7 respectively
c)14, 200 respectively
d) 200, 7 respectively

17. Assume that 45 % of the population favour a certain candidate in an election. If a random sample of size 200 is chosen, then the standard deviation of the number of members of sample that favour the candidate is
a) 6.12 b) 5.26 c) 8.18 d) 7.04

18. The probability is 0.02 that an item produce by a factory is defective. A shipment of 10000 items is sent to its ware house. Find the expected number E of defective items and the standard deviation.
a) 200.14
b) 100, 7
c) 14, 200
d) 200, 7

Career Avenues

19. In a manufacturing plant, the probability of making a defective bolt is 0.1. The mean and standard deviation of defective bolts in a total of 900 bolts are respectively.
    a) 90 and 9
    b) 9 and 90
    c) 81 and 9
    d) 9 and 819

20. If 3 is the mean and (3/2) is the standard deviation of a binomial distribution, then the distribution is
    a) $\left(\frac{1}{2}+\frac{3}{2}\right)^{12}$ b) $\left(\frac{3}{4}+\frac{1}{4}\right)^{12}$ c) $\left(\frac{1}{5}+\frac{4}{5}\right)^{12}$ d) $\left(\frac{4}{5}+\frac{1}{5}\right)^{12}$

Career Avenues

Solutions:

Bayes' Theorem, Probability Density Function and Statistics

1. S = {1, 2, 3, 4, 5, 6}

Let A = The event of getting an odd number

B = The event of getting a prime number

Then A = {1, 3, 5} and B = {2, 3, 5}

$A \cap B = \{3, 5\}$

we have to find the conditional probability of B when A has occurred

i.e. $P(B/A) = \frac{n(A \bigcap B)}{n(A)} = \frac{2}{3}$

2. (4 and an even number) = 1/6 i.e. P(A and B) =1/6. P(even number) =3/6 =1/2.

$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{1/6}{1/2} = \frac{1}{3}$

$P(B/A_1) = 1 - P(G/A_1) = 1 - 0.98 = 0.02$; $P(B/A_2) = 1 - P(G/A_2) = 1 - 0.95 = 0.05$

3. a)$P(A_1/B)$ = [(0.65)(0.02)]/[(0.65)(0.02) + (0.35)(0.05)] = 0.0130/(0.0130 + 0.0175) = 0.0130/0.0305 = 0.4262

b) $P(A_2/B)$ = [(0.35)(0.05)]/[(0.65)(0.02) + (0.35)(0.05)] = 0.0175/(0.0130 + 0.0175) = 0.0175/0.0305 = 0.5738

4.

A and B – bent AND will not work – 0.1

A = 0.4

Let A = event that relays have bent covers

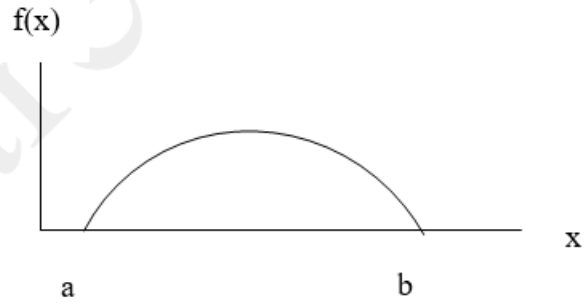
Let B = event that relays will not work

Given: P(A & B) = .10, P(A) = .40

The first formula of the conditional probability formulas, Bayes formulas, gives the following solution:

P(B/A) = P(A and B) / P(A) = 10/40 = 0.25

5.



For any function f(x) such that $f(x) \geq 0$, for $a \leq x \leq b$, and $\int_a^b f(x)dx = 1$ may be taken as the probability density function (p.d.f) of a continuous random variable in the range space $a \leq x \leq b$.

Procedure:

Step 1: In general, if X is a continuous random variable (r.v) with p.d.f. f(x) valid over the range $a \leq x \leq b$, then $\int_{\text{all } x} f(x)dx = 1$ i.e. $\int_a^b f(x)dx = 1$

Step 2: a). To determine k, we use the fact that in $f(x) = kx(16 - x^2)$, for $0 < x < 4$, then

$\int_0^4 kx(16 - x^2)dx = 1$

$\Rightarrow \quad k\int_0^4 16x - x^3)dx = 1$

$\Rightarrow \quad k = \frac{1}{64}$

Step 3: b) To Find P(1 < X < 2)

P(1<X<2)= $\int_1^2 f(x)dx$

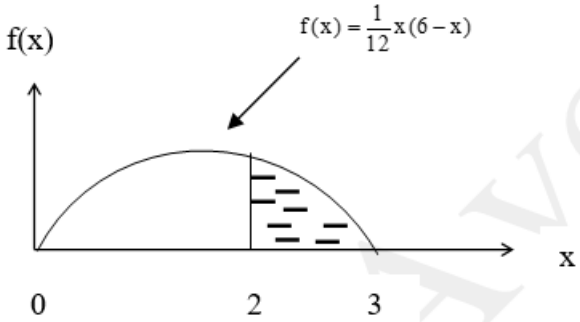
$= \quad \frac{1}{64}\int_1^2 (16x - x^3)dx = \frac{81}{256}$

Step 4: c). To find P($x \geq 3$)

$P(x \geq 3) = \frac{1}{64}\int_3^4 (16x - x^3)dx = \frac{49}{256}$

6.

X can take values from 0 to 3 only. We sketch f(x), and shade the area required.



$P(x > 2) = \int_2^3 \frac{1}{12}x(6 - x)dx$

$= \frac{1}{12}\int_2^3 (6x - x^2)dx$

$= \frac{1}{12}\left[3x^2 - \frac{x^3}{3}\right]_2^3$

= 0.722 (3 d.p)

The probability that the mass is more than 2 kg is 0.722.

7. i) Since X is a r.v, then $\int_{\text{all x}} f(x)dx = 1$

Therefore $\int_0^2 kdx + \int_2^5 k(2x-3)dx = 1$

$kx \;|_0^2 + k\left[x^2 - 3x\right]_2^5$

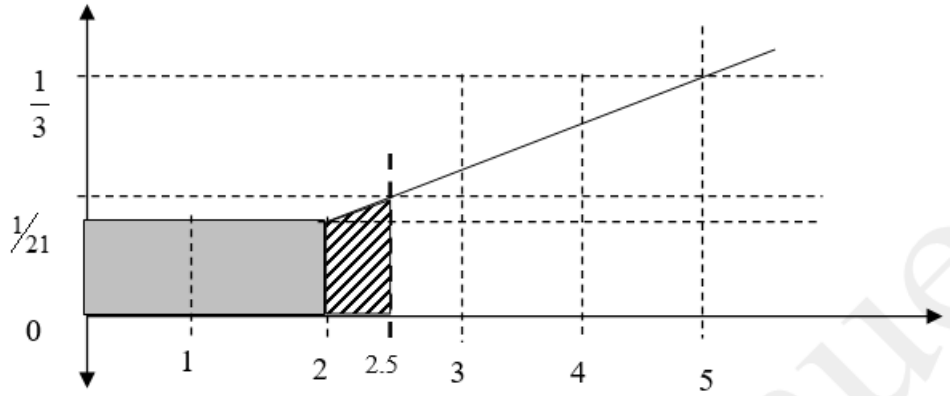
$2k + 19k = 1$

$\Rightarrow k = \frac{1}{21}$

ii) So the p.d.f of X is

$$f(x) = \begin{cases} \frac{1}{21} & 0 \leq x < 2 \\ \frac{1}{21}(2x-3) & 2 \leq x \leq 5 \\ 0 & otherwise \end{cases}$$

Career Avenues



iii) P($x \leq 1$) = area between zero and 1 = L × W= $1 \times \frac{1}{21} = \frac{1}{21} = 0.048$

iv) Find P(X>2.5) $= \int_{2.5}^{5} \frac{1}{21}(2x-3)dx$

$= \frac{1}{21}\left(x^2 - 3x\right)_{2.5}^{5}$

$= \frac{1}{21}\left[(25-15)-(6.26-7.5)\right]$

= 11.25/21 = 0.535

8.

$$E(X) = \int_{all\ x} x \quad f(x)dx$$

$$\Rightarrow \int_0^3 \frac{1}{16}\{x\}\, x^2 dx$$

$$= \frac{1}{16}\left[\frac{x^4}{4}\right]_0^3 = \frac{81}{64} = 1.265$$

9.

$$E(X) = \int_{all\ x} x \quad f(x)dx$$

$$E(x) = \int_1^3 \frac{2}{5}\{x\}(3+x)(x-1)dx$$

$$= \frac{2}{5}\left[\frac{x^4}{4} + \frac{2x^3}{3} - \frac{3x^2}{2}\right]_1^3$$

$$= \frac{608}{60}$$

$$= 10.13$$

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10.

a) $E(X) = \int_{all\ x} x \quad f(x)dx$

$= \int_0^3 \frac{1}{2}x^2 dx$

$= \frac{1}{2}\left[\frac{x^3}{3}\right]_0^3$

$= 4.5$

b) $E(X^2) = \int_{all\ x} x^2 f(x)dx$

$= \frac{1}{2}\int_0^3 x^3 dx$

$= \frac{1}{2}\left[\frac{x^4}{4}\right]_0^3$

$= \frac{81}{8} = 10.125$

c) $E(2X+3) = E(2X) + 3$

$= 2E(X) + 3$

$= 2(10.125)+5$

$= 25.25$ (from (b) above)

11. Without restriction, no. of arrangements = 5! = No. of possible outcomes = 120.
With restriction, no. of arrangements = $4! \times 2$ (i.e. considering A and B together) = 48.

Hence, P(A and B sit to together)= $\frac{48}{120} = \frac{2}{5}$.

12. Total number of balls = 4 + 7 + 5 = 16

Probability of getting no red ball in a single draw $= \frac{11}{16}$

Since all four draws are independent,

The required probability $= \frac{11}{16} \times \frac{11}{16} \times \frac{11}{16} \times \frac{11}{16} = \left(\frac{11}{16}\right)^4$

13. $P = 10\% = \frac{1}{10}$, $q = 1 - \frac{1}{10} = \frac{9}{10}$, $n = 10$

$P(0 \le x \le 1) = P(x = 0) + P(x = 1)$

$= {}^{10}C_0\ P^0\ q^{10} + {}^{10}C_1\ P^1\ q^{10}$

$= \left(\frac{1}{10}\right)^0 \left(\frac{9}{10}\right)^{10} + 10\left(\frac{1}{10}\right)^1 \left(\frac{9}{10}\right)^9 = \left(\frac{9}{10}\right)^{10} + 10\frac{(9)^9}{(10)^{10}}$

$= \frac{9^9}{10^{10}}(9+10) = 19\left(\frac{9^9}{10^{10}}\right) = \frac{19}{10}\left(\frac{9}{10}\right)^9$

Career Avenues

14. Sol :-Mean = n p = 240 (1)

standard deviation = $\sqrt{npq}$ = 12

$\therefore$ npq = 144

$\therefore$ 240 q = 144 from (1)

$\therefore$ q = $\frac{144}{240} = \frac{12}{20} = \frac{3}{5}$

$\therefore$ P = 1 - $\frac{3}{5} = \frac{2}{5}$ (since p + q = 1)

$\therefore$ n = $\frac{144}{\frac{2}{5}\times\frac{3}{5}} = \frac{144\times 25}{6} = 600$

Hence n = 600 , p = $\frac{2}{5}$, q = $\frac{3}{5}$.

15. A and B are two events and P(A) = P_1, P(B) = P_2 and P(A$\cap$B) = P_3. Find P($\overline{A}|\overline{B}$).

P(A) = P_1 , P(B) = P_2 , P(A$\cap$B) = P_3

$\therefore P(\overline{A}) = 1 - P_1$, $P(\overline{B}) = 1 - P_2$

$\therefore P(\overline{A}\cap\overline{B}) = P(\overline{A})\ P(\overline{B})$

$= (1 - P_1)(1 - P_2)$

$\therefore P(\overline{A} \mid \overline{B}) = \frac{P(\overline{A}\cap\overline{B})}{P(\overline{B})}$

$= \frac{(1 - P_1)(1 - P_2)}{1 - P_2}$

$= \frac{1 - P_1 - P_2 + P_1 P_2}{1 - P_2}$

$= \frac{1 - P_1 - P_2 + P_3}{1 - P_2}$ $(\text{since } P_1 P_2 = P_3)$

16. E=np=10000 x 0.02=200 Variance = npq =10000x0.02x(1-0.02) =196 S.D=14

17. Mean =np=200x0.45=90 Varianc =npq= 200x0.45x(1-0.45) =49.5 S.D=7.04

18. E=np = 100000 x 0.02 = 200

$r = \sqrt{npq} = \sqrt{10000\times 0.02\times(1-0.02)} = \sqrt{196} = 14$

19. No. of bolts = n = 900, p= 0.1,q=1-0.1=0.9, mean=np=900 x0.1=90

Variance= npq

S.D = $\sqrt{npq} = \sqrt{900\times 0.1\times 0.9} = 9$

20. np=3 and npq= 9/4 then q= ¾ , p=1-q = ¼ then n= 12

and binomial distibution = $(p+q)^n = \left(\frac{3}{4}+\frac{1}{4}\right)^{12}$

Career Avenues

Past GATE Questions

1. A fair dice is rolled twice. The probability that an odd number will follow an even number is **(G-2005, 1-Mark)**

   a) $\frac{1}{2}$

   b) $\frac{1}{6}$

   c) $\frac{1}{3}$

   d) $\frac{1}{4}$

2. A probability density function is of the form $p(x)=Ke^{-\alpha|x|}, x\in(-\infty,\infty)$ The value of K is **(G-2006, 1-Mark)**

   a) 0.5
   b) 1
   c) $0.5\,\alpha$
   d) α

3. Three companies X, Y and Z supply computers to a university. The percentage of computers supplied by them and the probability of those being defective are tabulated below

Company	% Computers Supplied	Probability of being defective
X	60%	0.01
Y	30%	0.02
Z	10%	0.03

   Given that a computer is defective, the probability that it was supplied by Y is **(G-2006, 2-Mark)**

   a) 0.1
   b) 0.2
   c) 0.3
   d) 0.4

4. An examination consists of two papers, paper 1and paper 2. The probability of failing in paper 1 is 0.3 and that in paper 2 is 0.2. Given that a student has failed in paper 2, the probability of failing in paper 1 is 0.6. The probability of failing in both the papers is **(G-2006, 2-Mark)**

   a) 0.5
   b) 0.18
   c) 0.12
   d) 0.06

5. A fair coin is tossed 10 times. What is the probability that on ONLY the first two tosses will yield heads? **(G-2009, 1-Mark)**

   a) $\left(\frac{1}{2}\right)^2$

   b) ${}^{10}C_2\left(\frac{1}{2}\right)^2$

Career Avenues

c) $\left(\frac{1}{2}\right)^{10}$

d) ${}^{10}C_2\left(\frac{1}{2}\right)^{10}$

6. A fair coin is tosse independently four times. The probability of the event the number of times heads show up is more than the number of times tails show up is **(G-2010, 2-Mark)**

a) $\frac{1}{16}$

b) $\frac{1}{8}$

c) $\frac{1}{4}$

d) $\frac{5}{16}$

7. A fair coin is tossed two times. The probability that the second toss results in a value that is higher than the first toss is **(G-2011, 2-Mark)**

a) 2/36
b) 2/6
c) 5/12
d) 1/2

8. A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is **(G-2012, 2-Mark)**

a) 1/3
b) 1/2
c) 2/3
d) 3/4

9. In a housing society, half of the families have a single child per family, while the remaining half have two children per family. The probability that a child picked at random, has a sibling is……. **(G-2014, 1-Mark)**

10. An Unbiased coin is tossed an infinite number of times. The probability that the fourth head appears at the tenth toss is …….. **(G-2014, 1-Mark)**

a) 0.067
b) 0.073
c) 0.082
d) 0.091

11. A fair coin is tossed repeatedly till both head and tail appear at least once. The average number of tosses required is…….. **(G-2014, 2-Mark)**

12. Parcels from sender S to receiver pass sequentially through two post-offices. Each post-office has a probability 1/5 of losing an incoming parcel, independently of all other parcels. Given that a parcel is lost, the probability that it was lost by second post-office is …… **(G-2014, 2-Mark)**

Career Avenues

13. Suppose A and B are two independent events with probabilities $P(A) \neq 0$ *and* $P(B) \neq 0$ Let $\overline{A}$ and $\overline{B}$ be their complements. Which one of the following statements is FALSE? **(G-2015, 1-Mark)**

    a) $P(A \cap B) = P(A)P(B)$

    b) $P(A/B) = P(A)$

    c) $P(A \cap B) = P(A) + P(B)$

    d) $P(\overline{A} \cap \overline{B}) = P(\overline{A})P(\overline{B})$

14. Two Random variables X *and* Y are distributed according to

$$F_{x,y}(x,y) = \begin{cases} (x+y)\, 0 \le x \le 1, 0 \le y \le 1 \\ 0 \; otherwise \end{cases}$$

    The probability P (X + Y ≤ 1)is …….. **(G-2016, 1-Mark)**

15. The second moment of a Poisson-distributed random variable is 2. The mean of the random variable is ……. **(G-2016, 1-Mark)**

16. The probability of getting a head in a single toss of a biased coin is 0.3. The coin is tossed repeatedly till a head is obtained. If the tosses are independent, then the probability of getting head for the first time in the fifth toss is……. **(G-2016, 1-Mark)**

17. Three fair cubical dice are thrown simultaneously. The probability that all three dice have the same number of dots on the faces showing up is (up to third decimal place)…… **(G-2017, 1-Mark)**

Career Avenues

Solution:-

Past GATE Questions

1. d)

P(Odd) = 3/6 = 1/2

P (Even) = 2/6 = 1/2

Both are independent to each other then,

P (Odd followed even) = P (Odd) x P (Even) = $\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}$

2. c)

Since Integration of pdf function from - $-\infty to+\infty is always eqauls to 1$

$\int_{-\infty}^{\infty} f(x)dx=1 here\ f(x)=\rho(x)=ke^{-\alpha|x|}, x\in(-\infty,\infty)$

Then $\int_{-\infty}^{\infty}\rho(x)dx=1$

$\int_{-\infty}^{\infty} Ke^{-\alpha|x|}dx=1$ by solving the integral we can get K =0.5 α

3. d)

S = Supply by y

D= defective

Probability that the computer was supplied by y

If the product is defective, then

$P\left(\frac{S}{d}\right)=\frac{P(S\cap d)}{P(d)}=\frac{0.006}{0.015}=0.4$ Since P(S $\cap$ d) = 0.3 x .0.02= 0.006

and p (d) = 0.6 x 0.01 + 0.3 x 0.02 + 0.1 x 0.03 = 0.015

4. c)

It s given that

Probability of failing in paper 1 = P (A) = 0.2

Probability of failing in paper 2 = P (B) = 0.3

$P\left(A/B\right)=0.6$

Probability of failing in both $P(A\cap B)=P(A/B)\times P(B)=0.6\times 0.2=0.12$

5. c)

6. d)

The only 5 events when the number of times of heads show up is more than the number of times tails show up are given below:

HHHH

HHHT

HHTH

HTHH

THHH

Since coin is tossed 4 times, the probability of each events is $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)=\frac{1}{16}$

All the above events are mutually exclusive, so total probability =

$\frac{1}{16}+\frac{1}{16}+\frac{1}{16}+\frac{1}{16}+\frac{1}{16}=\frac{5}{16}$

7. c)

In first toss, result can be 1,2,3,4,5

For 1, The second toss results can be 2,3,4,5,6

Career Avenues

For 2, The second toss results can be ,3,4,5,6
For 3, The second toss results can be ,4,5,6
For 4, The second toss results can be ,5,6
For 5, The second toss results can be 6

Then the required probability will be $= \frac{1}{6}\times\frac{5}{6}+\frac{1}{6}\times\frac{4}{6}+\frac{1}{6}\times\frac{3}{6}+\frac{1}{6}\times\frac{2}{6}+\frac{1}{6}\times\frac{1}{6}=\frac{5}{12}$

8. d)

P (no. of tosses is odd) = P (no. of tosses is 1, 3, 5, 7…)
P (no. of toss is 1) = P (Head in first toss) = ½
P (no. of toss is 3) = P (tail in first toss, tail in second toss and head in third toss) = ½ x ½ x ½ = 1/8

P (no. of toss is 5) = P (TTTTH) = $\left(\frac{1}{2}\right)^5 = \frac{1}{32}$

P (no. of tosses is odd) = $\frac{1}{2}+\frac{1}{8}+\frac{1}{32}..... = \frac{\frac{1}{2}}{1-\frac{1}{4}} = \frac{2}{4}$

9. Required probability = $\frac{\frac{1}{2}\times\frac{2}{3}}{\frac{1}{2}\times\frac{1}{3}+\frac{1}{2}\times\frac{2}{3}} = \frac{2}{3} = 0.666$

10. c)

It means 3 heads appears in 1st 8 trials
Probability of getting exactly 3 heads in 1st 9 trials ${}^9C_3 p^3 q^6$
p= probability of getting head =1/2
q= probability of getting head =1/2
then

${}^9C_3\, p^3 q^6 = {}^9C_3 \frac{1^3}{2}\frac{1^6}{2} = {}^CC_3\left(\frac{1}{2}\right)^9$

For the 10th trial head must be appeared = ${}^9C_3\left(\frac{1}{2}\right)^9\times\frac{1}{2} = \frac{84}{1024} = 0.082$

11. To get both head and tail atleast once, we have to toss minimum two times.
The random variable x represents first toss starting with head

x	2	3	4	5	……
P(X)	$\frac{1}{2^2}$	$\frac{1}{2^3}$	$\frac{1}{2^4}$	$\frac{1}{2^5}$	……

then = $E\ x = \sum x.p\ \ x = 2\times\frac{1}{2^2}+3\times\frac{1}{2^3}+4\times\frac{1}{2^4}+5\times\frac{1}{2^5}+....=3/2$

Career Avenues

The random variable y represents first toss starting with tail.

y	2	3	4	5	……
P(Y)	$\frac{1}{2^2}$	$\frac{1}{2^3}$	$\frac{1}{2^4}$	$\frac{1}{2^5}$	……

then $E\ x = \sum y.p\ y = 2\times\frac{1}{2^2}+3\times\frac{1}{2^3}+4\times\frac{1}{2^4}+5\times\frac{1}{2^5}+.... = 3/2$

Then average number of toss required = $E\ x + E\ y = \frac{3}{2}+\frac{3}{2} = 3$

12. Probability to lost post – office 1 = $\frac{1}{5}$

Probability to lost post – office 2 = $\frac{4}{5}\times\frac{1}{5} = \frac{4}{25}$

Then total probability of lost = $\frac{1}{5}+\frac{4}{25} = \frac{9}{25}$

Required probability = $\frac{\frac{2}{25}}{\frac{9}{25}} = \frac{4}{9} = 0.444$

13. c)

$P\ A\cup B = P\ A + P\ B - P\ A\cap B$

Since $P\ A\cap B$ is not always equals to zero so,

$P\ A\cup B\ P\ A + P\ B is\ flase.$

14. $P\ X+Y\le 1 = \int_{x=0}^{1}\int_{y=0}^{1-x} f_{x,y}\ x,y\ dx\,dy = \frac{1}{3} = 0.33$

15. In Poisson distribution

Mean = First Moment = λ

And second Moment = λ^2

Given that second moment is 2, then

$\lambda^2+\lambda = 2$ then by solving we can get $\lambda = 1$

16. P(H) =0.3

P(T) =0.7

Then the probability of getting head for the first time in 5th toss is = P(T) P(T) P(T) P(T) P(H) = 0.7 x 0.7 x 0.7 x 0.7 x 0.3 = 0.072

17. When three dice are thrown, then total number of possible cases = 6x6x6=216

Favourable cases of all three dice have same number are,

{(1,1,1) (2,2,2) (3,3,3) (4,4,4) (5,5,5) (6,6,6)} = 6

Then required probaility = $\frac{6}{216} = \frac{1}{36} = 0.028$

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